

ALLPLAN 2022

Basics Tutorial

Basics Tutorial

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Before you start ...

This tutorial gives you a quick and practical introduction to all important tools for designing and modifying in Allplan 2022.

This tutorial contains several examples in the form of exercises, showing how to design in 2D and how to get started in 3D modeling.

Requirements

This guide assumes that you are familiar with and have a working knowledge of Windows and Allplan 2022.

The basics are covered in the manual. In particular, you should know

- How to start and close Allplan 2022
- How to create a project
- How to make drawing files current, open them in edit or reference mode, or close them
- How to use the tools for zooming; in particular, how to display the entire drawing on the screen and how to zoom in on details

Work through the exercises in the sequence specified because tools that are presented in detail in the earlier exercises are only referred to by name in later exercises.

Your feedback

We are always trying to improve the overall quality of our program documentation. Your comments and suggestions are important to us, and we welcome feedback.

Please do not hesitate to contact us to express criticism or praise concerning the documentation. Feel free to contact us as follows:

Documentation

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81829 Munich, Germany

Email: dokumentation@allplan.com

Sources of information

The documentation on Allplan consists of the following parts:

- The **Help** is the main source of information for learning about and using Allplan.
While Allplan is running, you can get Help on the current tool by selecting F1. You can also select  **What's This** in the  **Help** drop-down list (right side of the title bar) or use the Shift+F1 keyboard shortcut and click the icon on which you need Help.
- The **Manual** consists of two parts. The first part shows how to install Allplan. The second part provides an overview of basic concepts and basic terms in Allplan as well as showing you how to make entries in Allplan.
- The **Basics Tutorial** guides you step by step through the most important tools for designing and modifying elements in Allplan.
- The **Architecture Tutorial** guides you step by step through the process of designing a building, analyzing the building data in reports, and printing the results.
- The **Engineering Tutorial** guides you step by step through the process of creating and printing key plans, general arrangement drawings, and reinforcement drawings.
- The **New Features in Allplan 2022** guide provides information about what's new in the latest version.
- Each volume in the **Step-by-Step** series deals with a specific concept or series of tools or tasks in Allplan, deepening the knowledge and understanding of this special subject. The areas covered include data exchange, system administration, geodesy, presentation, 3D modeling, and so on. Serviceplus members can download these guides as PDF files from the Training – Documentation area of Allplan Connect (<https://connect.allplan.com>).
- You can also find numerous publications on social networks.

Further help

Tips for Efficient Usage

The  **Help** drop-down list (right side of the title bar) provides **Tips for Efficient Usage**. You can find practical tips and tricks showing you how to use Allplan efficiently and how to carry out everything with ease.

Hello Allplan!

Hello Allplan! shows how to get started with Allplan. Go to the Hello Allplan! website (<https://www.allplan.com/hello-allplan/>), where you can find more information, a project that you can import into Allplan, and videos that complement the **Hello Allplan!** tutorial.

User forum (for Serviceplus customers)

Allplan forum in Allplan Connect: Users exchange information, valuable tips relating to everyday work, and advice on specific tasks. Register now at connect.allplan.com

On the internet: Solutions to frequently asked questions

You can find solutions to numerous questions answered by Technical Support in the comprehensive knowledge database at <https://connect.allplan.com/support/solutions.html>

Feedback on the Help

If you have suggestions or questions on the Help, or if you come across an error, send an email to: dokumentation@allplan.com

Training, coaching, and project support

The type of training you are given is a decisive factor in the amount of time you actually spend working on your own projects: A professional introduction to the programs and advanced seminars for advanced users can save you up to 35% of your editing time!

A tailor-made training strategy is essential. Our authorized seminar centers offer an extensive range of programs and are happy to work out a custom solution with you that will address your own needs and requirements:

- Our **sophisticated, comprehensive seminar program** is the quickest way for professional users to learn how to use the new system.
- **Special seminars** are designed for users who want to extend and optimize their knowledge.
- **One-on-one seminars** are best when it comes to addressing your own particular methods of working.
- One-day **crash courses**, designed for office heads, convey the essentials in a compact format.
- We are also happy to hold seminars on your premises: These include not only Allplan issues but also analyses, process optimization, and project organization.

To get more detailed information about the current training program, visit our website (<https://www.allplan.com/training>) and consult our online seminar guide, where you can find both face-to-face training and online training.

Unit 1: introduction

This unit briefly introduces the seven exercises in this tutorial.

You create a separate project for these exercises and define basic settings which apply to all the exercises.

At the end, a short troubleshooting section makes sure you succeed.

Objectives

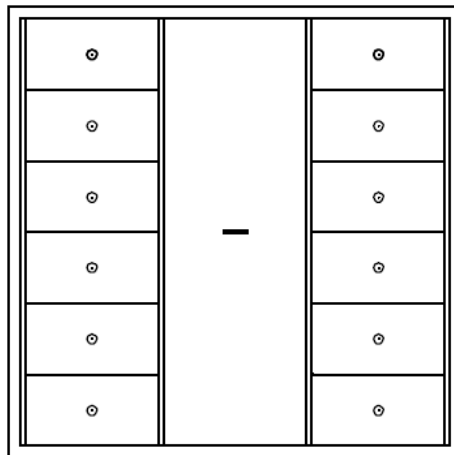
The Actionbar configuration (on page 22) is the default setting in Allplan 2022.

You can use the **Draft** role for all exercises in this tutorial. Exercises 1 to 6 require the **Design** task with the **2D Objects**, **2D Areas**, **Quick Access**, **Filter**, and **Change** task areas.

The last exercise requires the **Modeling** task with the **3D Objects**, **Boolean Operators**, and **Work Environment** task areas.

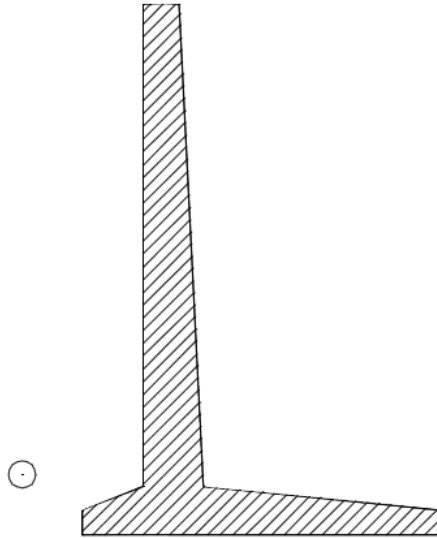
Exercise 1: file cabinet with drawers

- Precision drafting with the  **Point snap options**
- Applying the tools of **direct object modification**
- Basic edit tools
- Modifying the distance between parallel lines
- Stretching entities
- Copying and mirroring elements



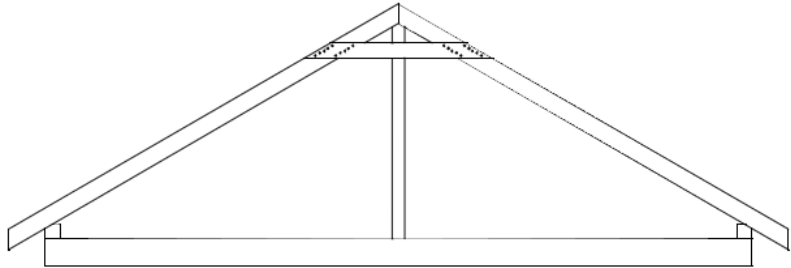
Exercise 2: retaining wall with a drainage

- Delta point
- Hatching and hatching definition
- Polyline entry tools



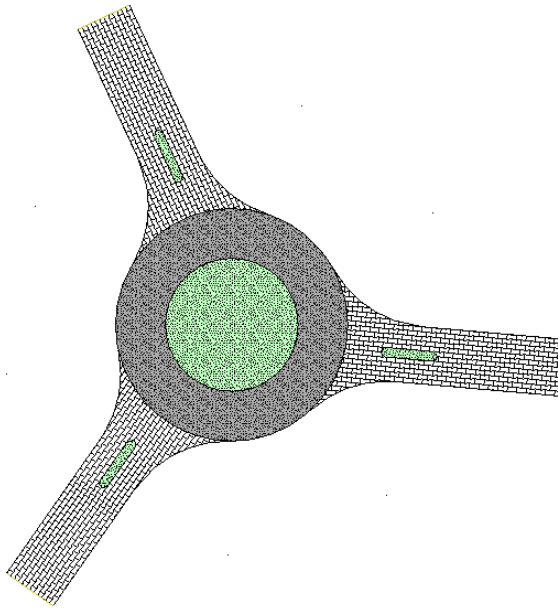
Exercise 3: purlin roof

- More tools for editing elements
- Creating labels with text leaders



Exercise 4: rotary

- Creating a circle
- Area detection and island detection
- Patterns and pattern definition
- **Direct object modification** – copy and rotate



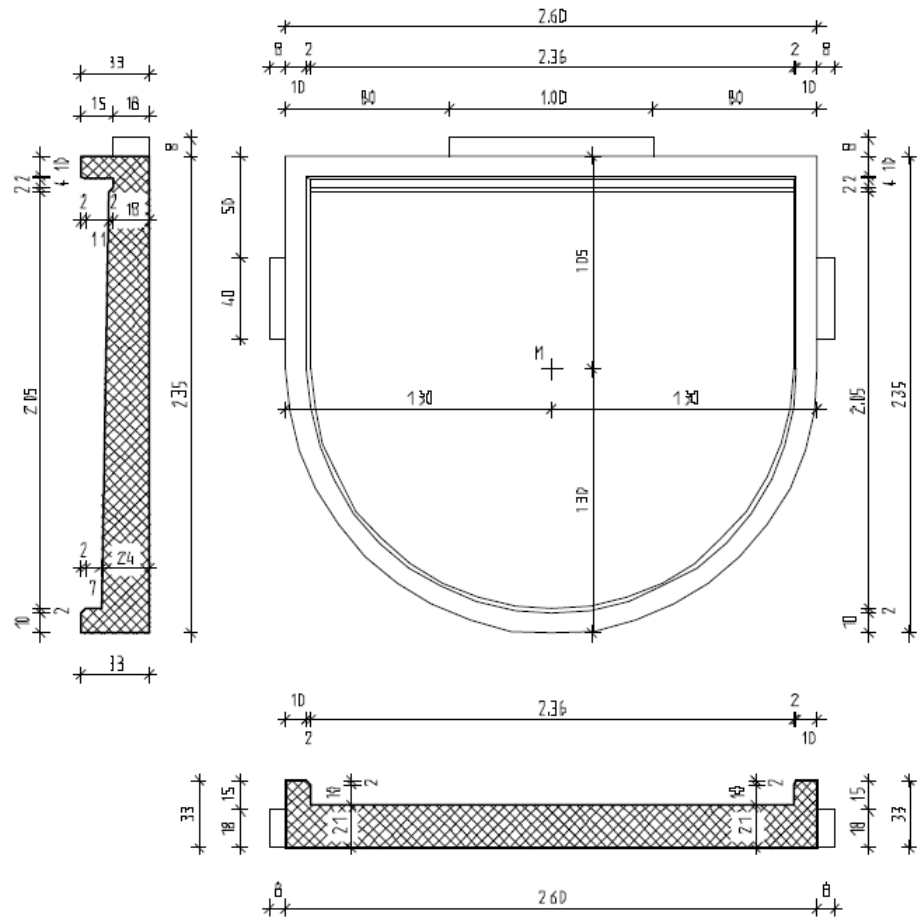
Exercise 5: title block

- More tools for editing elements
- Creating and saving symbols
- Retrieving a symbol from the library

Index	Type of Change	Date / Name
Contents		
Precast Balcony, Type 12		
Project		
New Condominium With Underground Parking		
Client	Client Street, Munich	Date XX.XX.20XX
		Drawn by: Name
Architect	Architects Street, Munich	Checked by: Name
		Scale: 1:50/25
Engineer	Consulting Engineers Street, Munich	Layout number XXX

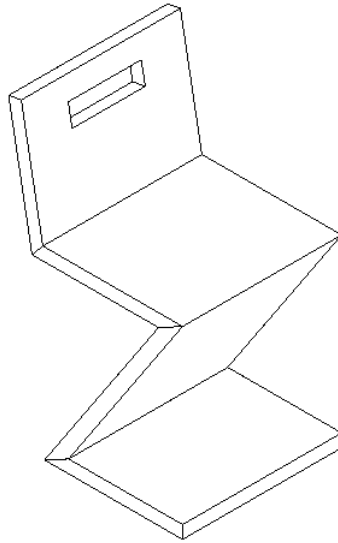
Exercise 6: precast balcony

- Creating and modifying dimension lines
- Hatching and hatching definition
- Print preview



Exercise 7: Rietveld chair

- Introduction to modeling
- Using a work plane
- Excursus: surfaces



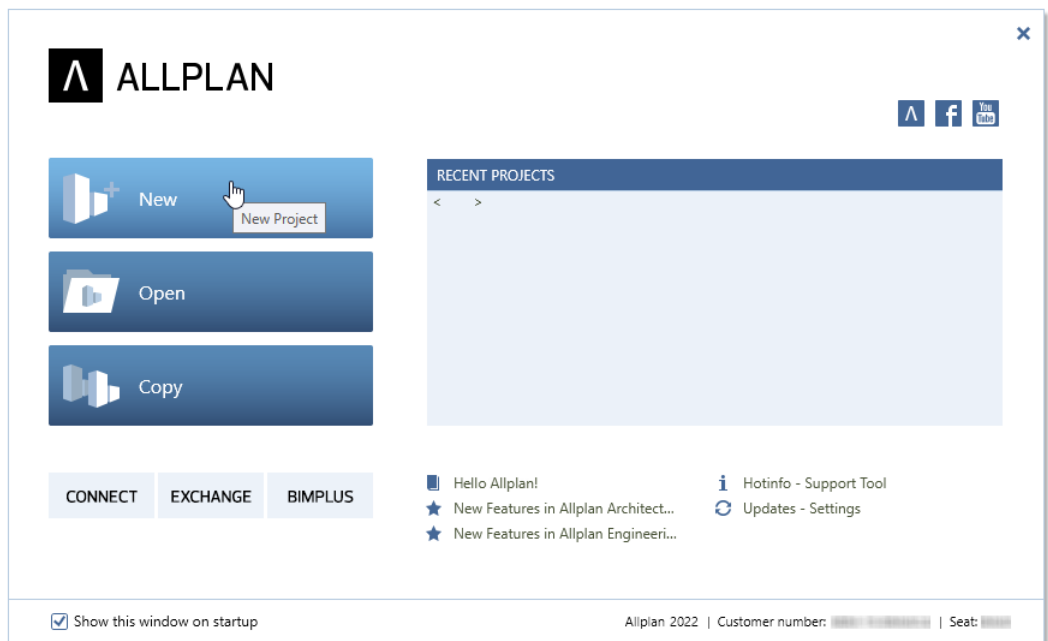
Creating the project

You use drawing files and NDW files in Allplan 2022. Drawing files are organized by **project**.

You will start by creating a project for the exercises in this tutorial.

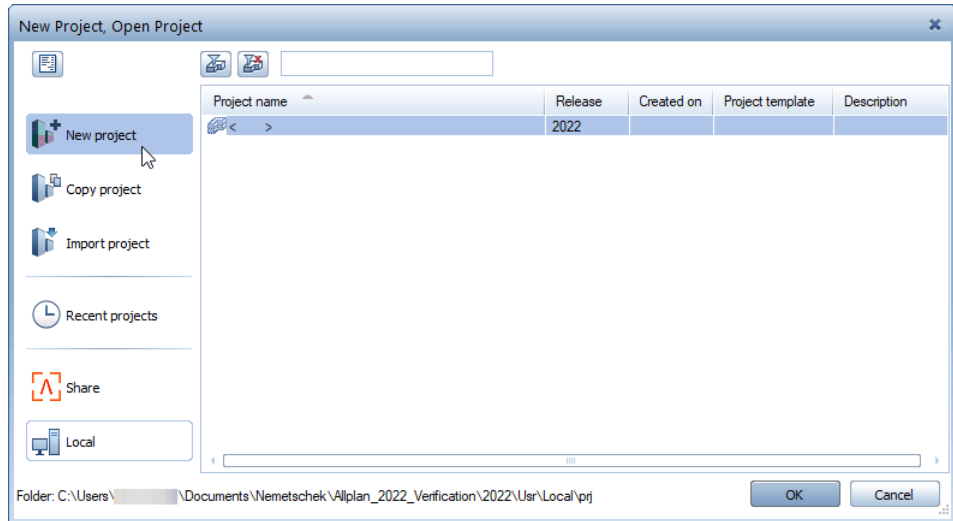
To create a project

- 1 After having started Allplan 2022, you can create a new project directly from the welcome screen. Click the corresponding tool and go to step 4.

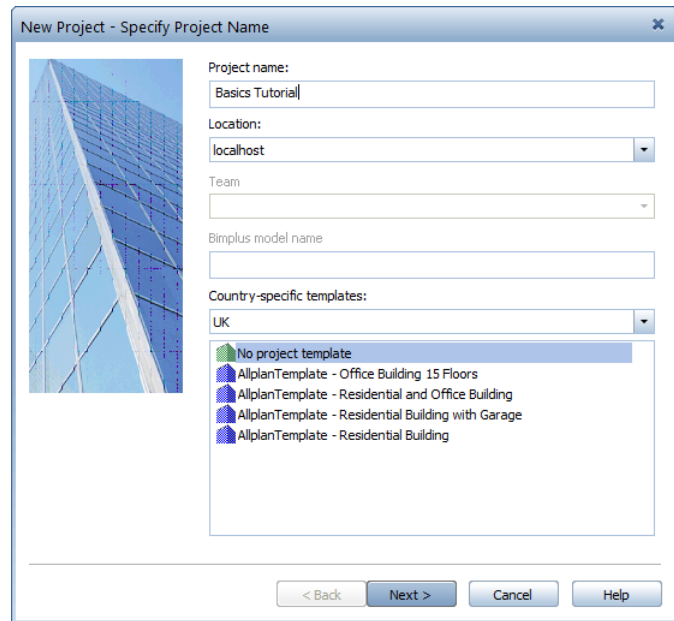


- 2 If you have turned off the welcome screen, click  **New Project**, **Open Project...** on the Quick Access Toolbar.

- 3 The **New Project, Open Project** dialog box opens. Click  **New Project**.



- 4 *New Project – Specify Project name*
Enter the project name: **Basics Tutorial**.
- 5 Use **Location** to select the place where you want to save the new project. If you have the **Workgroup Manager** option, you can see all computers that were added as servers to the workgroup. If you have the **Allplan Share** option, you can also select BIMPLUS.
- 6 If you want to save the project on BIMPLUS: Select the team and enter the Bimplus model name.
- 7 You can select the country for the project templates in **country-specific templates**.
The new project does **not require a project template**. Therefore, you can ignore this setting.

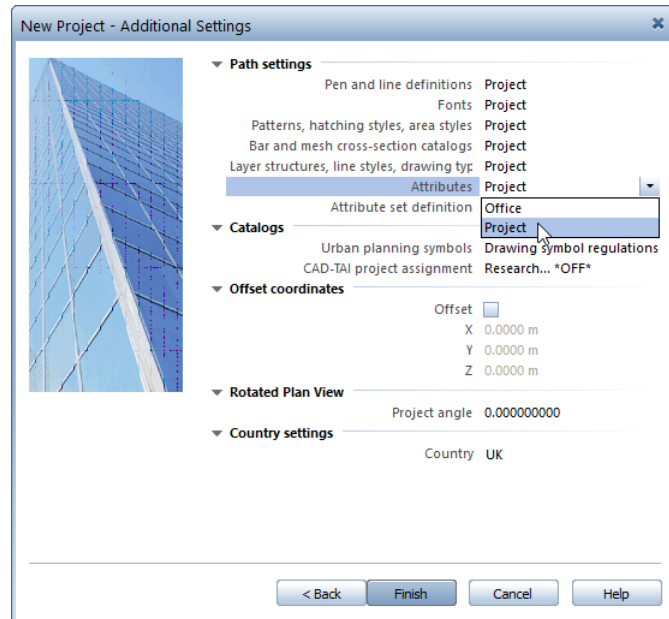
8 Click **Next >**.

- 9 You will define new patterns and hatching styles as you go along. You will use project-specific settings so that you do not inadvertently change the office standard.

Note:

Organizing projects is described in detail in the Allplan Help and in the Architecture Tutorial.

Change all **path settings** to **project** and click **Finish** to confirm the dialog box.



Allplan creates the **Basics Tutorial** project and opens it automatically.

Path settings:

The path settings define which pen definitions, line definitions, hatching definitions, fonts, and material catalogs you use. You usually work with the office standard.

Office:

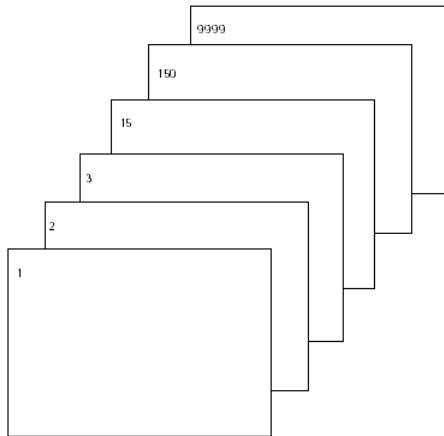
Choose this option if you want to use the same settings to work on different projects. In a network environment, the office standard is the same on all computers and can be changed only by users with special privileges.

Project:

Choose this option if you want to use the settings for this project only. The settings can be different from those in the office standard.

Understanding drawing files

In Allplan, the actual design and data creation happens in *drawing files*. These are the equivalent of the transparencies used in conventional building design. Drawing files can be used to give projects a structure. In IT terms, a drawing file is a conventional file stored on your hard drive. You can display and edit up to 1024 drawing files at once – in other words, you can have several files open simultaneously. A project can contain up to 9999 drawing files. Without layers, the individual building elements (such as walls, stairs, labeling) are drawn in different drawing files and superimposed like transparencies.



To edit the drawing files, you must open them. You can do this in the **Open on a project-specific basis: drawing files from fileset/building structure** dialog box.

Drawing file status

By selecting the drawing file status, you define the drawing file in which you draw and which drawing files are visible or can be changed. The following illustration shows the different drawing file statuses. You can find explanations in the table.



Number	Drawing file status	Comment
1	Current or active	The current or active drawing file is the one in which you draw. There must always be one current or active drawing file.
2	Open in edit mode	Elements in drawing files open in edit mode are visible and can be modified. Up to 1024 drawing files can be open simultaneously (regardless of whether they are current, in edit mode, or in reference mode).
3	Open in reference mode	Elements in drawing files open in reference mode are visible, but they cannot be modified. You can configure the program to use the same color for all elements in reference drawing files. To do this, select the  Options - Desktop environment and open the Display page. You cannot open empty drawing files in reference mode.
4	Not selected	Elements in these drawing files are not visible.
5	Empty	Empty drawing files have no data type icon.
6	Assigned temporarily	The drawing file is temporarily assigned to the fileset. This assignment will be lost as soon as you switch to a different fileset.

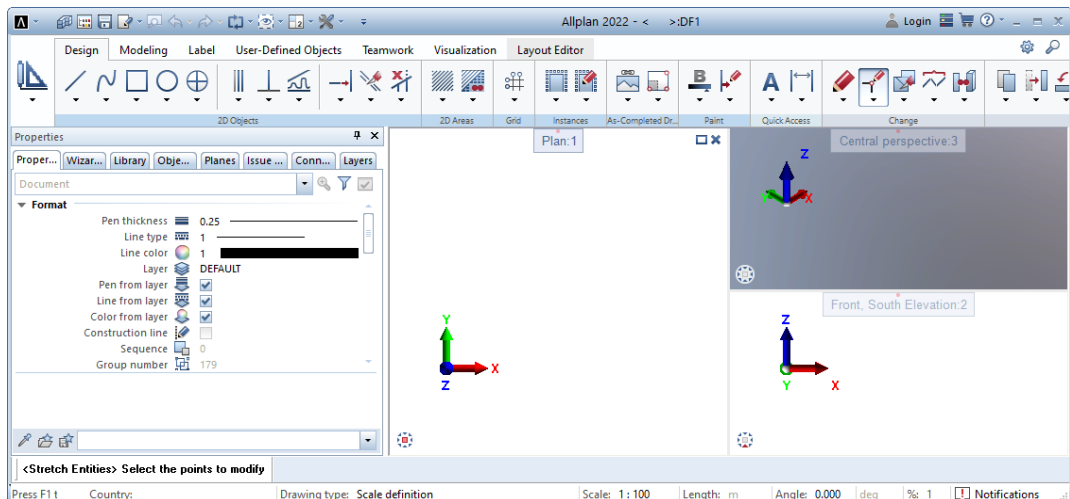
7	Open in reference mode	The drawing file has been opened by another user in the workgroup environment.
8	Open in reference mode	The drawing file has been opened by another user in the workgroup environment; the color red indicates that the drawing file has changed. You can apply the changes by selecting Update drawing file on the shortcut menu. In the  Options – Desktop environment page, you can configure the program to inform you of changes in reference drawing files.
9	Linked drawing file	The drawing file contains model data or views and sections linked with other drawing files. Allplan links drawing files when you create views and sections with the tools in the Reinforcement Views task area. You can use the shortcut menu to list all reference drawing files of the current drawing file or open its reference drawing files in edit or reference mode.
10	Views and sections created automatically	The drawing file contains views and sections that you created by using the shortcut menu in the "Derived from building structure" area or views and sections that you created with the tools in the Sections task area of which the results were saved in this drawing file. These views and sections are usually linked with other drawing files. Allplan considers the model data in these linked drawing files.
11	Update locked	By using the shortcut menu in the "Derived from building structure" area, you can lock the update of drawing files with views and sections created with the Update automatically option not being selected. Allplan cannot update the result until the drawing file is unlocked or the Update automatically option is selected. After having confirmed a prompt, you can create a new view or section in the drawing file.
12	Views and sections placed manually	The drawing file contains views and sections created with the tools in the Sections task area. These views and sections might be linked with other drawing files. Allplan considers the model data in these linked drawing files.

Basic settings

Define the settings that you will use in the exercises.

Actionbar Configuration

The Actionbar configuration is the default setting in Allplan 2022. When you work with this configuration, you can find the **Actionbar** above the workspace. In addition, you can see the **Properties**, **Wizards**, **Library**, **Objects**, **Planes**, **Issue Manager**, **Connect**, and **Layers** palettes on the left side.



Click the **Allplan icon** on the left side of the title bar to open important tools such as save, copy, import, and export. The title bar also includes the **Quick Access Toolbar**. By using a drop-down list, you can select the tools that you want to display on this toolbar. In addition, you can show and hide the menu bar, define the sequence of the tools on the Quick Access Toolbar, and click **Customize User Interface...** to open the **Customize** dialog box – **Actionbar** tab.

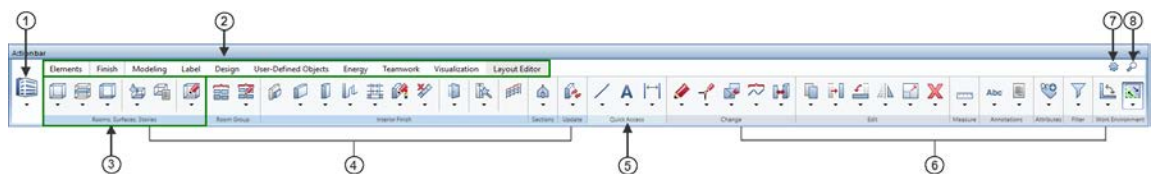
Contents and structure of the Actionbar

The **Actionbar** contains all Allplan tools grouped by tool groups. The groups of tools are combined into task areas, which are combined into tasks. You can find the tasks required for each discipline in a role.

If you position the cursor over an icon for a short time, a tooltip will appear. Use **Customize User Interface – Customize** tab to specify whether this is a *Plain tooltip* or an *Advanced tooltip*. **Plain tooltips** contain a short description of the tool, **Extended tooltips** consist of a short description of the tool; they can also contain further information, illustrations and/or links to short explanatory videos. **Customize user interface** can be found in the title bar of the **Allplan** viewport under  **Customize quick access toolbar**.

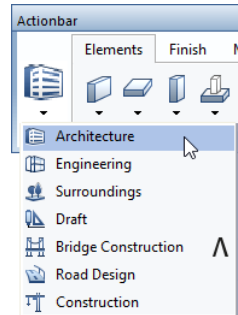
The **Actionbar** is docked to the top of the working area. If you want, you can drag the Actionbar to the bottom of the workspace and dock it there. You can also make the **Actionbar** float anywhere on your screen. By double-clicking, you can dock it to the place where it was docked last.

Structure of the Actionbar



- 1 – Role
- 2 – Tasks arranged on tabs
- 3 – Task area
- 4 – Varying task areas
- 5 – **Quick Access** task area
- 6 – Fixed task areas
- 7 –  **Actionbar Configurator**
- 8 –  **Find**

Selecting the role



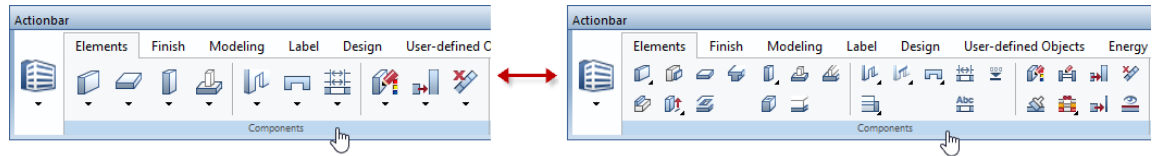
You start by selecting a **role** (1). The drop-down list contains all the roles that are available to you, both those you purchased (license-dependent) and self-configured. The **tasks** (2) that are available to you change with the selected role. To open a task, click the corresponding tab. Each task is subdivided into appropriate areas. You can find areas in different colors, indicating varying and fixed **task areas** (3). The varying task areas (4) change with the selected task, such as the **Components** task area of the **Elements** task. You can find the fixed task areas (6) in all roles and tasks, such as the **Change** and **Filter** task areas. The **Quick Access** task area (5) contains tasks with tools used frequently.

The first time you open Allplan the task areas of the **Actionbar** are collapsed. The flyout menus of the visible tools contain all the tools in the collapsed area.

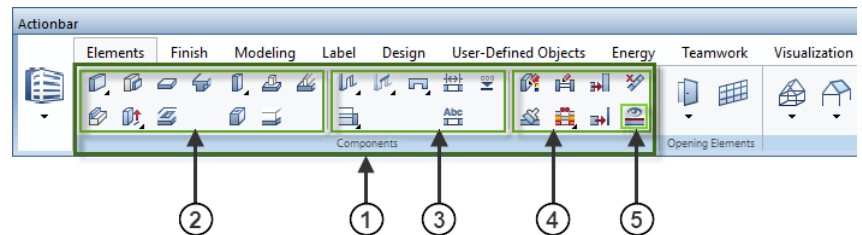
When you point to the name line of a task area, the cursor changes to: 

You can maximize or minimize a task area by double-clicking within the name line of a task area. A maximized task area shows more tools, which can also have flyout menus.

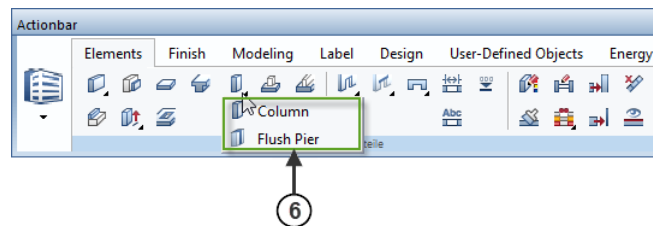
Note: You can maximize or minimize **all** task areas of the task currently selected by selecting and holding the Ctrl key while double-clicking within the name line of a task area. You can expand or collapse **all** areas **across tasks and roles** by selecting and holding Ctrl+Shift while double-clicking within the name line of a task area. How many task panes are displayed expanded depends on the width of your Allplan window. If the window is not wide enough, Allplan starts on the left side, expanding as many task areas as possible.



Structure of a task area in detail



- 1 - Task area
- 2 - **Create** group of tools
- 3 - **Create in context** group of tools
- 4 - **Modify in context** group of tools
- 5 - Tool



- 6 - Tool menu = flyout menu of a tool

An expanded task area (1) contains one or more groups of tools (2/3/4). Different groups of tools are separated by vertical lines. The tools are grouped by topic. Some tools have flyout menus (6) where you can find similar tools.

Palette window

The palette window displays the palettes as tabs. Palettes are important controls of Allplan, making the user interface simple and easy to use. You can float or dock each palette individually. In addition, you can arrange the palette window or individual, free-floating palettes around the edge of the workspace or make them float anywhere on your screen. You can even configure Allplan to automatically show or hide the palette window or palettes arranged around the edge.

Properties palette

When the **Properties** tab is open, you have the following options:

Drop-down list at the top

Tools at the top and bottom

Properties

Select active elements

 Zoom in on selected objects

 Filter step by step

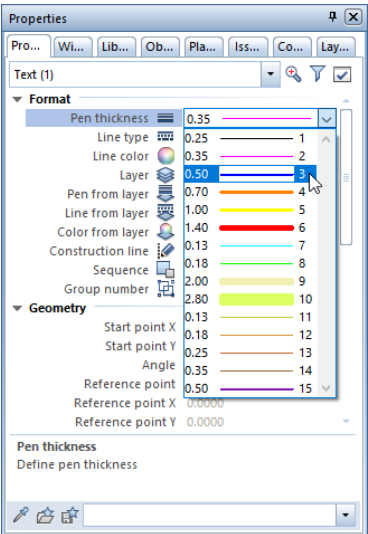
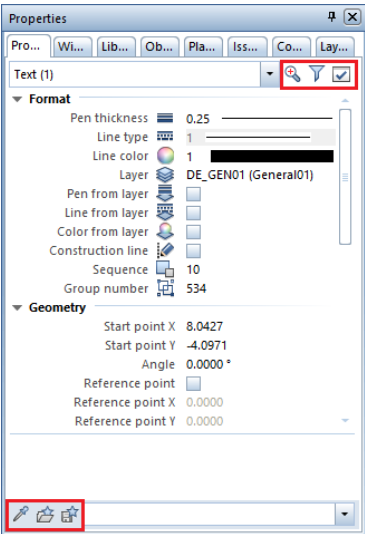
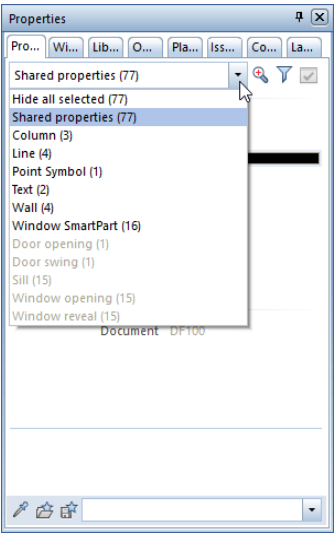
 Change the properties of the selected object

 Match parameters

 Load favorite

 Save as a favorite

Modify properties



Wizards palette

When the **Wizards** tab is open, you have the following options:

Drop-down list at the top

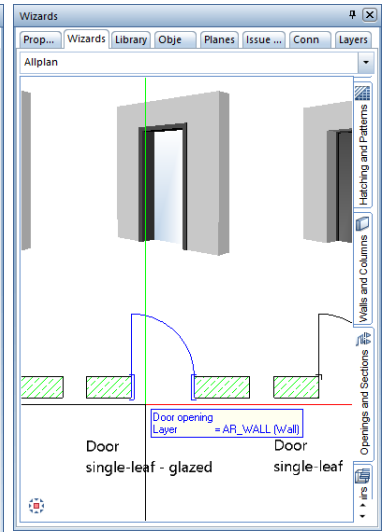
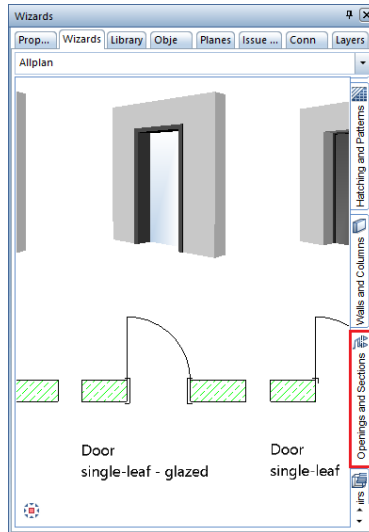
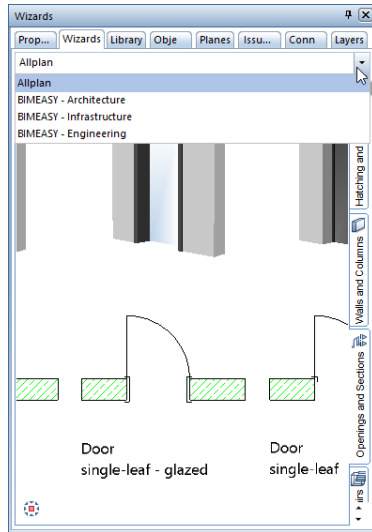
Tabs on the right side of the palette

Available tools

Select a wizard group

Select a wizard

Select a tool

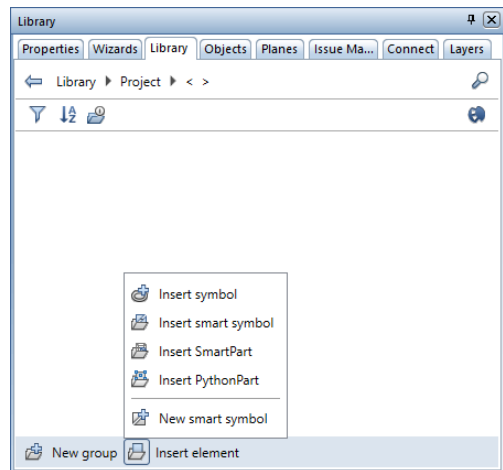
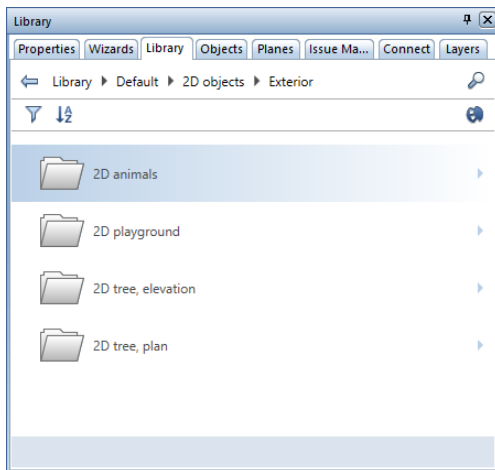


library palette

The **Library** palette provides a  **Filter** that you can use to show or hide specific types of library elements (symbols, smart symbols, SmartParts, and PythonParts).

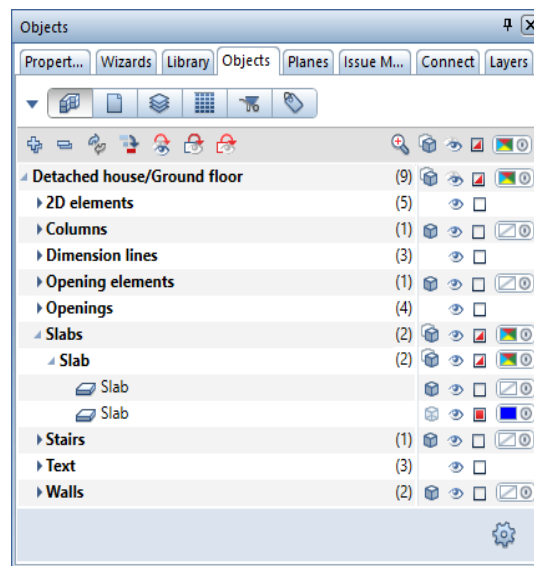
After having opened a folder, you can see all subfolders with library elements (symbols, smart symbols, SmartParts, and PythonParts) if you have not filtered out library elements. You can select the required objects. You can also add your own objects to the corresponding library folders.

 **Connect (online library)** takes you to the **Connect** palette to log in to Connect. If you are already signed in, the online library will open immediately.



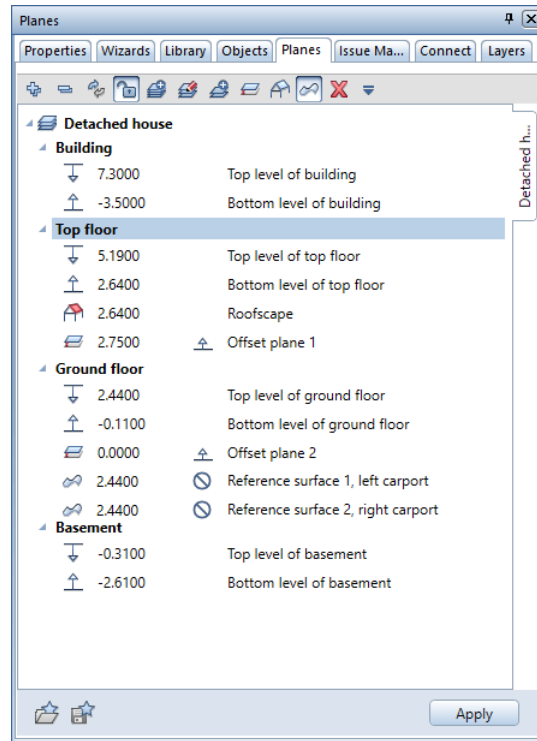
objects palette

The **Objects** palette lists all objects and elements in the currently open drawing files (**current** or **open in edit mode** or **open in reference mode**). You can sort these objects by  **Topology**,  **Drawing file**,  **Layer**,  **Material**,  **Trade**, or  **Attribute**. Here you can selectively show or hide selected objects, or select or deselect them. Furthermore, it is possible to set the transparency of 3D objects as well as to apply color coding (both have an effect only in the **Animation** view type).



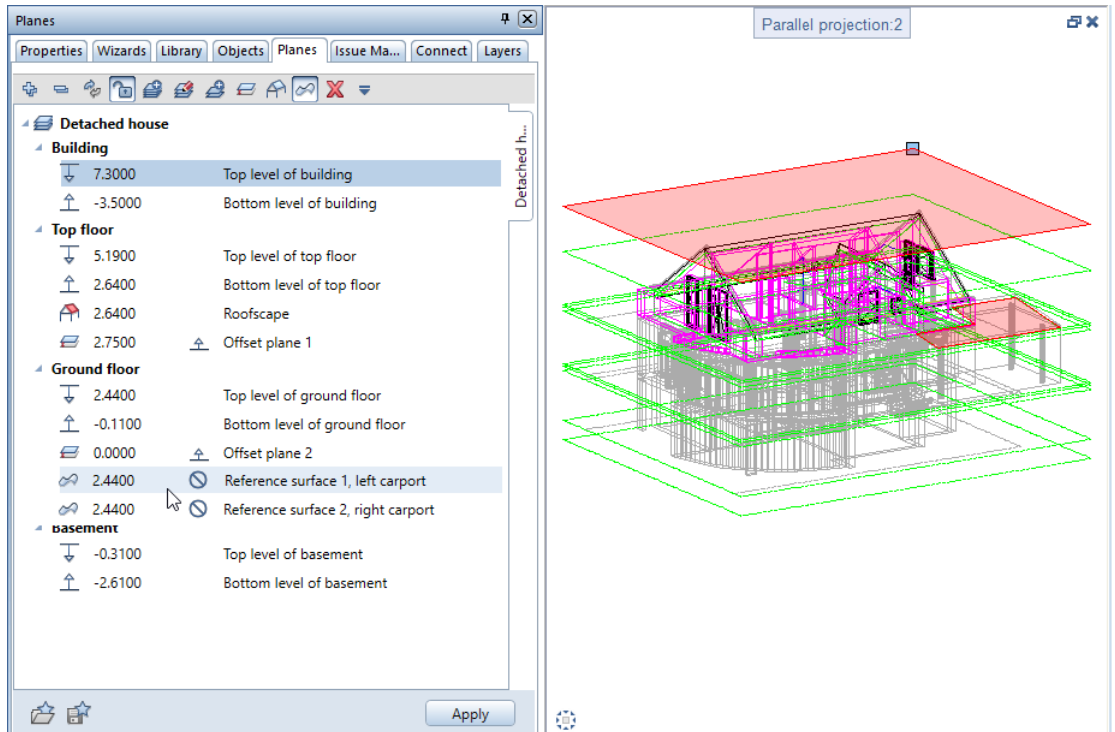
Planes palette

The **Planes** palette displays all plane models in the active project. Each plane model has its own tab. While creating components, you can keep track of the default planes and all other objects on which the heights of the components can be based.



To edit a plane model, click  **Modification mode on/off**. If modification mode is on () , you can make entries in the **Planes** palette. Allplan displays the planes of the plane models in all viewports. When you point to or select an entry of the plane model in the tree structure, this entry is also highlighted in the detection color in the viewport. So, you can immediately check the position of the plane and see the effects of changes.

The tools in the **Planes** palette are similar to those provided by the **Floor Manager** dialog box of the building structure. For example, you can use the **Insert pair of planes**, **Insert or replace roofscape**, **Insert or replace reference surface**, and **Insert offset plane** tools. Use **New model** to create a new plane model.

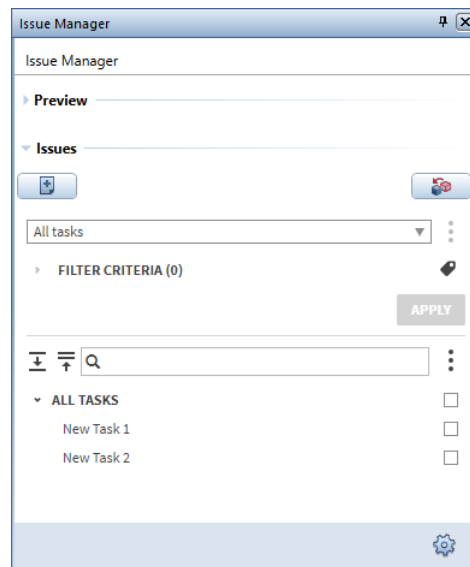


Issue Manager palette

You can use the **Issue Manager** palette to communicate with all those involved in a Bimplus project. In Allplan, you can access the issues of the currently loaded Allplan project directly from Bimplus. In addition, you can use Allplan to create new issues in Bimplus or edit existing issues. You can also import or export issues in BCF format or export issues to an Excel file from Allplan.

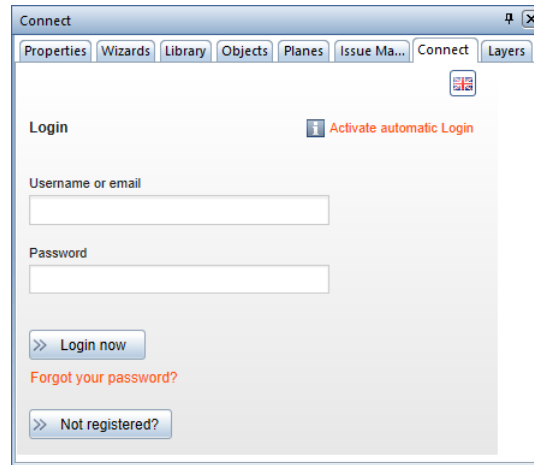
This is only possible if you have used the Allplan workstation to sign in to Bimplus and if the Allplan project is linked with a Bimplus project, that is to say, the Allplan project data has been uploaded to Bimplus *at least once*.

Note: See Handling projects with Allplan Bimplus for more information about handling projects in a BIM-compliant manner with Bimplus, the web service offered by ALLPLAN GmbH.



Connect palette

The **Connect** palette takes you directly from Allplan to content on Allplan Connect. Click **ALLPLAN Webservice**, enter your access data in the **ALLPLAN** dialog box and click **Login**.

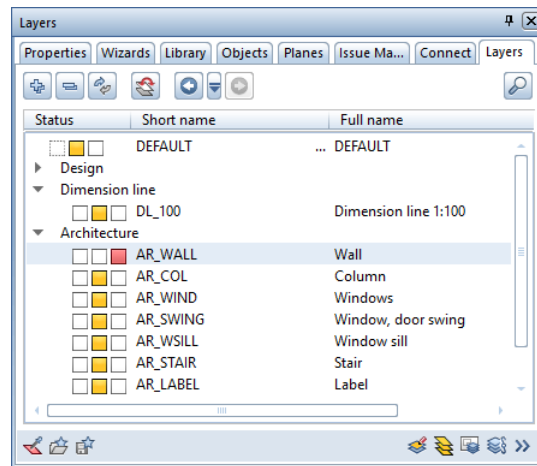


If you have already signed in using  **Login** in the title bar or **Issue Manager** palette, you can access Allplan Connect content directly without logging in again.

Note: As login data, use either your Allplan Connect Login Data or – if available – the e-mail address provided in connection with your registration with Bimplus and the password you chose when registering. Signing out is only possible via  **Login** in the title bar.

Layers palette

By using the **Layers** palette, you can access the layer structure quickly and easily. The palette displays the entire layer hierarchy. You can define the visibility of layers, change the layer status, select the current layer, and choose privilege sets and print sets. When you select the  **Match current layer** button at the bottom of the palette, you can click an element to use its layer as the current layer. Click  to save the current layer setting as a favorite file (*.lfa); click  to load a favorite file. You can sort the layers alphabetically by clicking on the column labels **Short name** or **Full name** in the list header.



Note:

You can find the **Customize User Interface...** tool in the drop-down list of the Quick Access Toolbar. On the **Palettes** tab, you can arrange and customize the palettes to suit your needs. You can show and hide the palettes as you need.

As an alternative, open the shortcut menu of a palette and select **Customize...**

Settings on the Actionbar

You can use the **Draft** role for all exercises in this tutorial.

To make settings on the Actionbar for the following exercises

Tip: You can expand or collapse all task areas of the task currently selected by selecting and holding the Ctrl key while double-clicking within the name line of a task area. The width of the Allplan window defines how many task areas can be expanded. If the window is not wide enough, Allplan starts on the left side, expanding as many task areas as possible.

- 1 Select the  **Draft** role.
- 2 Open the tab of the **Design** task.
- 3 Expand the **2D Objects** task area by double-clicking within the name line of this task area.

Note: The **Actionbar** is docked to the top of the working area. If you want, you can drag the Actionbar to the bottom and dock it there. You can also make the **Actionbar** float anywhere on your screen. By double-clicking, you can dock it to the place where it was docked last.

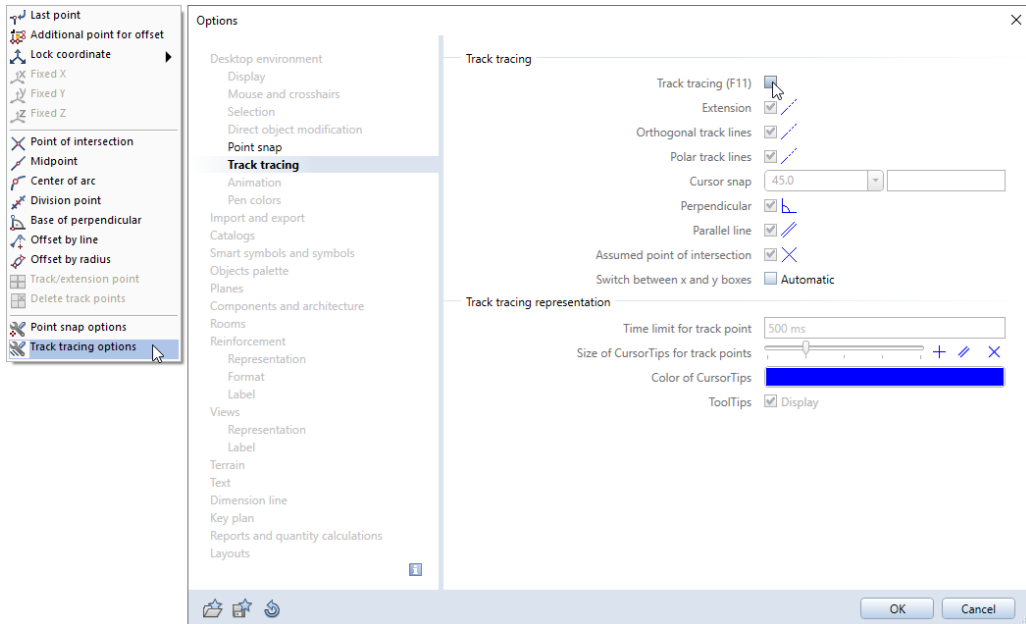
Track tracing

Track tracing helps you design intuitively. You work with fixed measurements in most of the following exercises. Therefore, you can turn off track tracing, which is on by default.

Tip: You can quickly turn track tracing on and off at any time while entering points. Just select the **F11** key or click  **Track line** in the dialog line.

To turn track tracing on and off

- 1 Click  **Line** (**Draft** role – **Design** task – **2D Objects** task area or **Architecture** role – **Elements** task – **Quick Access** task area).
- 2 Right-click in the workspace and select  **Track tracing options** on the shortcut menu.
- 3 Turn off the **Track tracing** option



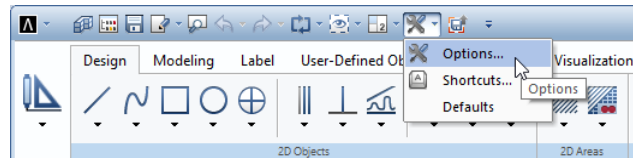
- 4 Click **OK** to confirm the settings; select  **Line** tool.
- 5 Repeat these steps if you want to turn track tracing on again.

Options

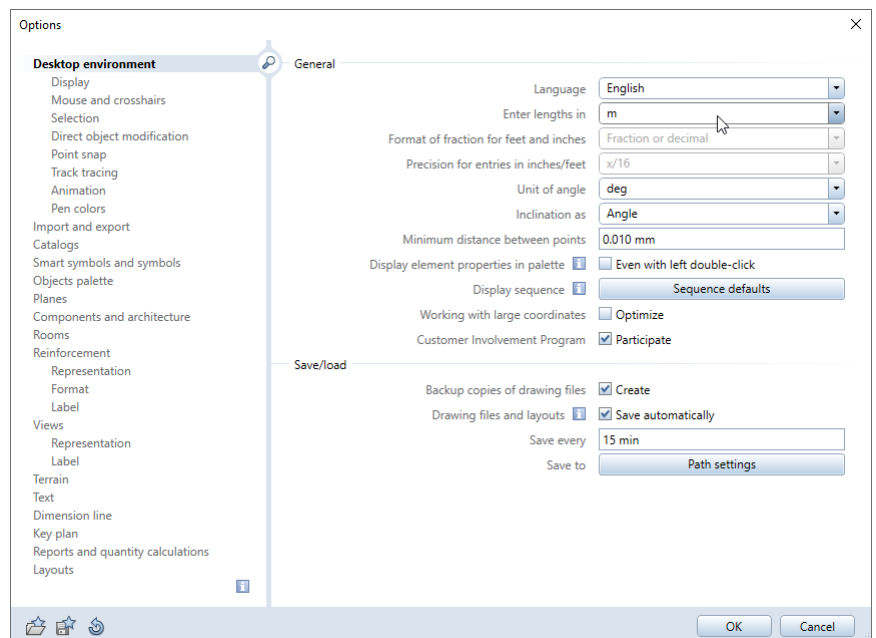
You will use the unit **m** for the following exercises.

To define settings in the options

- 1 Open the  **Default Settings** drop-down list on the Quick Access Toolbar on the title bar and click  **Options**.



- 2 The **Options** dialog box opens. Click **Desktop environment** in the left area.
- 3 Check the **Enter lengths in** option in the **General** area on the right side. If it is not **m**, click the button and select **m**.



- 4 Click **OK** to confirm the settings.

Pen settings

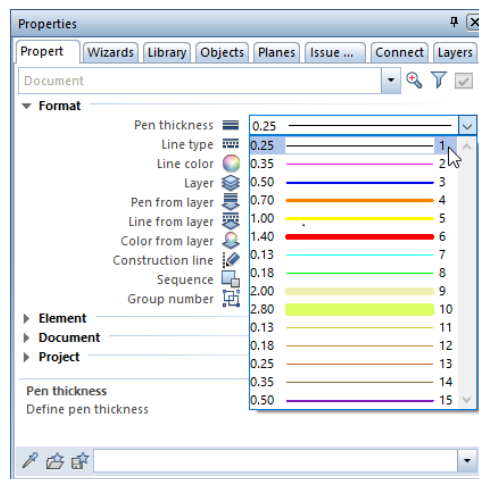
Before you start drawing, define the line thickness (pen) and the line type in the **Properties** palette. You can change these settings at any time.

Each element can be given one of Allplan's 256 line colors or element colors. However, the way elements look depends on the setting of the **Color stands for pen** option in  **Show/Hide** ( **View** drop-down list on the Quick Access Toolbar):

- When the **Color stands for pen** option is selected, the element automatically appears in the color that is linked with the current pen thickness (default setting).
- When the **Color stands for pen** option is not selected, the element appears on the screen in the line color selected.

To define the pen and line type

- 1 Switch to the **Properties** palette, go to the **Format** area, click the **Pen thickness** drop-down menu, and choose **0.25 mm**. You can see the selected pen.



- 2 Click the **Line type** drop-down list and choose **1** (a continuous line).
 - 3 Click the **Line color** drop-down list and choose **1** (black).
-

Draw all exercises in this guide with these basic settings, even if this is not explicitly specified.

Allplan provides two different options for structuring drawing files:

- The building structure
- The fileset structure

You can use these two structures in parallel. The building structure is particularly useful for logically structuring a building.

The exercises in this tutorial do not build on one another. Therefore, use a separate drawing file for each exercise.

Controlling what's on your screen

Tip: The  **Actionbar Configurator** contains the **Window** task area, where you can find all the tools mentioned here. You can drag this task area onto your Actionbar.

Allplan provides various tools that you can use to control how your model and its design elements appear on the screen. Thus, you can always choose the tool best suited to the task at hand.

You can access these tools from various places in Allplan. For example, you can use the  **View** and  **Window** drop-down lists on the **Quick Access Toolbar**. You can also use the shortcut menu or the viewport toolbar. You can even use the keyboard and mouse to control what's on your screen.

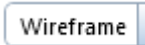
Tools for displaying the model

By using the tools on the viewport toolbar, you can not only move freely on the screen but also display any view. You can zoom in on any section or detail of your drawing as closely as you want. You can even use different view types to display the entire model or selected components.

Most of these tools are 'transparent' tools; in other words, you can use them while another tool (for example, **Line**) is active.

You cannot see the viewport toolbar until you point to the lower border of the viewport, guaranteeing as large a workspace as possible. When you use multiple viewports, each viewport has its own viewport toolbar.

Tool	Use
Left area:	
 View flyout menu	You can use this tool to choose between plan view and any of the predefined standard views.
 Zoom All	You can use this tool to select the display scale so that you can see all the elements in the visible files.
 Zoom Section	Note: If you have loaded a view by using  Save, Load View , you can see this view only.
 Navigation Mode	You can use this tool to zoom in on a section. To do this, enclose the elements you want to zoom in a selection rectangle.
 Previous View	You can use this tool to turn navigation mode on or off in the active viewport. In this mode, you can use the mouse to view a 3D model.
 Next View	Note: You can move in sphere mode or in camera mode (while selecting and holding CTRL KEY).
 Save, Load View	You can use this tool to restore the previous view or display scale (if you had selected a different view or scale before you selected the current setting).
 3D View	You can use this tool to restore the next view or display scale (if you have already selected a subsequent view or scale).
 Element Selection	You can use this tool to save the current view under a name of your choice or retrieve a view you saved beforehand.
 Drawing File Selection	You can use this tool to display 3D models in three-dimensional space in a perspective view by entering an eye point (observer) and a target point. You can choose between parallel projection and central projection for the perspective view. You can also use this tool to create a view based on the building structure.
 Always on Top	You can use this tool to select the design entities you want to display in the active viewport. The program temporarily hides all the other design entities.
or 	You can use this tool to temporarily hide drawing files that are currently visible in the active viewport.
	You can use this tool to place the viewport so that it is always on top (that is, in front of) the other ones.
	Note: You can only use this tool if you have <i>not selected</i> the Connected option and the viewport is <i>not maximized</i> .

Tool	Use
	Exposure (only for the Animation and RTRender view types) Right area: You can use this box to control the brightness in viewports of the Animation or RTRender view type. You can enter a value between -25 and 25. Important! This setting <i>only</i> changes the way elements look in the active viewport. It has <i>no</i> effect on rendering.
	Section Display You can use this tool to display your design in an architectural section for which you have already defined the  Clipping Path.
	Display Scale You can use this tool to select the scale for displaying the model on the screen. The display scale governs the ratio between the model on the screen and its real-life dimensions. The scale therefore changes automatically if you change the size of sections on the screen. You can see the current display scale on the viewport toolbar in the lower border of a viewport.
	View Type You can use this list box to select one of the predefined view types (Wireframe, Hidden, Animation, Sketch or RTRender) for the active viewport. Of course, you can also select a view type you defined yourself. Click  to modify various settings of the view types. The settings apply to all the viewports that use this view type. Click New view type to define and save your own view types. When Layout Editor is open, you can switch between Design view and Print view (= preview of resulting printout). Note: You can find more tools for controlling what's on your screen in the  View and  Window drop-down lists on the Quick Access Toolbar and on the shortcut menu (in navigation mode only).

How to

Sometimes, things will not immediately work out as required. This list helps you succeed.

What if ...

- **... I have selected the wrong tool?**
Select the ESC key and click the correct icon.
- **... I make a mistake as I go along?**
Select the ESC key to cancel (you might have to do this several times).
Click  Undo.
- **... I have inadvertently deleted the wrong elements?**
If  Delete is still active, right-click twice.
If no tool is active, Click  Undo.
- **... I have unintentionally opened a dialog box or entered wrong values?**
Click Cancel.

And what if ...

- **... the workspace is empty although there is data?**
 - Click  Zoom All (viewport toolbar).
 - Click  Plan.
- **... the workspace is suddenly divided into a series of different viewports?**
Click  1 Viewport ( Window drop-down list on the Quick Access Toolbar).
- **... specific kinds of elements such as text or hatching do not appear in the workspace?**
Click  Show/Hide ( View drop-down list on the Quick Access Toolbar) and check that the relevant element type is selected.

Tip: Check whether the relevant layer is visible.

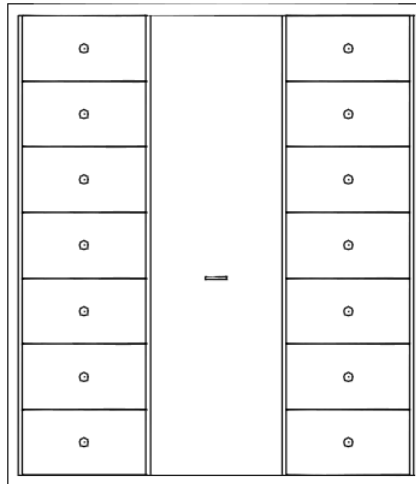
Unit 2: designing and modifying 2D elements

This unit presents the basic 2D tools in Allplan 2022. In particular, you will learn

- How to precisely place points by means of point snap, offset entry, and other tools
- How to work with track tracing and direct object modification
- How to modify existing elements
- How to apply hatching and patterns – you will familiarize yourself with the polyline entry tools, which are used by countless Allplan tools.
- How to modify and redefine hatching styles and patterns
- How to connect text and design with a leader
- How to create a title block and save it as a symbol
- How to dimension components

Exercise 1: file cabinet with drawers

In this exercise, you will design a file cabinet with drawers. You will then modify the height of the file cabinet.



You will use the tools in the **2D Objects** task area.

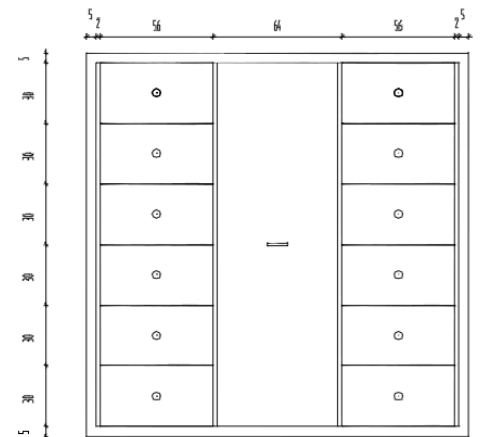
Task 1: designing the file cabinet

The first exercise shows how to draw rectangles and how to copy and mirror elements. In addition, you will learn how to use the **Reference point**, **Point of intersection**, and **Midpoint** tools for precision drafting.

Tools:

-  Point snap options
Point snap and offset entry
-  Offset Polyline
-  Circle
-  Midpoint
-  Copy
-  Options – desktop environment – direct object modification
-  Copy and Mirror
-  Delta point

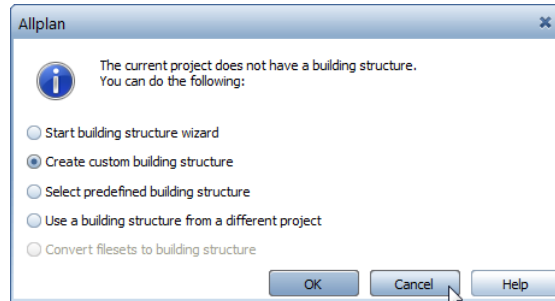
Objective:



Drawing the file cabinet as a rectangle

To draw the file cabinet as a rectangle

- 1 Click  **Open on a Project-Specific Basis** (Quick Access Toolbar).
- 2 You do not need to create a building structure for this tutorial. As the exercises in this tutorial do not build on one another, you will use a separate drawing file for each exercise. Therefore, click **Cancel**.

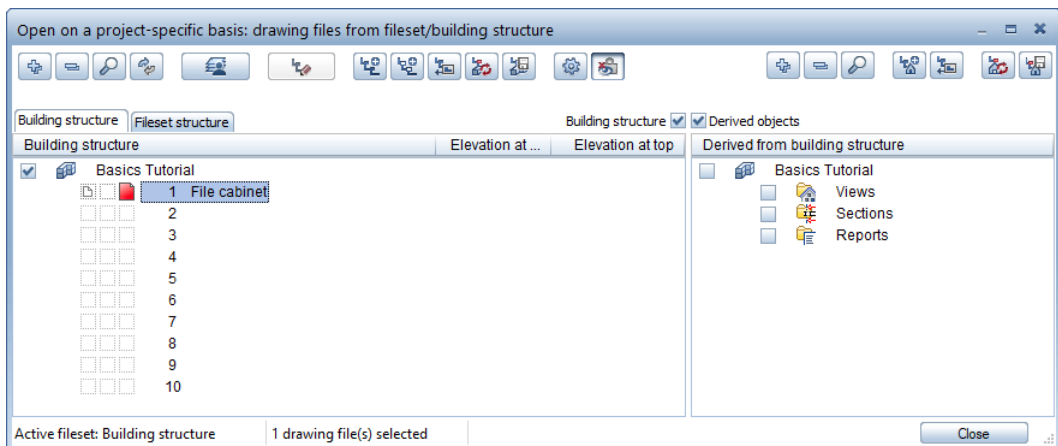


The **Open on a Project-Specific Basis** dialog box: Drawing files from drawing/building structure are opened on the **Building structure** tab.

- 3 The **Basics Tutorial** has 10 drawing files.
Click drawing file number **1** and click a second time inside the selection or select F2.

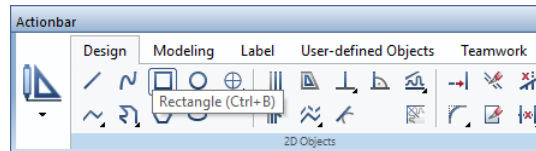
You can now enter a name for the drawing file.

- 4 Type **File cabinet** and select the Enter key to confirm.



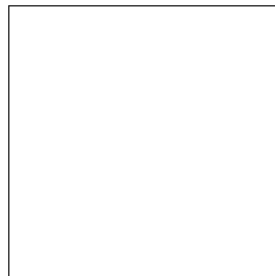
- 5 Click **Close**.

- 6 In the  **Window** drop-down list (Quick Access Toolbar), click on  **1 Viewport** to initially get the display of the following construction in the  floor plan view only. Furthermore, in the  **View** drop-down list, deselect the  **Show Coordinate System** setting.
- 7 Click  **Rectangle** (2D Objects task area).



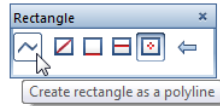
- 8 Select  **Based on diagonal line** in the input options.
Note:  Make sure that **Create rectangle as a polyline** is not selected in the input options because you will edit some lines of the rectangle later.
- 9 Click in the workspace to place the first point of the rectangle.
- 10 The rectangle is 1.8 m long in the x-direction. Therefore, enter  **dx = 1.8** in the dialog line. Select the Tab key to go to  **dy**.
- 11 The rectangle is 1.8 m high in the y-direction. Therefore, enter  **dy = 1.8** in the dialog line and select the Enter key to confirm. The file cabinet appears as a rectangle in the workspace.

Tip: To switch between , , and  in the dialog line, use the Tab key or Shift+Tab.



- 12 Select Esc to close the  **Rectangle** tool.

A note on Create rectangle as a polyline



You can use the  **Create rectangle as a polyline** option in the input options to do the following:

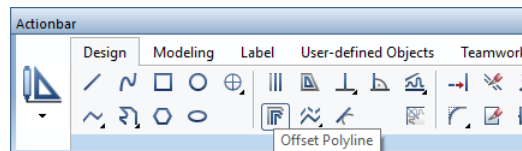
- If  **Create rectangle as a polyline** is selected, you create the rectangle as *one* connected element, which you can select with a single mouse click.
- If  is not selected, the rectangle consists of individual lines that you can select separately by clicking or as an **entity group** by selecting and holding the Shift key while clicking.

Drawing the file cabinet by using offset polyline

The next step is to draw the frame of the file cabinet by means of the **Offset Polyline** tool. **Point snap** helps you place points with great precision.

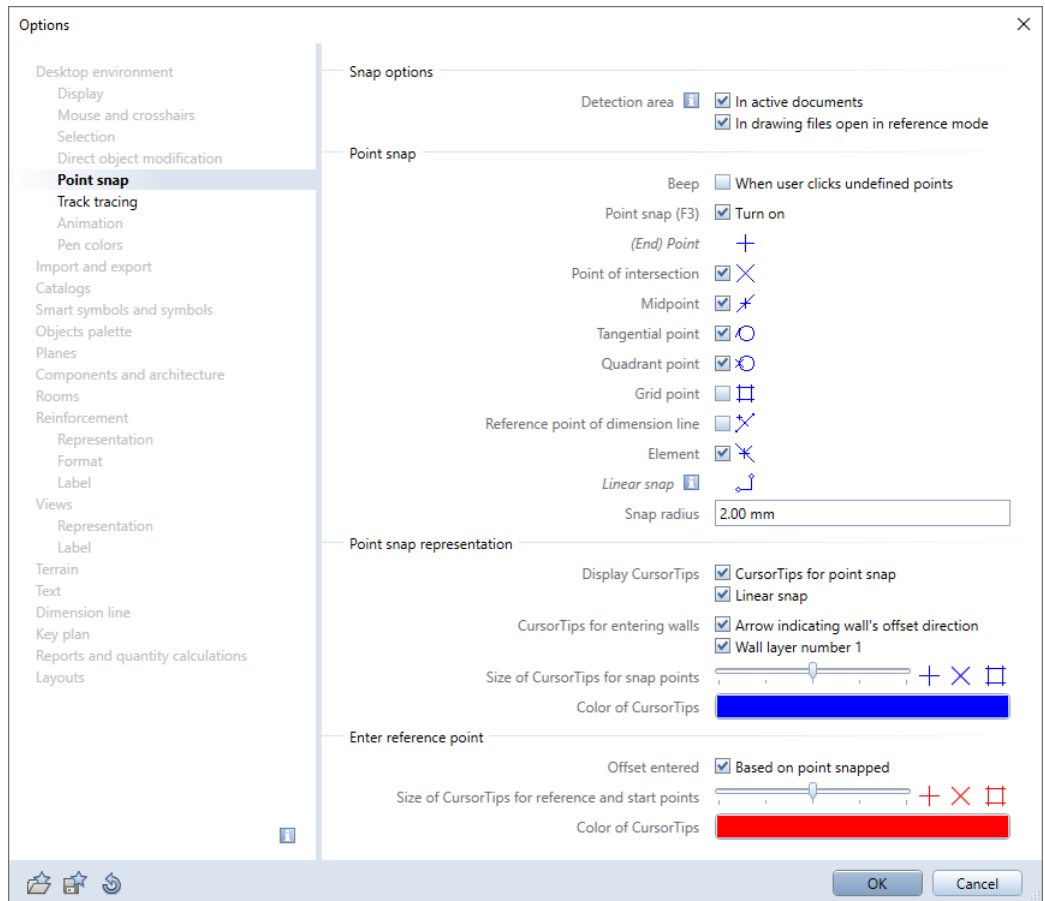
To draw the file cabinet by using "Offset Polyline"

- 1 Click  **Offset Polyline** (2D Objects task area).



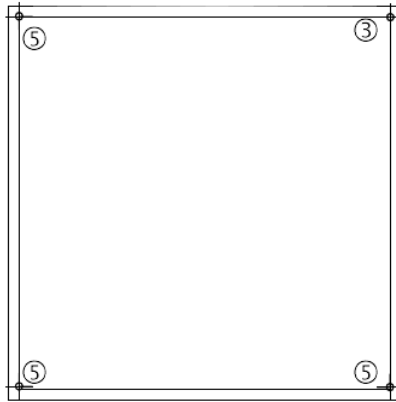
- 2 Enter the following values in the dialog line; select the Enter key to confirm each value.
Number of parallel lines: 1
Offset: 0.05
- 3 Click the upper-right corner of the rectangle.
Right is selected in the input options.

- 4 Open the shortcut menu by right-clicking in the workspace. Select  **Point snap options**; select all options on this page except **Grid point** and **Reference point of dimension line**.



As soon as you point to a point, the system will snap to this point. The point snapped is marked with a red X.

- 5 To draw the new rectangle outside the existing one, click the corners of the file cabinet *in a counterclockwise direction*. To close the polyline, make sure that the last corner you click coincides with the first one.



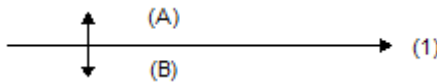
- 6 Select the Esc key to close the  **Offset Polyline** tool.

A note on the direction in which you enter the offset polyline

When you use  **Offset Polyline**, pay attention to the connection between the setting in the input options and the direction in which you enter the polyline:

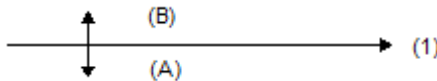
- When you select **right**, you must enter the points *in a counterclockwise direction* to draw the outer rectangle. By entering the points *in a clockwise direction*, you draw the inner rectangle.
- When you select **left**, it is the other way round.

Right setting:



(1) Direction
(A) Negative offset
(B) Positive offset

Left setting:



(1) Direction
(A) Negative offset
(B) Positive offset

Designing drawers

Create a drawer by using the **Rectangle** tool. Allplan provides a number of tools to help you place points with great precision. In the following section, you will design the drawer by snapping to points and entering offsets.

To design a drawer

- 1 Click  **Rectangle**.

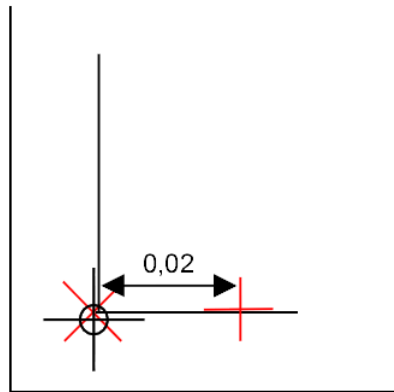
Note: Check that  **Create rectangle as a polyline** is *not selected* in the input options. Otherwise, the rectangle can be addressed as a single entity only. You will copy individual lines of the rectangle later. Therefore, make sure the lines can be selected individually.

- 2 Check that  **Delta point** is selected in the dialog line.
- 3 To specify the rectangle's starting point, point to the lower-left corner of the inner cabinet line.

The system snaps to this point, which is indicated by a blue CursorTip at the crosshairs. A red x appears on this corner, and the  **x-coordinate** and  **y-coordinate** boxes are highlighted in yellow in the dialog line.

- 4 Select the TAB key to go to the  **x-coordinate** box and enter **0.02**.

A red point symbol (+) moves to the right.



- 5 Click the corner or select ENTER to confirm.
You have defined the first point of the drawer.
- 6 Enter the coordinates of the diagonally opposite point of the rectangle in the dialog line:
 dx = 0.56
 dy = 0.30
Select ENTER to confirm.

dx=0.56

dy=0.30



- 7 This completes the first drawer. You will design the other drawers based on this first one.
- 8 Select ESC to close the  Rectangle tool.

A note on placing points by means of point snap and offset entry

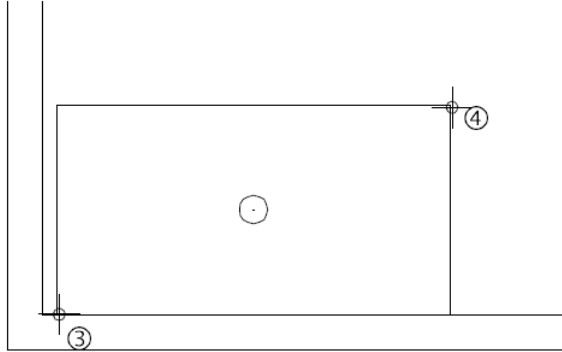
- Point to a point (do not click!):
Allplan snaps to this point; the boxes are highlighted in yellow in the dialog line.
- Enter the relative coordinates dx and dy in the dialog line.
- Select Enter to confirm: This places the point.

Creating the knob

Create the knob of the drawer by using the **Circle** tool. To position the knob exactly, you will use the **Midpoint** option.

To draw a knob

- 1 Click  **Circle**.
- 2 The **Circle** context toolbar opens. Click  **Circle based on center** and  **Enter full circle**.
- 3 To define the first point, open the shortcut menu and click  **Midpoint**.
- 4 First click the lower-left corner of the drawer.
- 5 Then click the upper-right corner of the drawer.
This defines the center of the circle.
- 6 Enter a radius of **0.02** in the dialog line and select ENTER to confirm.



- 7 Select ESC to close the tool.
-

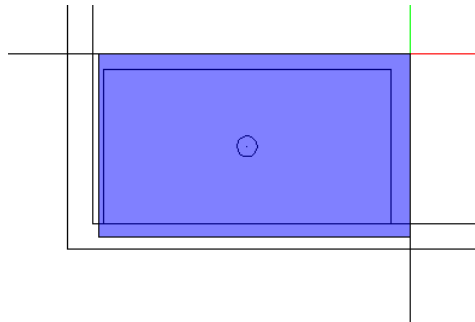
Copying the drawer

You will create the other drawers by **copying** the first one.

To copy the drawer

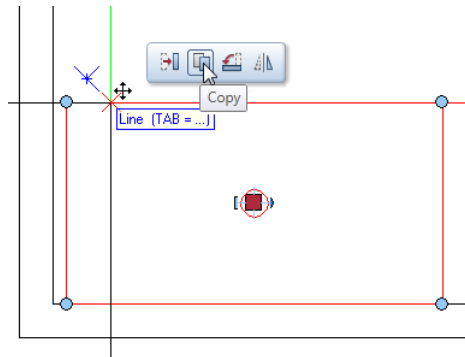
- 1 Enclose the entire drawer in a selection rectangle that you open from the lower left to the upper right (positive x-direction).

Tip: You can select elements by enclosing them in a selection rectangle. The default setting is  **Select Elements Based on Direction** (Work Environment task area): By opening the selection rectangle in the positive x-direction, you select only the elements that are fully bounded by the selection rectangle. By opening the selection rectangle in the negative x-direction, you select all elements that are fully bounded or partially bounded by the selection rectangle.

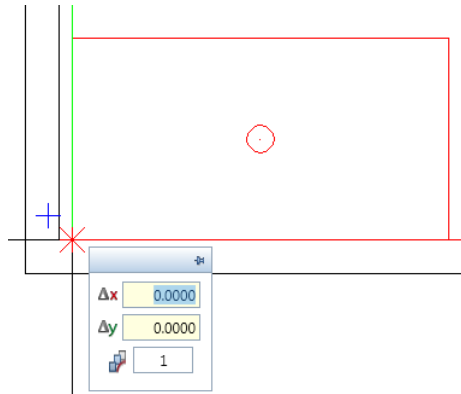


The knob is included because it is within the selection rectangle.

- 2 Point to a line of the drawer and click  **Copy** on the context toolbar.

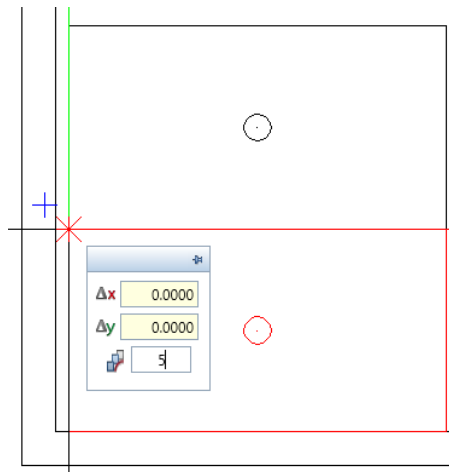


- 3 *From point or enter offset:*
Click the lower-left corner of the drawer.

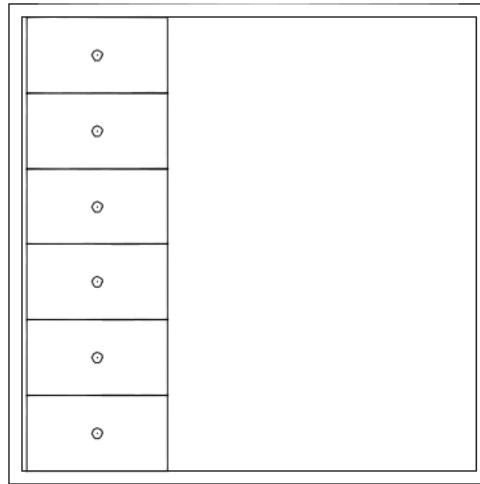


The drawer with the knob is attached to the crosshairs (at the corner you just clicked).

- 4 Go to the coordinate dialog box and enter **5** for the  **Number of** copies (do *not* select ENTER to confirm!).
- 5 *To point or enter offset*
To define the drop-in point, point to the upper-left corner of the drawer so that Allplan snaps to this point.



- 6 Click the drop-in point snapped.
Allplan creates the drawers on the left.
- 7 Select ESC to close direct object modification.

**Note:**

Use the **Work Environment** task area (**Actionbar** – all roles and all tasks) to define how and which elements are selected by the selection rectangle:



Selects the elements that are fully bounded by the selection rectangle.



Selects the elements that are fully bounded or partially bounded by the selection rectangle.



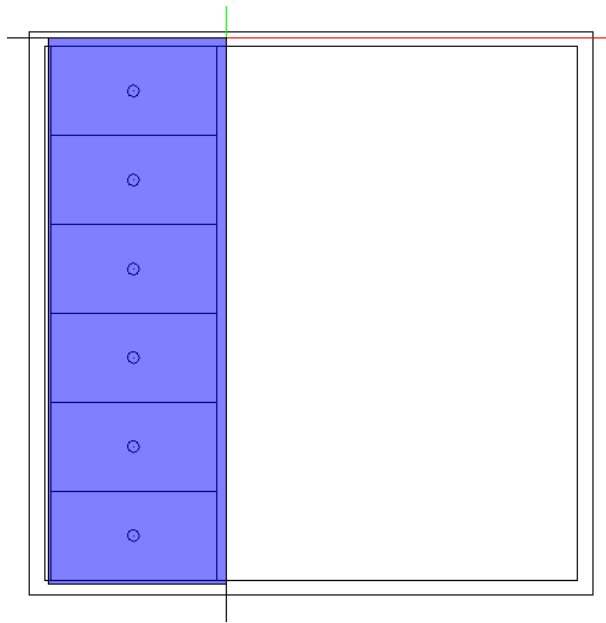
Selects the elements that are partially bounded by the selection rectangle.

Mirroring drawers

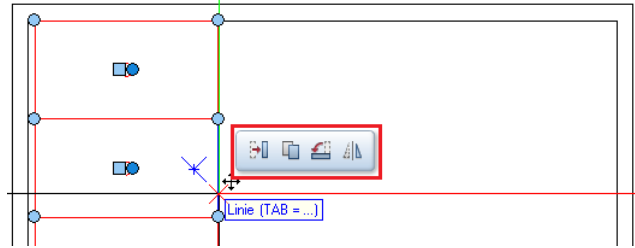
The next step is to copy and mirror the drawers to the right side by means of the **Copy and Mirror** tool. The center axis of the file cabinet will serve as the mirror axis.

To mirror the drawers to the right

- 1 Select the drawers by enclosing them in a selection rectangle that you open from the lower left to the upper right (positive x-direction).



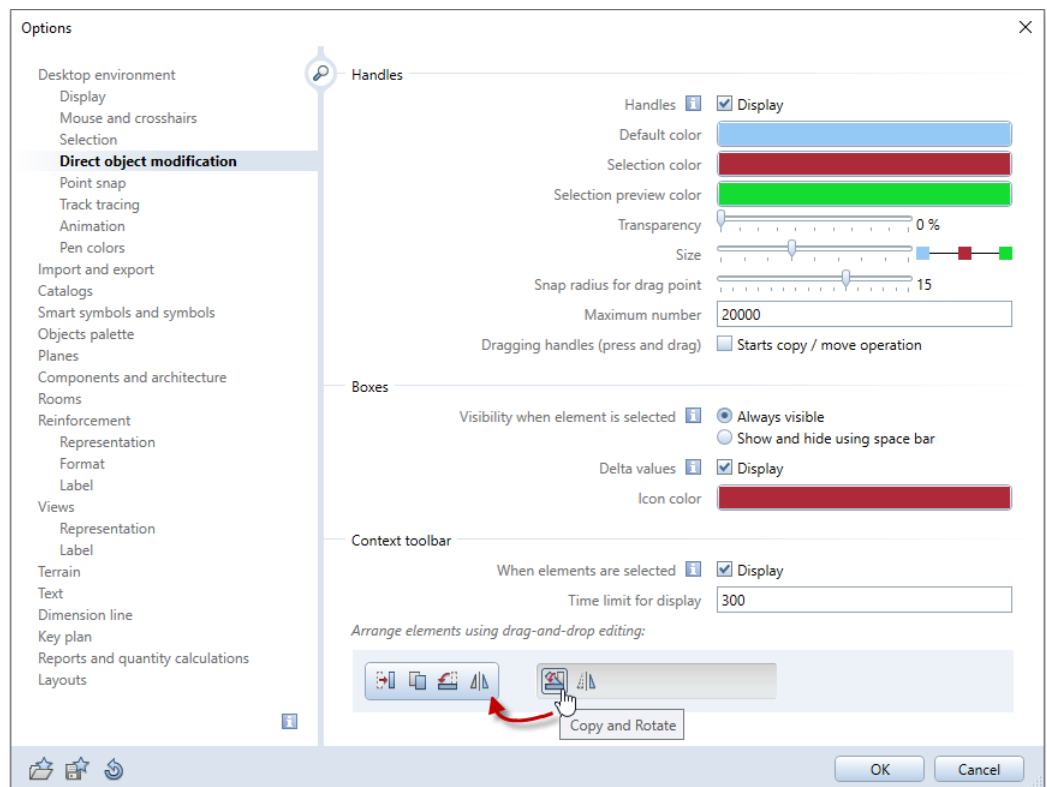
- 2 Point to a selected element, for example, a line.
- 3 The context toolbar for direct object modification opens, providing four tools:



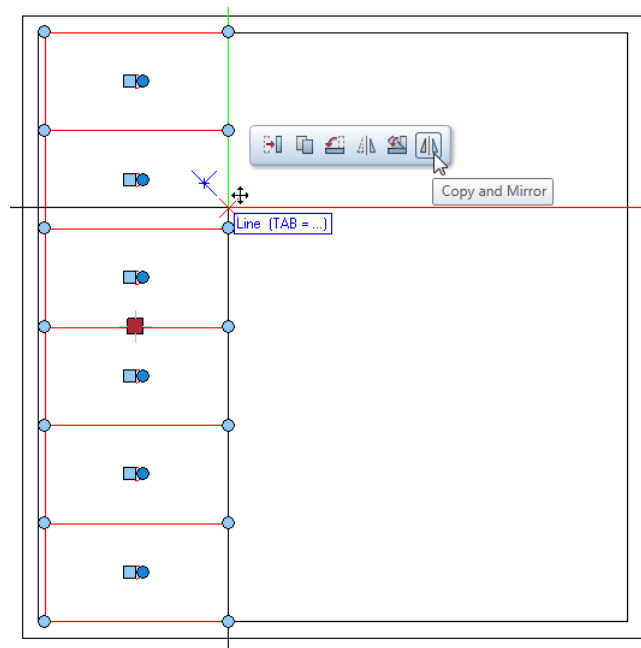
You can add two more tools to this context toolbar.

To do this, click  **Options** ( **Default Settings** drop-down list on the Quick Access Toolbar).

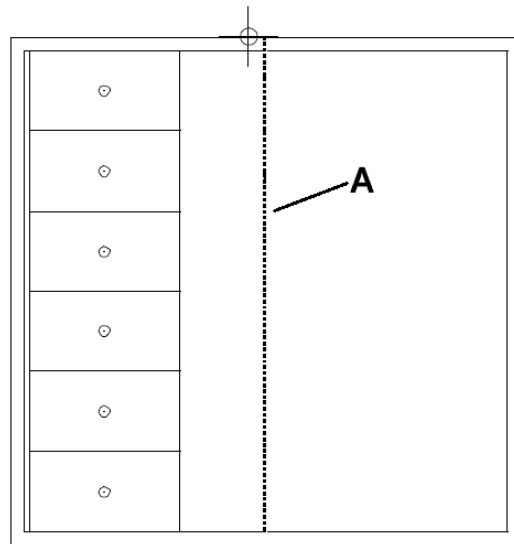
- 4 Open the **Desktop environment – Direct object modification** page and drag the  **Copy and Rotate** and  **Mirror without Copy** tools one after the other onto the context toolbar.



- 5 Click **OK** to close the **Options** dialog box.
- 6 Select the drawers again by enclosing them in a selection rectangle.
- 7 Point to a selected element, for example, a line.
The context toolbar opens. As you can see, it contains six tools.
- 8 Click  **Copy and Mirror** on the context toolbar.

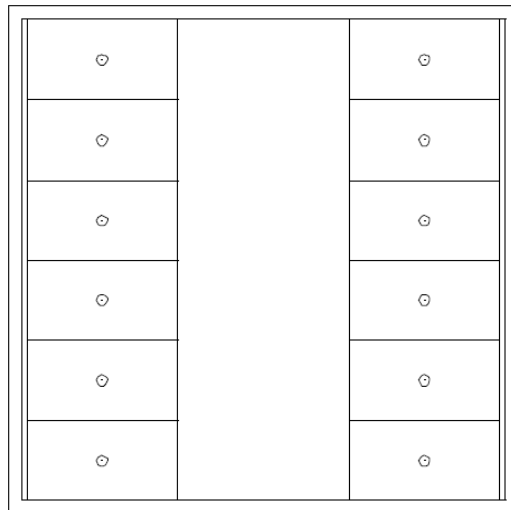


- 9 Define the center axis of the file cabinet as the mirror axis.
Point to the top line of the file cabinet, open the shortcut menu, and click  **Midpoint**.
Allplan snaps to the midpoint; this point defines the first point of the mirror axis (see illustration).
- 10 To define the second point of the mirror axis, point to the bottom line of the file cabinet and, on the shortcut menu, click  **Midpoint** again.



A = Mirror axis

Allplan copies the drawers to the right.



11 Select ESC to close direct object modification.

Creating a knob for the door in the middle

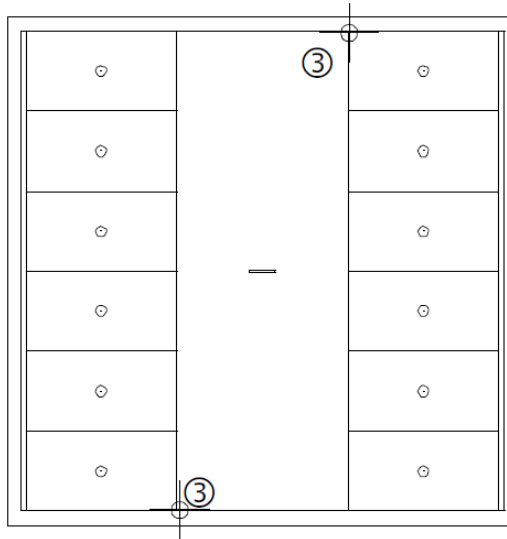
Finally, you will draw a knob for the door in the middle. To do this, you will use the **Midpoint** and **Based on center** options.

To create a knob for the door in the middle

- 1 Click  **Rectangle**.
- 2 Click **Based on center** in the input options.



- 3 Open the shortcut menu and choose  **Midpoint**. Then click two diagonally opposite corners of the door in the middle. This defines the center of the rectangle.
- 4 Enter **0.1** for the length; select ENTER to confirm.
- 5 Enter **0.01** for the width; select ENTER to confirm.



- 6 Select ESC to close the tool.

Task 2: modifying the file cabinet

Based on the file cabinet designed, you will create a new cabinet that is 2.1 m high. This cabinet has seven drawers. Start by copying the design to a new drawing file. Then you will modify the design. In this section, you will find out about the two most important modification tools: **Parallel to Element** and **Stretch Entities**.

Tools:



Copy, Move Elements between Documents



Stretch Entities

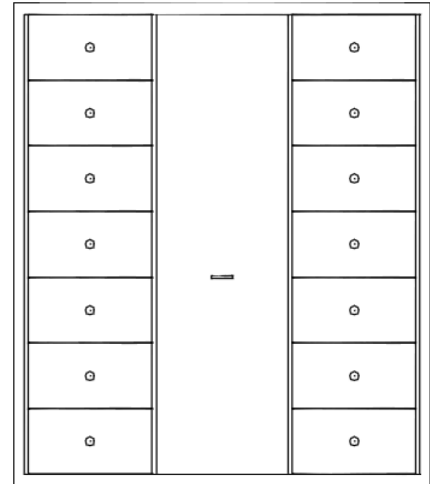


Parallel to Element



Brackets

Objective:



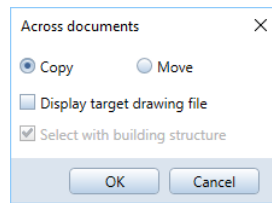
Copying a drawing file

Begin by copying the file cabinet you created in the last exercise to a new drawing file.

To copy the drawing file with the file cabinet

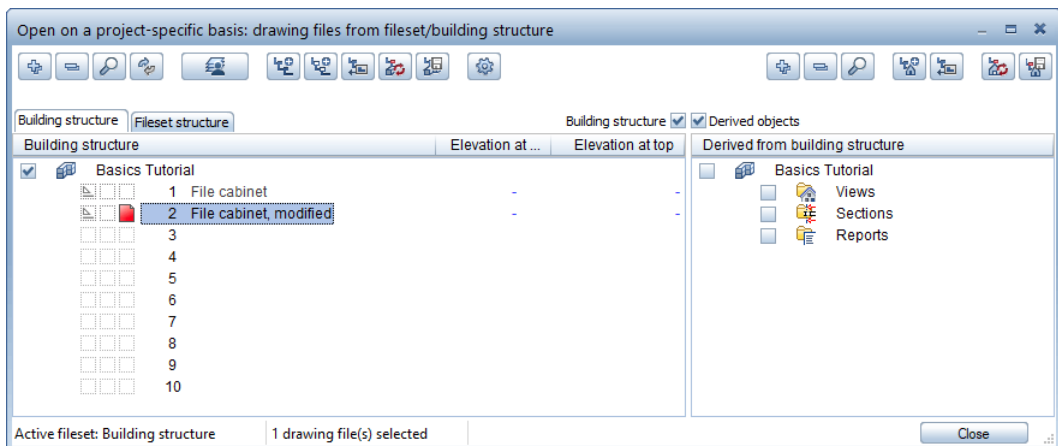
➡ Only drawing file **1 File cabinet** is open.

- 1 Click **Copy, Move Elements between Documents...** in the drop-down list of the Allplan icon on the title bar.
- 2 Select **Copy** and click **OK** to confirm.



- 3 Select an empty drawing file (for example, drawing file **2**) and click **OK** to confirm.
- 4 The dialog line prompts you to select the elements that you want to copy to the new drawing file.

You want to copy all elements in the drawing file. Therefore, right-click in the workspace twice or click **All** in the input options. This copies the file cabinet to the new drawing file.
- 5 Click  **Open on a Project-Specific Basis** (Quick Access Toolbar) and select the drawing file to which you have just copied the file cabinet.
- 6 Enter a name for drawing file **2**, for example, **File cabinet, modified**.



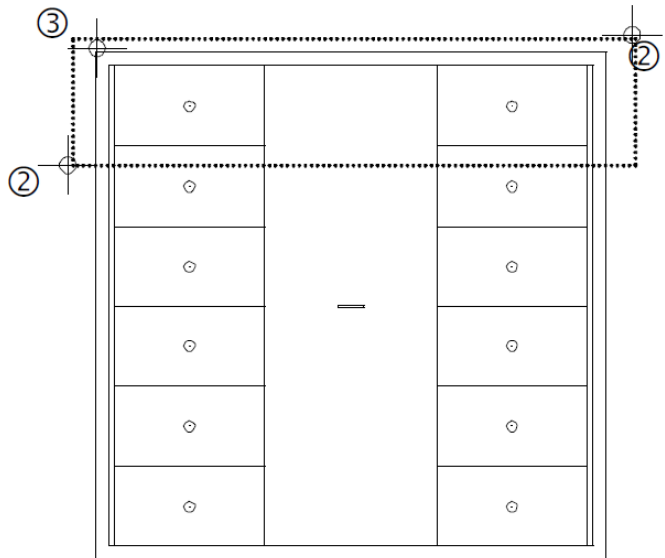
- 7 Make drawing file **2** current, close drawing file **1**, and close the dialog box.
- 8 Click  **Zoom All** (viewport toolbar) to display the entire file cabinet on the screen.

Stretching entities

In this section, you will modify the upper two corners of the file cabinet, giving the file cabinet a new height of 2.1m. In addition, you will add two drawers by using the **Copy** tool. To do this, you will use **direct object modification**.

To stretch entities

- 1 Right-click in the workspace and select the  **Stretch Entities** tool on the shortcut menu.
- 2 Select all points that you want to modify. Make sure that you select the two top drawers together.



- 3 *From point:*
Click the upper-left corner of the file cabinet.

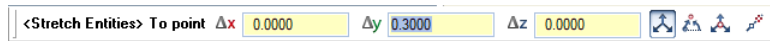
4 *To point:*

The file cabinet is 2.1 m high; in other words, you must lengthen the file cabinet by 0.3 m in the y-direction. Click  **Delta point** in the dialog line and enter  $\Delta y = 0.30$.

Tip: You can also enter the values in the dialog line without clicking a starting point:

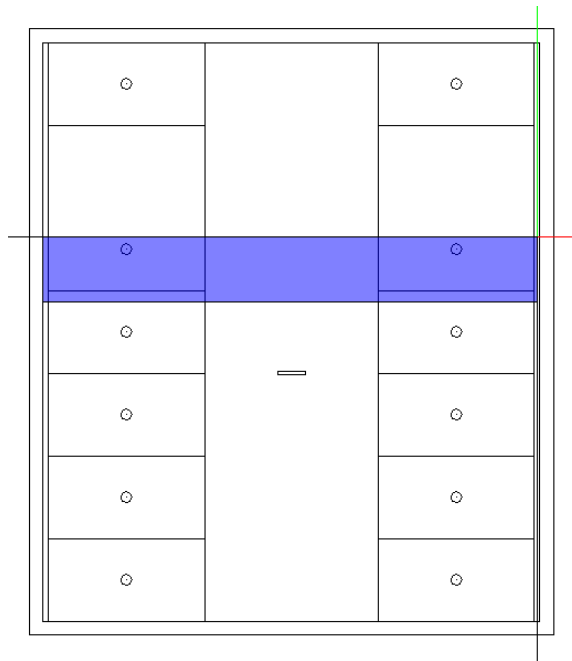
 $\Delta x = 0$

 $\Delta y = 0.30$

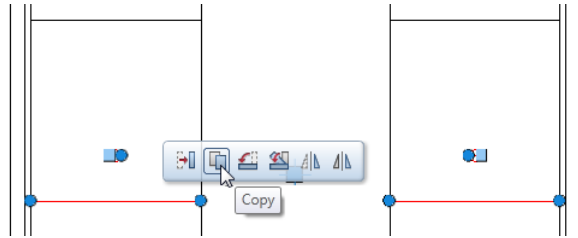


Select ENTER to confirm.

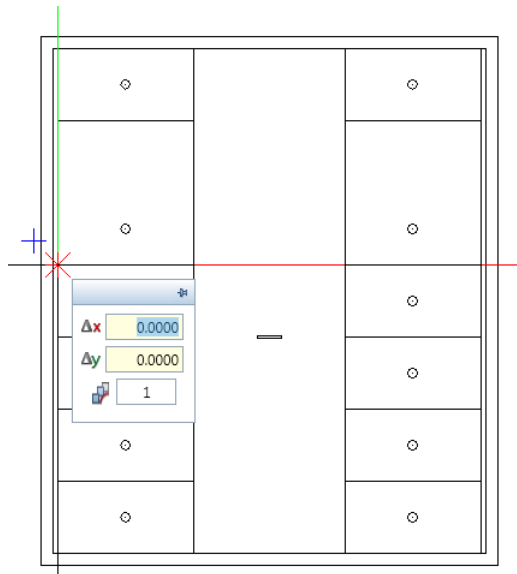
- 5 Select the Esc key to close  **Stretch Entities**.
- 6 Select the elements that make up the two incomplete drawers (two lines and circles each) by enclosing them in a selection rectangle that you open from the lower left to the upper right (positive x-direction).



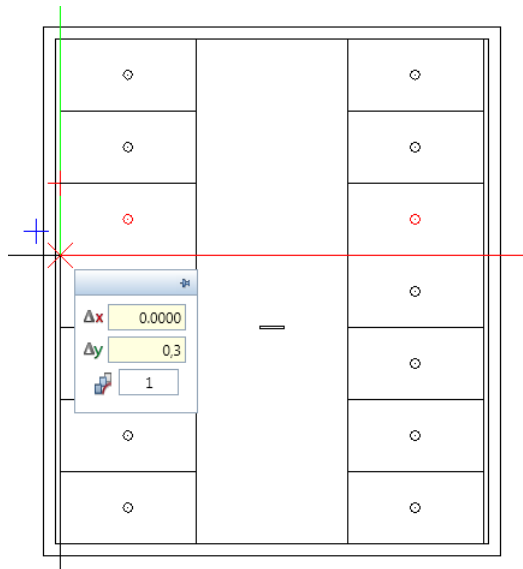
- 7 Point to one of the two selected lines and click  **Copy** on the context toolbar.



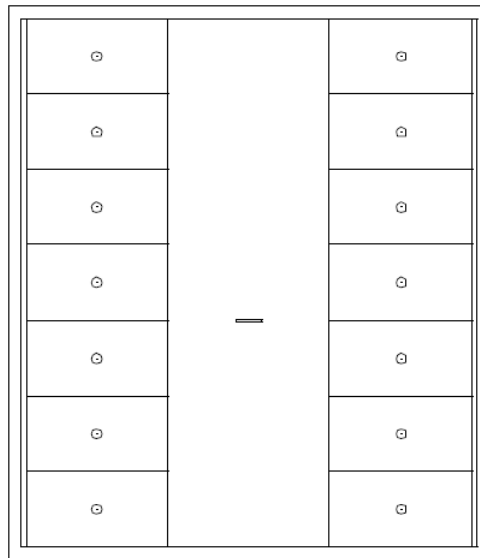
- 8 *From point:*
Click the lower-left corner of the incomplete drawer on the left side.



- 9 *To point or enter number of copies:*
Select the Tab key to switch to  in the coordinate dialog box and enter **dy** = 0.30.



10 Select ENTER to confirm.



11 Select ESC to finish.

A note on selecting and modifying several elements and regions together by using the brackets

As an alternative to the selection rectangle, you can use the brackets to select elements one after the other. Do the following:

- Select an edit tool.
- Open the  **Brackets** by right-clicking in the workspace.
- Click the elements one after the other or open selection rectangles around the elements that you want to select.
- To exclude an element from the selection, simply click it again.
- Close the brackets.

Adding a frame

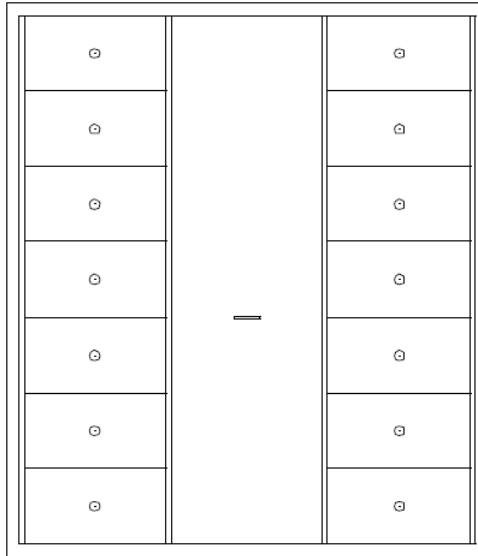
To finish, you can enhance the file cabinet by adding a frame to the door in the middle. To do this, you will use the **Parallel to Element** tool.

To add a frame

- 1 Click  **Parallel to Element** (2D Objects task area).
The dialog line prompts you to select an element.
Click the inner cabinet edge on the left side.
- 2 *Through point or offset:*
Enter **0.6** in the dialog line and select ENTER to confirm.
- 3 *Which side?*
Click to the right of the line.
- 4 *Number:*
Enter **1** and select ENTER to confirm.
- 5 The  **Parallel to Element** tool is still active. The program computes the distance to the next line based on the new element.
- 6 Check that the value in the dialog line is **0.6**; select ENTER to confirm.

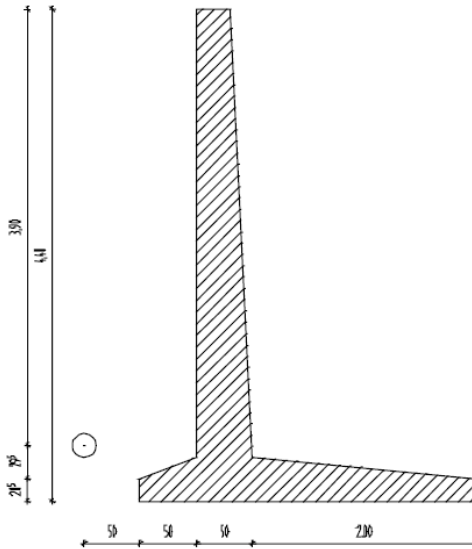
7 Number:

Check that the number in the dialog line is **1**; select ENTER to confirm.

**8 Select ESC to close the tool.**

Exercise 2: retaining wall with a drainage

In the following exercise, you will design a cross-section of a retaining wall with a drainage.



You will use the tools in the **2D Objects**, **2D Areas**, **Filter**, and **Change** task areas.

Task 1: designing a retaining wall with a drainage

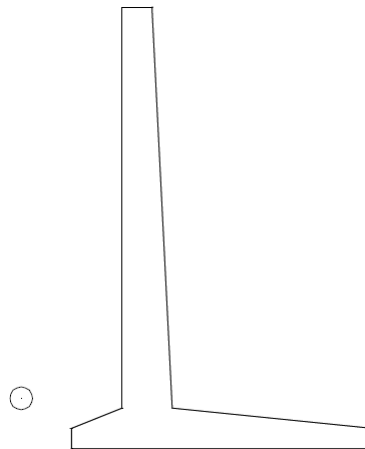
In this section, you will learn how to use delta points to create lines that are parallel to neither the x-axis nor the y-axis. By means of delta points, you can place a point at a specific distance from an existing point.

To enter delta points, use  **Delta point** in the dialog line.

Tools:

-  Line
-  Delta point
-  Circle
-  Track line

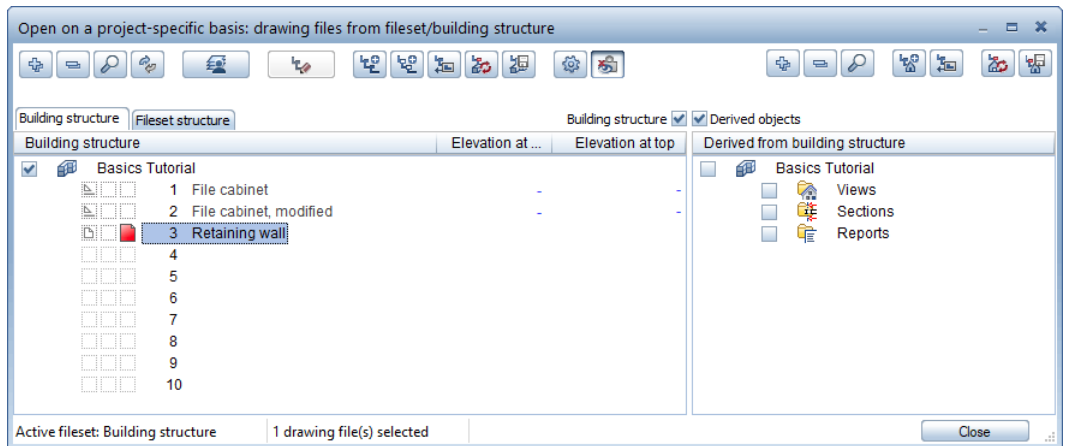
Objective:



Retaining wall of angular shape

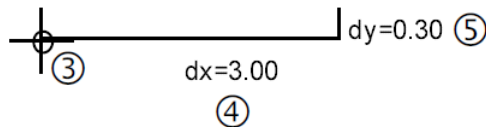
To draw the retaining wall

- 1 Click  **Open on a Project-Specific Basis** and open an empty drawing file. Name it **Retaining wall** and close all the other drawing files.



Tip: Make sure that the **Element** option is selected in the  **Point snap options**. To check this, right-click to open the shortcut menu. Click  **Point snap options**.

- 2 Click  **Line** in the **2D Objects** task area.
- 3 The **Line** context toolbar box opens. Select the  **Polyline** option and click the starting point of the line.
- 4 *<Line> To point*
Enter  **dx = 3.00** in the dialog line; select ENTER to confirm.
- 5 *<Line> To point*
Enter  **dy = 0.30** in the dialog line; select ENTER to confirm.



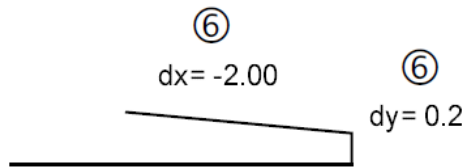
The next point is not perpendicular to the previous point. However, you know the offsets in the x-direction and y-direction. Use  **Delta point** to place this point.

Tip: Select the TAB key to go to the next box in the dialog line.
Select ENTER to confirm the values.

- 6  **Delta point** is already selected in the dialog line.
Enter the following values:
 $\Delta x = -2.00$
 $\Delta y = 0.20$



- 7 Select ENTER to confirm.



To place the next point, use  **Delta point** again.

- 8 Enter the following values in the dialog line:
 $\Delta x = -0.2$
 $\Delta y = 4.0$



- 9 Select ENTER to confirm.



10 You can enter the next two lines in two ways:

As these two lines are perpendicular to the previous point, you can create them by entering values directly in the dialog line or by means of track lines.

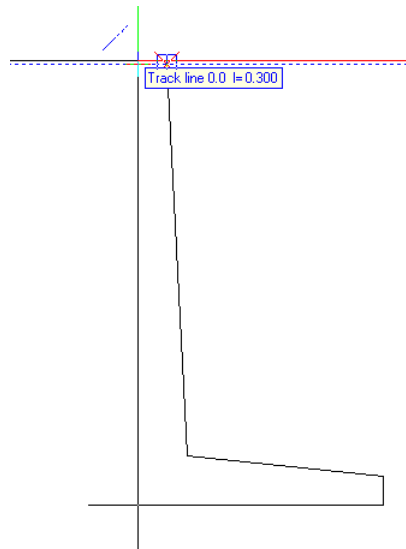
First option:

Draw the horizontal line by entering the length in the x-direction in the dialog line: $\Delta x = -0.30$ - ENTER.

Draw the vertical line by entering the length in the y-direction in the dialog line: $\Delta y = -4.00$ - ENTER.

Second option:

Turn on track tracing by clicking  **Track line** in the dialog line. Point to the end of the last line you created. Slowly move the crosshairs to the left. The 0.0-degree track line appears.

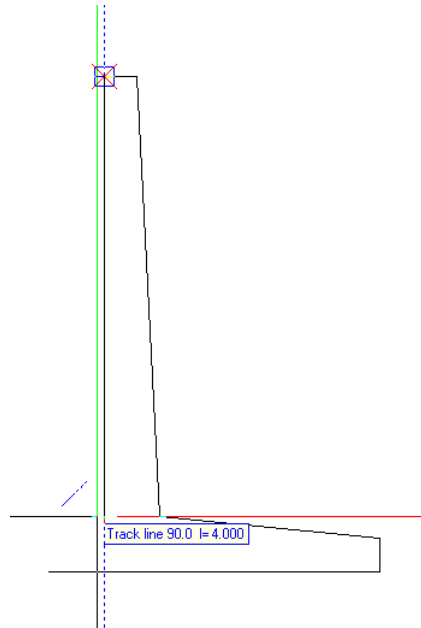


As soon as Allplan displays $l = 0.300$ for the offset, click this point or enter 0.3 m for  **Offset to reference point** in the dialog line. Then select ENTER to confirm.



Note: Track lines display a preview of the current length. This length is a multiple of the **grid length** that you can enter next to  **Rasterize length** in the dialog line. If you cannot define the required length by means of track lines, it is a good idea to change the **grid length**.

To draw the second line, slowly move the crosshairs vertically downward. The 90.0-degree track line appears.

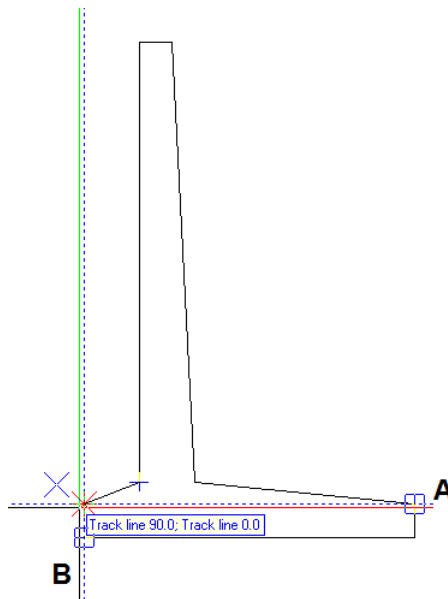


As soon as Allplan displays $l = 4.000$ for the offset, click this point  **Offset to reference point** in the dialog line. Then select ENTER to confirm.



- 11 You can also use track tracing to place the next point. Point to point **A** and wait at least 500 milliseconds. The program creates a track point from the point snapped.

- 12 Then point to point **B** (= first point of the design). Wait until Allplan has identified points **A** and **B** as track points and marked them with blue squares.
- 13 Start at point **B** and move the crosshairs vertically upward as far as the point where the 90.0-degree track line intersects the 0.0-degree track line. Click this point.



See also:



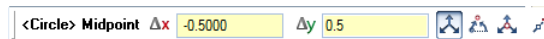
Track tracing is described in detail in the Allplan Help.

- 14 Turn track tracing off by clicking  **Track line** in the dialog line.
- 15 Finish creating the retaining wall by clicking the starting point of the first line (point **B**).
- 16 Select ESC twice to close the **Line** tool.

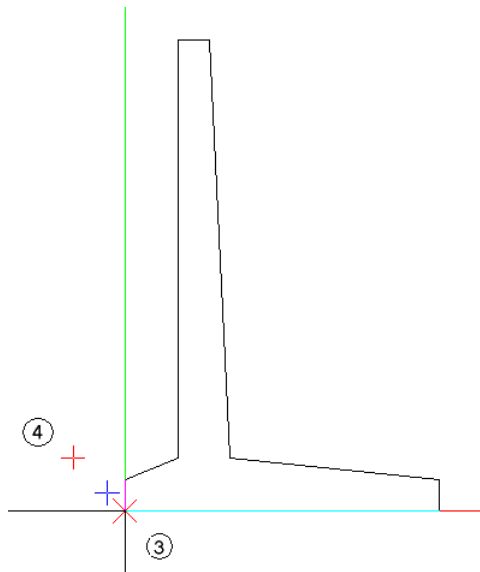
Drainage

To design the drainage

- 1 Click  **Circle** in the **2D Objects** task area.
- 2 The **Circle** context toolbar opens. Click  **Circle based on center** and  **Enter full circle**.
- 3 Point to the lower-left corner of the retaining wall. This point is marked with a red X.
- 4  **Delta point** is selected in the dialog line. Enter the following values:
 $\Delta x = -0.5$
 $\Delta y = 0.5$



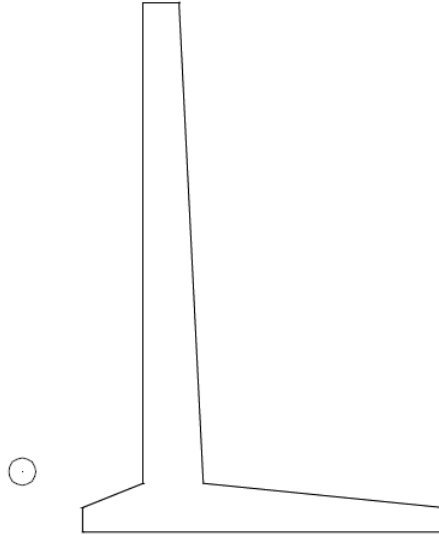
Select ENTER to confirm.



This defines the center of the circle.

- 5 Enter a radius of **0.1** in the dialog line; select ENTER to confirm.

6 Select ESC to close the tool.



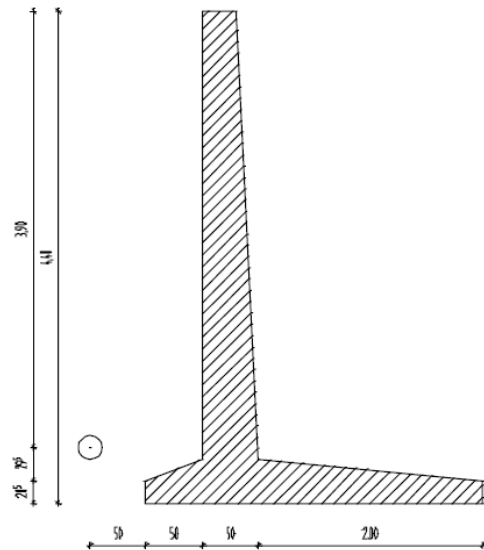
Task 2: hatching

In this section, you will apply hatching to the retaining wall. You will also learn about the principles of entering polylines. The polyline entry tools are used by just about all functions that expect you to define polylines or polygonal-bounded areas (for example, hatching, pattern, fill).

Tools:

-  Hatching
-  Filter by Element Type
-  Area detection
-  Modify Format Properties
- Hatching defaults

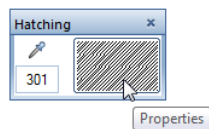
Objective:



Applying hatching to the retaining wall

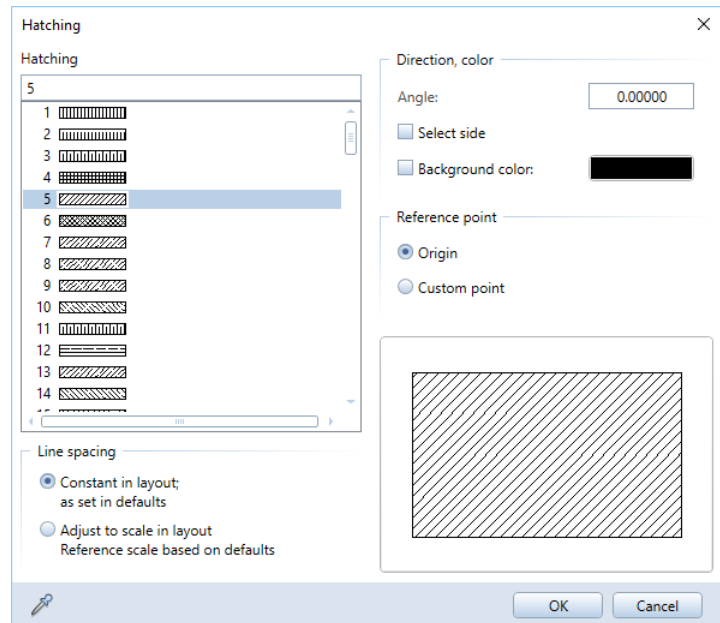
To apply hatching to the retaining wall

- 1 Click  **Hatching** (2D Areas task area).
- 2 Click **Properties** on the **Hatching** context toolbar.



3 Select hatching style **5** and define the following parameters:

- **Line spacing area:**
Constant in layout, as defined in defaults
- **Reference point area:**
Origin



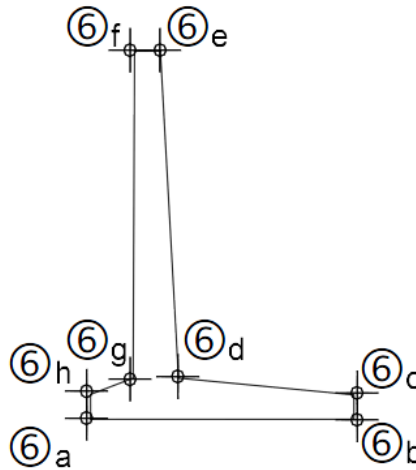
4 Click **OK** to confirm the settings.

5 Click **Single** in the input options.

Tip: When you click **Multi** in the input options, you can enter as many areas as you want. After you have selected ESC to finish entering the polyline, Allplan applies hatching to these areas in one go.

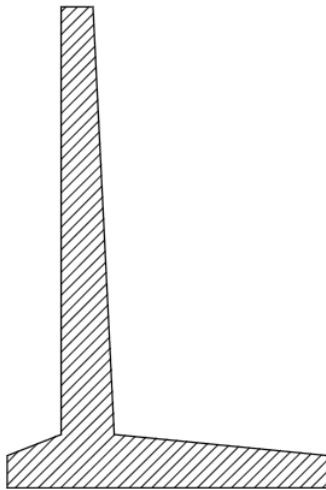


- 6 To define the area for hatching, click the corners of the retaining wall one after the other.



- 7 To close the polyline, select ESC after you have clicked the last point or click the first point again.

Allplan applies the selected hatching to the retaining wall.



- 8 Select ESC to close the  Hatching tool.

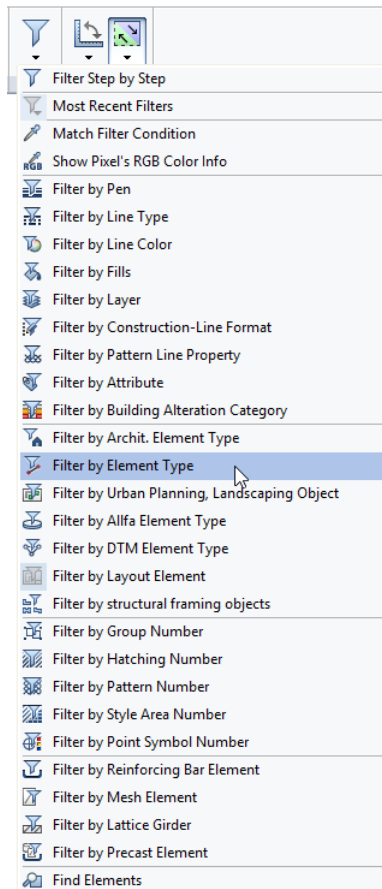
Copying the outline of the retaining wall

In addition to clicking each corner of the polyline (as described), you can enter polygonal-bounded areas in other ways. By using the retaining wall as an example, the following exercise shows you how.

Start by copying the wall so that you have a copy for experimenting. In doing so, you will use a filter to copy the retaining wall without the hatching.

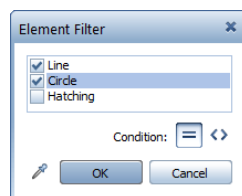
To copy the outline of the retaining wall

- 1 Right-click in the workspace and select  **Copy** on the shortcut menu.
- 2 *<Copy> Select the elements you want to copy*
The **Filter** task area has not been expanded yet. Point to the icon in this area and click within the blue part at the bottom of this area. The drop-down list opens, displaying all tools in the **Filter** task area.
Click  **Filter by Element Type**.

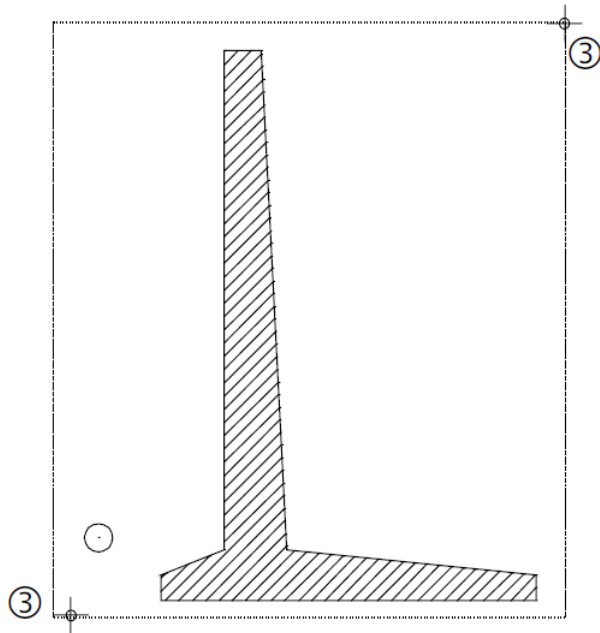


- 3 Select **Line** and **Circle**; click **OK** to confirm.

Tip: You can also use the properties of the outline as a filter. Click  **Match parameters** and click the required element.



- 4 <Copy> *Select the elements you want to copy* <=Line =Circle>
Enclose the retaining wall in a selection rectangle.

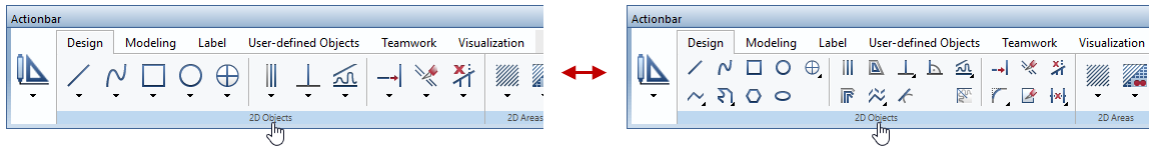


As you have selected the **Line** and **Circle** filters, only the outlines of the retaining wall and the drainage appear in the selection color.

- 5 <Copy> *From point*
Specify the starting point for the copy; place the retaining wall in the workspace. The position is irrelevant. However, make sure that the two retaining walls do not overlap.
- 6 Select ESC to close the  **Copy** tool.
- 7 Click  **Zoom All** to display both walls on the screen.

Minimized task areas do not show all tools. To open the flyout menu of one of these tools, click the downward arrow. You can then see all the tools in the collapsed area.

You can maximize or minimize a task area by double-clicking within the name line of a task area.



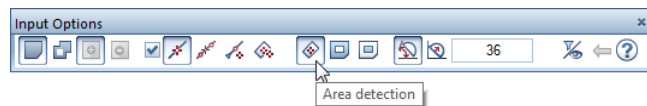
A maximized task area shows more tools, which can also have flyout menus.

Applying hatching by means of area detection

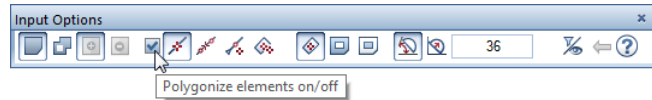
The next step is to apply hatching to the copy of the retaining wall. To do this, you will use a tool that automatically detects closed, delimited areas.

To apply hatching by means of area detection

- 1 Click **Hatching** (2D Areas task area or **Repeat** drop-down list on the Quick Access Toolbar).
- 2 **Hatching style 5** is still selected on the **Hatching** context toolbar. If it isn't, click **Properties** and select hatching style 5. Click **OK** to confirm.
- 3 Click **Single** in the input options.
- 4 Select **Area detection** in the input options.



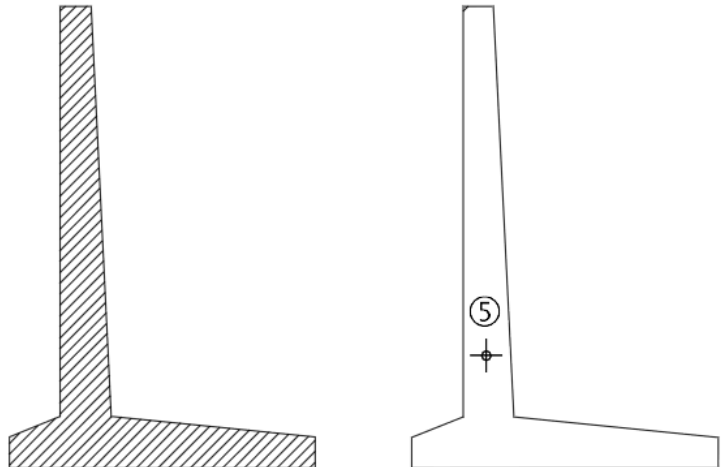
Note: You can select  **Area detection** only when **Polygonize elements** is turned on.



- 5 Click a point within the retaining wall.

Allplan automatically detects the outline, polygonizing the entire retaining wall.

 **Single** is selected in the input options. Therefore, you can immediately see the hatching.



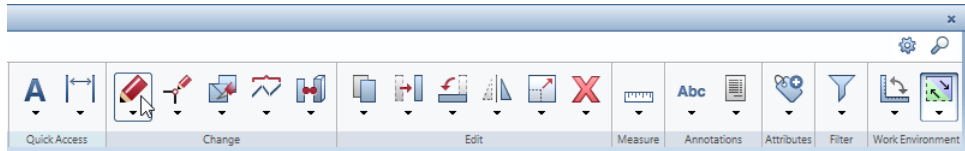
- 6 Select ESC to close the  **Hatching** tool.

Modifying hatching

Modify the hatching pen.

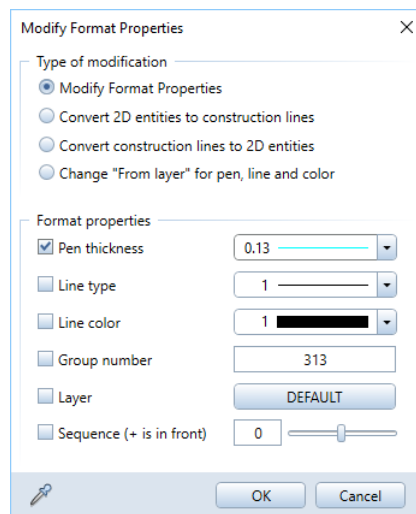
To modify the hatching pen

- 1 Click  **Modify Format Properties** in the **Change** task area.



- 2 To change the pen, open the **Pen thickness** list box and select pen number **7** with a thickness of 0.13 mm.

This automatically selects the **Pen thickness** check box.

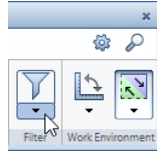


- 3 Click **OK** to confirm.

The dialog line prompts you to select the elements that you want to draw with the new pen. Here, too, use a filter because you want to modify the hatching only.

- Point to the icon in the **Filter** task area and click within the blue part at the bottom of this area.

Tip: You can combine as many filters as you want.



- Click  **Filter by Element Type** in the drop-down list.
- Select **Hatching** and click **OK** to confirm.
- Enclose the two retaining walls in a selection rectangle. As you have used a filter, Allplan modifies the hatching only.
- Select ESC to close  **Modify Format Properties**.

Hatching defaults

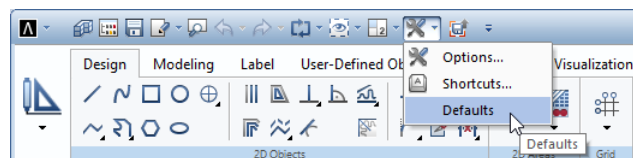
Allplan 2022 comes with a wide range of ready-made hatching styles. You can also define your own hatching styles or modify existing hatching styles.

If you have worked your way through the exercises step by step, you selected the **Project** paths for patterns and hatching styles when you created the project for this tutorial. In other words, any changes you make to defaults (for example, hatching, pattern) affect the current project only.

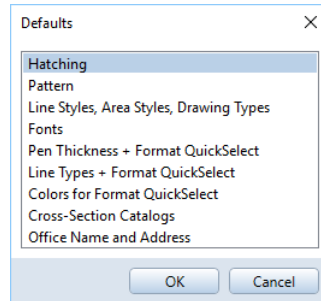
If the **Office** path is selected, you run the risk of modifying the office standard. This means that any changes you make affect all projects based on the office standard.

To define or modify hatching styles

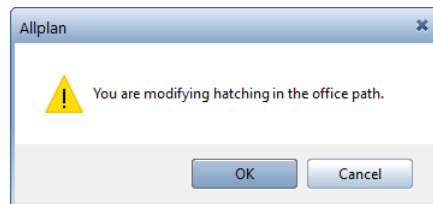
- Open the  **Default Settings** drop-down list on the Quick Access Toolbar and select **Defaults**.



- 2 The **Defaults** dialog box opens. Select **Hatching** and click **OK**.

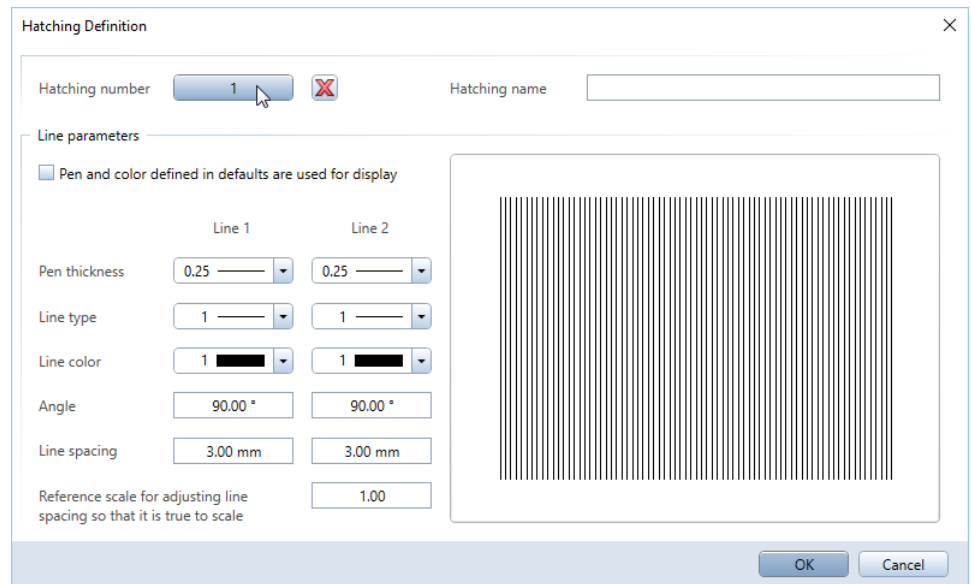


Note: If the **You are modifying the hatching in the office path** message appears, the settings you are about to make will modify the patterns and hatching styles in the **Office** path.

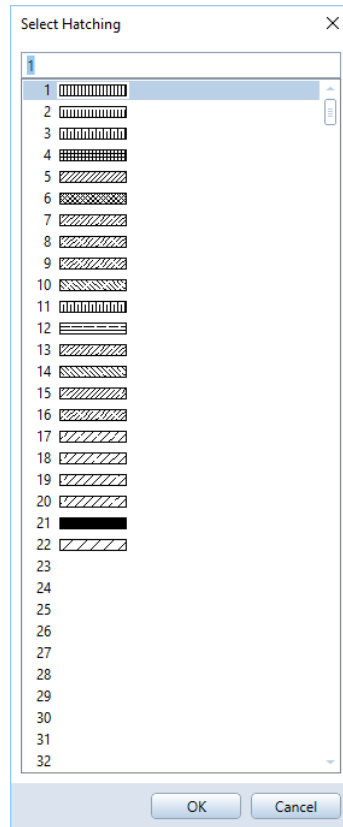


In this case, click **Cancel** and change the path to **Project**, which is described in the following section.

- 3 Click the button next to **Hatching number** in the top part of the **Hatching Definition** dialog box.



- 4 To modify an existing hatching style, select the required number.
To define a new hatching style, select an unassigned number.



- 5 Make the required settings in the **Hatching Definition** dialog box.

Note: You can use the **Pen and color defined in defaults are used for display** option to specify whether you want to use the pen in the **Properties** palette or the pen defined in this dialog box.

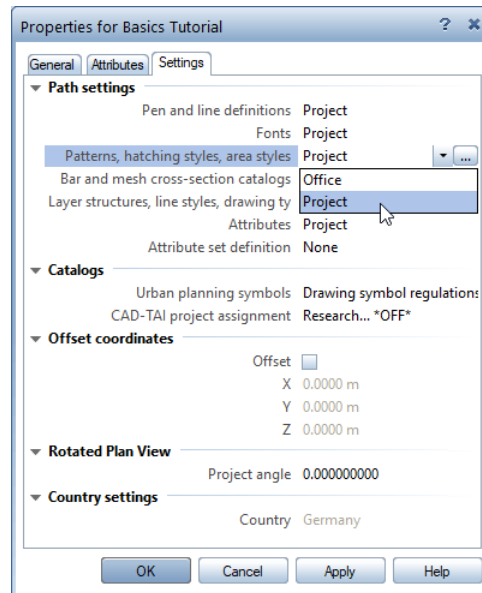
- 6 Click **OK** to confirm the dialog box. If you have changed the defaults, confirm the following prompt by clicking **Yes**.
-

The following section shows how to switch the path settings for patterns and hatching styles to project. You only need to do this when you see the **You are modifying the hatching in the office path** message after you have selected the hatching defaults.

To switch the path settings for patterns and hatching styles to project

Tip: You can also access the path settings by clicking  **New Project, Open Project...** on the Quick Access Toolbar. Open the shortcut menu of the **Basics Tutorial** project and click **Properties...**

- 1 Click  **ProjectPilot** in the drop-down list of the Allplan icon on the title bar.
- 2 Open the **Projects** folder.
- 3 Right-click the **Basics Tutorial** project and select **Properties**.
- 4 Open the **Settings** tab, go to the **Path settings** area, and select **Project** for Patterns, hatching styles, area styles.

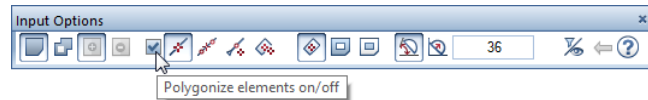


- 5 Click **OK** to confirm.
Allplan copies the office standard to this project.
- 6 Close **ProjectPilot** by clicking **Exit** on the **File** menu.

Polyline entry tools

When working with Allplan 2022, you will find that the polyline entry tools help you a lot. This feature is used by countless Allplan tools that expect you to define polylines or polygonal-bounded areas (for example, fills, patterns, slabs, and roof outlines).

The polyline entry tools, which are included in the **Input Options**, open automatically when you select a tool for which they are available.



Select the check box in the input options to turn on the polyline entry tools.

Input options for entering polylines, overview

The **Input Options** open whenever you select a tool that uses polyline entry tools (for example, pattern, hatching, room). You can use these options to specify how the polyline entry tools handle architectural lines and how these tools behave when you create polylines based on existing elements.



Entering areas

Single

Use this to create **single**, discrete areas.

Multi

Use this to create areas composed of several polygons. Hatching, patterns, or fills get the same group number; rooms are handled as a single entity. Consequently, you can define separate rooms, which Allplan then analyzes as a single room.

Plus, Minus

When you select  **Multi**, you can use  **Plus** and  **Minus** in the input options to specify whether each new polygon you enter will be added to or subtracted from the overall area.

Polygonizing existing elements

Polygonize elements on/off

When the check box is not selected, Allplan ignores elements when you click them. In this mode, Allplan detects points only.

When the check box is selected, Allplan polygonizes the elements you click. You can use the options next to this check box to specify the type of polygonization.

Polygonize entire element

This uses the entire element that you clicked. The starting point defines the direction of polygonization. If the last point in the polyline coincides with the starting point or end point of the element, you do not need to specify the direction.

Use this option when the outline consists of entire elements.

Define area of element to polygonize

With this option, the program prompts you for the area with every element you click (from point, to point).

Use this option when the outline consists of segments.

Enter reference point

With this option, the program prompts you for the reference point with every element you click. This option uses a point on the element clicked with a defined offset to the reference point. Click to define a new reference point and then enter the offset to the reference point. Use this option when you want to specify the outline based on existing elements (when you enter a dormer, for example).

 Area detection using additional point

 **Area detection using additional point** combines areas bounded by lines and polylines to form a polygon. Allplan uses the inner boundaries or outer boundaries depending on whether you click the additional point inside or outside the outline.

By selecting  **Element filter**, you can configure the program to ignore architectural lines when detecting areas.

 Area detection

You can use  **Area detection** to automatically detect the outlines of closed polygons. You can use closed areas delimited by design entities of any kind as an outline polygon simply by clicking anywhere within the area. Allplan automatically detects and polygonizes the entire outline. The boundary elements can have points in common; they can intersect or touch. You can turn this automation feature on and off at any time.

Note: The **Minimum distance between points** setting in the  **Options** on the **Desktop environment** page also applies to the  **Area detection** tool. To make sure that Allplan detects outlines with small gaps, you can increase the minimum distance between points temporarily.

 Island detection, **Inverse island detection**:

 **Island detection** detects closed outlines within an area and automatically cuts them out.

 **Inverse island detection** does not cut out closed outlines but fills these outlines with the selected surface element. It is the area around the "island" that remains empty.

You can use these tools only together with  **Area detection using additional point** and  **Area detection**.

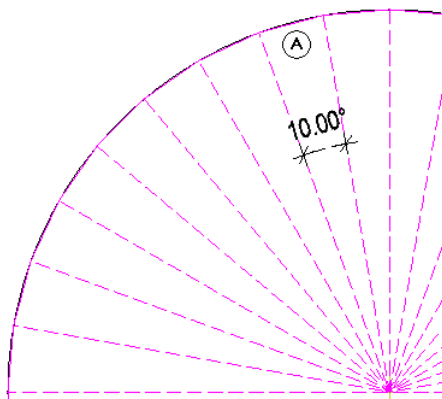
Number of segments, Rise



Number of segments

The polygonization value is interpreted as the number of segments.

The value for  **Rise** defines the number of segments used to approximate a curve. In the case of a circle, for example, **120** means that a full circle is approximated by a 120-sided polygon. The higher the required degree of accuracy or the larger the radius, the greater the number of segments should be used to approximate a circle. You can enter a value between 36 and 360.

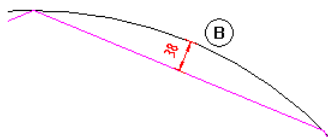


(A) Segments in circle = 36; this produces an angle of 10°



Rise

The polygonization value is interpreted as the rise. The value you enter for  **Number of segments** defines the maximum rise of the secant relative to the arc (in mm). As a result, the curve is polygonized so that the maximum offset of the polyline's segment to the curve is less than or equal to the value you specified. This setting produces more accurate results than the number of segments.



(B) Rise (38 mm or less)

Element filter



Element filter

Ignore plan lines of architectural elements

Ignore 2D surface elements (hatching, patterns, fills, bitmap areas, smart fit placements) when using area detection

When you select the  **Element filter**, Allplan ignores lines of architectural elements and 2D surface elements when you use

 **Area detection** or  **Area detection using additional point**. Use this option if you want to automatically apply surface elements like hatching, patterns to adjacent outlines that are separated by arcs, splines, or curves.

Here is some background information: Allplan polygonizes curves based on the number of segments specified.

When you enter a second (third...) area, **Area detection** can take a long time or produce incorrect results because Allplan detects both the outline of the surface (2D line) and the boundary line of the polyline of the first area.

Back, Help



Back

Undoes the last point you entered.

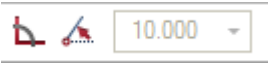


Help for entering polylines

Displays help for the **polyline entry tools**.

Additional tools in the dialog line

When entering points, you can find the following drawing aids in the dialog line:

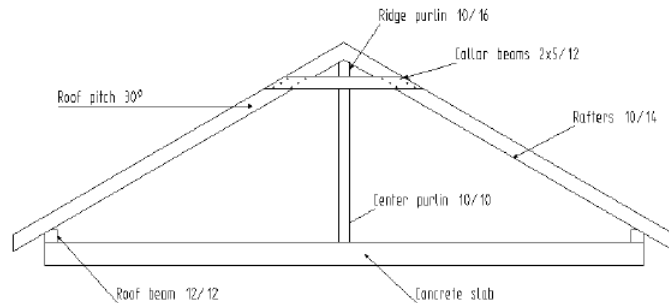


Icon	Tool	Use
	Enter at right angles	The line can be drawn only at right angles to the current system angle.
	Enter using cursor snap	The line can be drawn only at specific angles.
	Angle	Define the cursor snap angle. You can see the current angle.

Note: While you are entering a polyline, it can happen that you inadvertently click a point. By clicking  **Back** in the input options, you can undo the last point entered.

Exercise 3: purlin roof

In this exercise, you will design a purlin roof. In addition, you will label the roof design and add text leaders.



You will use the tools in the **Draft** role – **Design** task – **2D Objects**, **Filter**, and **Quick Access** task areas.

Task 1: designing a purlin roof

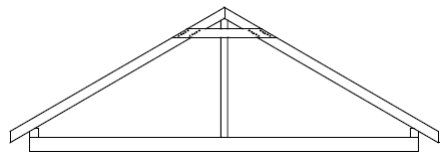
You will learn how to use the **Intersect 2 Entities**, **Polar Coordinates**, and **Division Point** tools.

Tools that were covered earlier in previous exercises (for example, rectangle, parallel to element, brackets) are not described in detail in this exercise.

Tools:

-  **Intersect 2 Entities**
-  **Division point**
-  **Track line**

Objective:



Slab and rafters

The first part of this exercise involves designing the slab, the roof beams, and the rafters. You will draw the slab as a rectangle and create the rafters as lines and parallel lines. First, you will design the rafter on the left side and then copy it to the right side.

To draw the slab and the rafters

- 1 Click  **Open on a Project-Specific Basis** and open an empty drawing file. Name it **Purlin roof** and close all the other drawing files.
- 2 Click  **Rectangle** (2D Objects task area) and select  **Based on diagonal line** in the input options
- 3 Draw the concrete slab as a rectangle.
 x-coordinate = **5.74** (length);  y-coordinate = **0.22** (width)

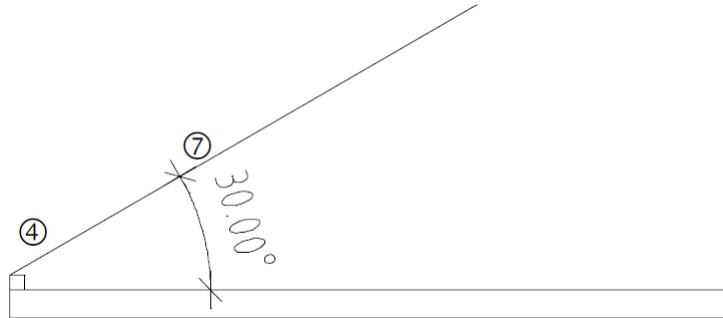


- 4 The  **Rectangle** tool is still open. Click the upper-left corner of the concrete slab and create a roof beam:
 x = **0.12** and  y = **0.12**

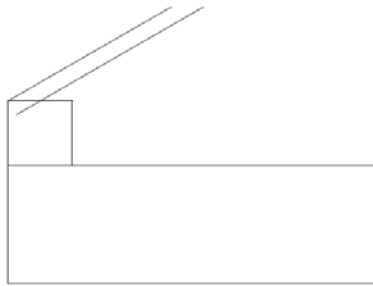


- 5 Click  **Line** in the 2D Objects task area.
- 6 Click  **Individual lines** on the **Line** context toolbar. To define the starting point of the line, click the upper-left corner of the beam (see illustration).
You will create the roof overhang later.
- 7 The roof pitch is 30°.
To draw a line at this angle, click  **Enter using cursor snap**.
- 8 Enter **30** to define the angle.
Now you can draw the line only at an angle of 30° (and in steps incremented by 30°).

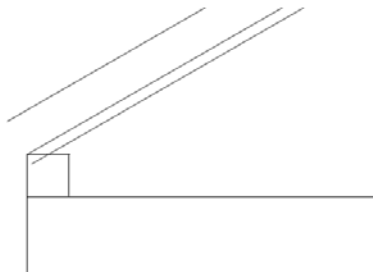
- 9 Draw the line as shown in the illustration; place its end point by clicking. Make sure that the line is long enough. You can delete redundant segments later.



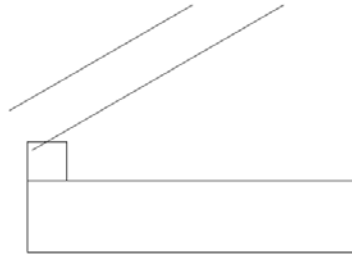
- 10 The rafter rests on the roof beam. Click  **Parallel to Element** (2D Objects task area) and create the bottom edge of the rafter. Enter **0.03** to define the offset.



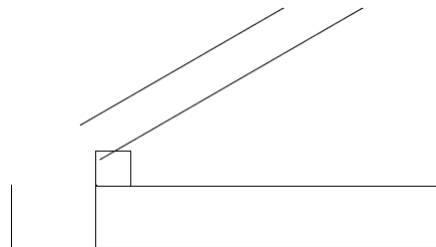
- 11 The  **Parallel to Element** tool is still open. To create the top edge of the rafter, enter **-0.14** for the offset (opposite direction!). Select ESC to close the tool.



- 12 Right-click the line in the middle and select **Delete** on the shortcut menu to remove the reference line.



- 13 The next step is to create the vertical end of the rafter. Click  **Parallel to Element** again. To define the reference element, click the left edge of the slab and enter **0.30** for the offset (= roof overhang).

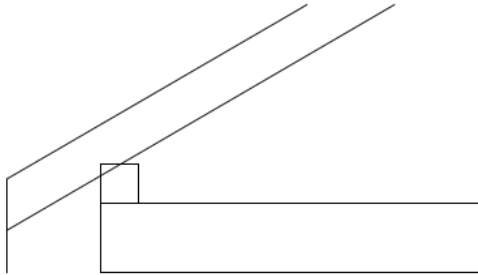


- 14 Lengthen the top and bottom edges of the rafter as far as the point where they intersect the vertical edge. To do this, use the  **Intersect 2 Entities** tool.

Right-click the top edge of the rafter and select  **Intersect 2 Entities** on the shortcut menu.

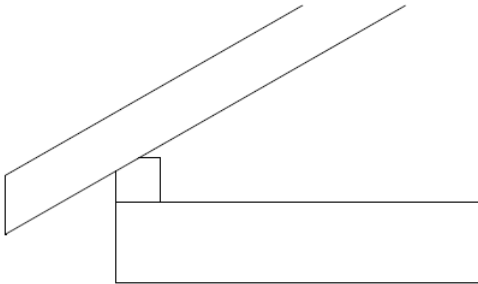
- 15 To define the second element, click the vertical edge of the rafter.

- 16 Click the bottom edge of the rafter and then the vertical edge.



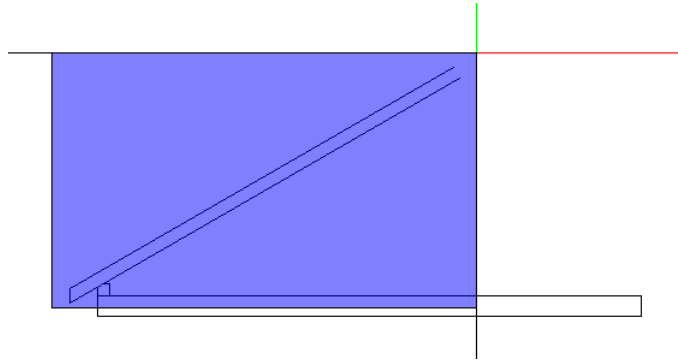
Allplan intersects the lines. The next step is to delete the redundant line segments.

- 17 Right-click one of the lines that you want to delete and select  **Auto-Delete Segment**. Click the protruding line segments.

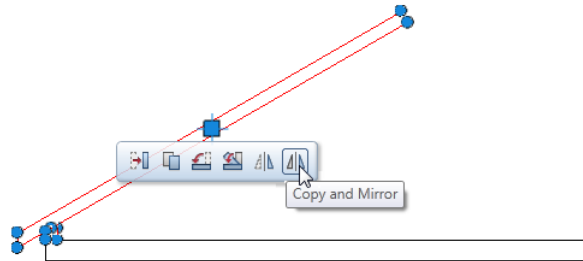


The left rafter is complete. To create the right rafter, you will mirror the left rafter about a vertical line which passes through the middle of the roof beam.

- 18 Open a selection rectangle from the lower left to the upper right (positive x-direction) to select the left rafter and the left roof beam.



- 19 Point to a line of the rafter and click  **Copy and Mirror** on the context toolbar.

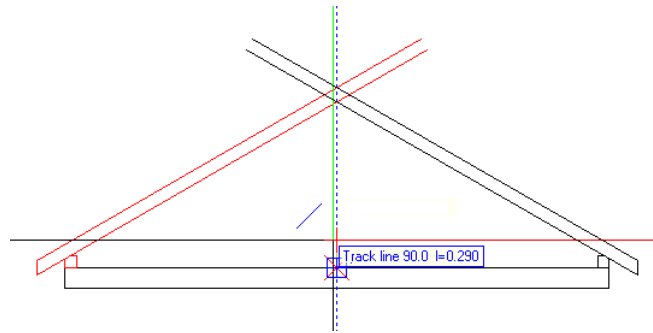


- 20 To enter the mirror axis, you can use track tracing. Select the F11 key to turn on track tracing.

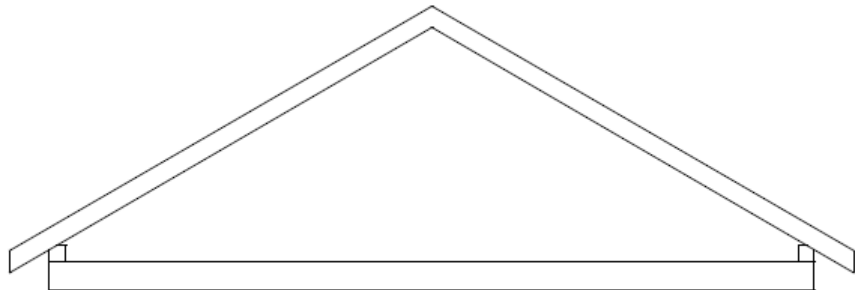
- 21 *Place point 1 for mirror axis:* The first point of the mirror axis is the center of the beam. Select  **Midpoint** on the shortcut menu and click the top edge of the beam. Click the center of the beam which is marked with a red cross.

2nd point of mirror axis: By means of track tracing, you can display the track line that is perpendicular to the first point of the mirror axis (= midpoint of the top edge of the beam). Move the crosshairs roughly at a 90-degree angle above or below the first point of the mirror axis. The 90.0-degree track line appears. Click this line wherever you want.

Allplan creates a vertical mirror axis, mirroring and copying the selected elements.

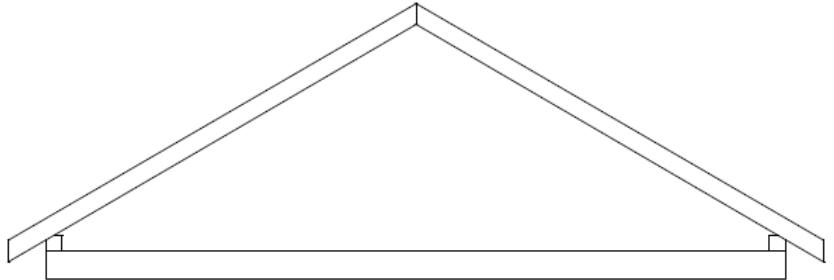


- 22 Select ESC to finish.
- 23 To delete the protruding line segments, open the shortcut menu of one of these line segments and click  **Auto-Delete Segment**.
- 24 Click all protruding line segments. The result should look like this:



25 To draw the line between the two rafters, click  **Line** and select  **Individual lines**.

26 Draw a vertical line as shown in the illustration.



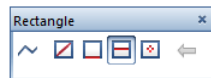
27 Select ESC to close the  **Line** tool.

Ridge purlin and collar beam

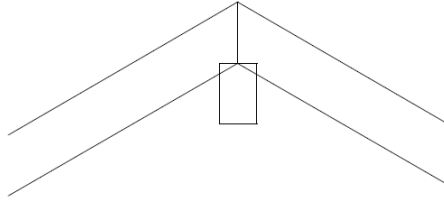
In this section, you will draw the ridge purlin, the center purlin, and the collar beam. Begin by drawing the ridge purlin as a rectangle. Then create the collar beam and the center purlin by intersecting two elements and drawing a parallel line.

To draw the ridge purlin and the collar beam

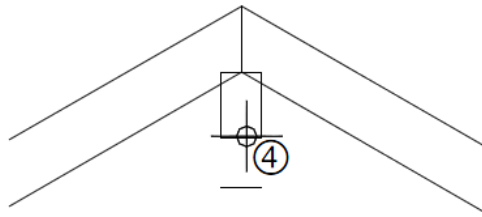
- 1 Click  **Rectangle** and select **Based on center line** in the input options.



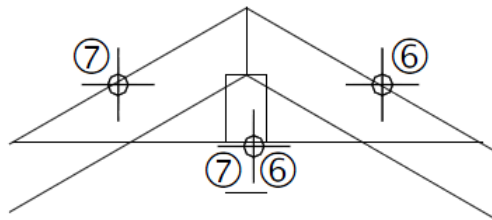
- 2 *Starting point:* Click the bottom point where the two rafters intersect.
End point: Click  **Delta point** in the dialog line and enter the **y**-direction: **-0.16**.
Point or half the width: Enter half the width of the ridge purlin: **0.05**.



- 3 Use the elements of the ridge purlin to create the center purlin and the collar beam.
- 4 Draw the bottom edge of the collar beam based on the bottom edge of the ridge purlin. Click  **Parallel to Element** and enter **0.12** for the offset.

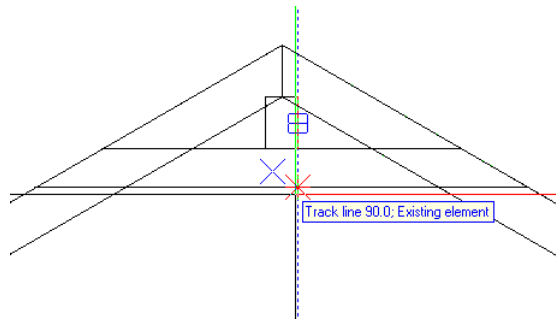


- 5 The bottom edge of the ridge purlin is to intersect the two outer edges of the rafters.
- 6 To achieve this, point to the bottom edge of the ridge purlin. Open the shortcut menu and click  **Intersect 2 Entities**. This selects the first element. To define the second element, click the right outer edge of the rafter.
- 7  **Intersect 2 Entities** is still open. The bottom edge of the ridge purlin is to intersect the outer edge of the left rafter.

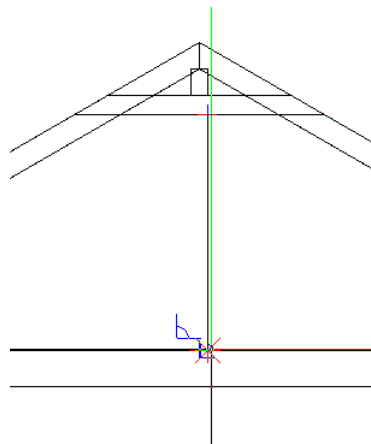


- 8 Use the same approach to ensure that the bottom edge of the collar beam intersects the outer edges of the two rafters.
- 9 Use track tracing to join the two vertical edges of the ridge purlin with the top edge of the slab.
Select the  **Line** tool and click  **Individual lines**.
- 10 Point to the vertical, right edge of the ridge purlin and move the crosshairs in a vertical direction.

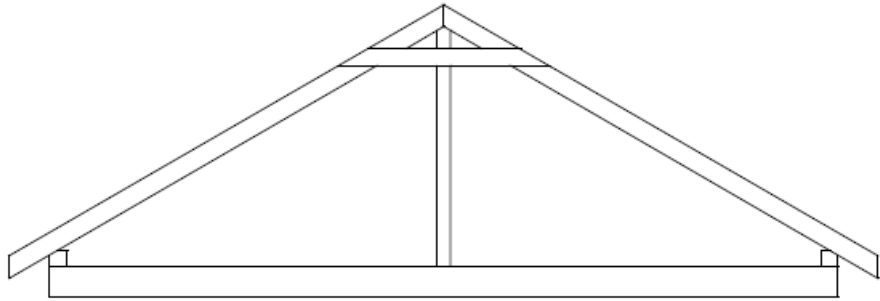
The 90.0-degree track line appears. Move the crosshairs along this track line until Allplan displays the point where the track line intersects the collar beam. Click this point.



- 11 Follow the track line as far as the point where the track line intersects the top edge of the slab and click this point.



- 12 Repeat steps 10 and 11 for the left edge of the ridge purlin.
- 13 Use  **Auto-Delete Segment** (shortcut menu of a line segment that you want to delete) to delete redundant line segments.
The design should look like this:



- 14 Select ESC to close the  **Auto-Delete Segment** tool.

Construction lines

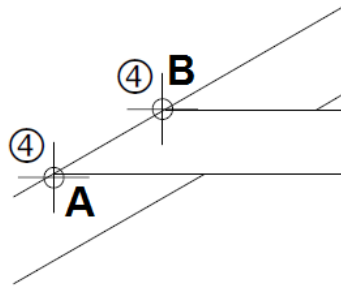
You will use six nails to fasten each rafter to the collar beam. First, create a grid consisting of lines in construction-line format. To do this, use the **Division point** tool, which you can find on the shortcut menu when a tool is active (for example, **Line**). This grid will help you place the nails later. You can use the **Division point** tool to identify division points of lines or other design entities. The nails will be placed on the points where the gridlines intersect.

Tip: The color and line type of construction lines are based on the setting in the **Options – Desktop environment – Display** page.

To draw horizontal construction lines

- 1 Draw the gridlines as construction lines.
To turn on construction-line mode, select the  **Construction line** check box in the **Properties** palette.
- 2 Click  **Line** in the **2D Objects** task area.
Click  **Individual lines** on the **Line** context toolbar.
- 3 To define the starting point of the line, click  **Division point** on the shortcut menu.

- 4 Click the endpoints of the line that you want to divide.



A = Starting point

B = End point

- 5 *Click division point:* Enter the number of divisions in the dialog line:
n = 6.

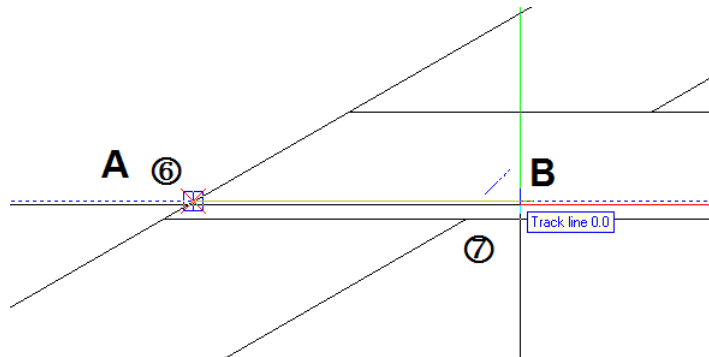


Allplan temporarily displays the division points on the screen.

- 6 To define the division point where the line is to start, enter its number (1) in the dialog line and select ENTER to confirm.
Allplan starts to count at point A (= the starting point of the line you divided).
You can also define the division point by clicking it.

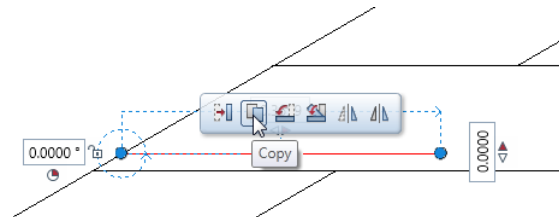
Tip: By entering -1, -2, and so on, you can also identify division points that are on the extension of the division line.

- 7 *To point or enter length:* As the line is horizontal, you can use the 0.0-degree track line to define the end of the line. The exact length of the line is not important. However, make sure that it projects beyond the right edge of the rafter.



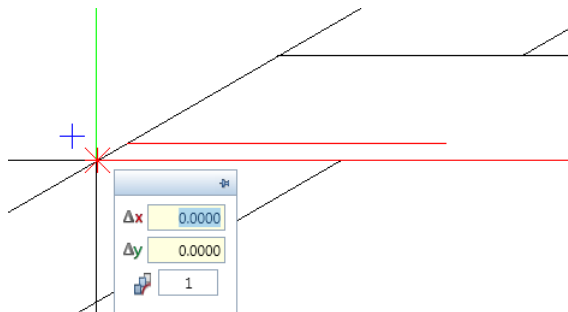
A = Division point 1
B = 0.0-degree track line

- 8 Select ESC to close the  **Line** tool.
- 9 Create four equidistant copies of the construction line and place them above the first one. Click the construction line and select the  **Copy** tool on the context toolbar.



10 *From point or enter offset:*

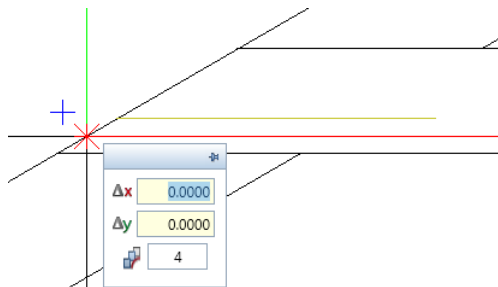
Click the point where the outer edge of the rafter intersects the bottom edge of the collar beam (see illustration).



11 *To point or offset:*

Enter the  **Number** of copies in the coordinate dialog box: **4**.

12 Click the point where the outer edge of the rafter intersects the construction line (see illustration).



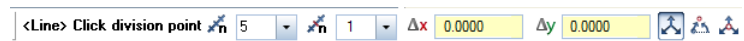
13 Select ESC to close the **Copy** tool.

Use **Division point** again to draw the sloping construction lines. Use **Polar coordinates** to define the direction of the construction line.

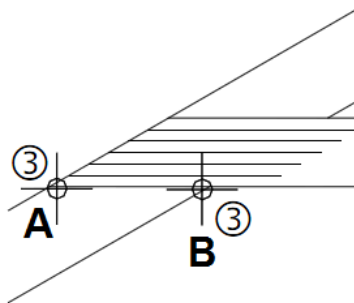
To draw sloping construction lines

➡  **Construction line** mode is still turned on.

- 1 Click  **Line** in the **2D Objects** task area.
- 2 Click  **Individual lines** on the **Line** context toolbar.
- 3 Define the starting point of the sloping construction line:
 - a) Click  **Division point** (shortcut menu).
 - b) Click the end points of the line (see illustration).
 - c) Enter **5** for the number of divisions.
 - d) Click division point 1.

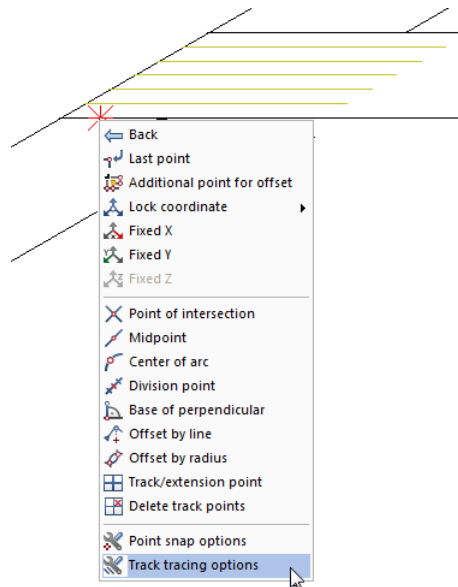


This defines the starting point of the sloping construction line.

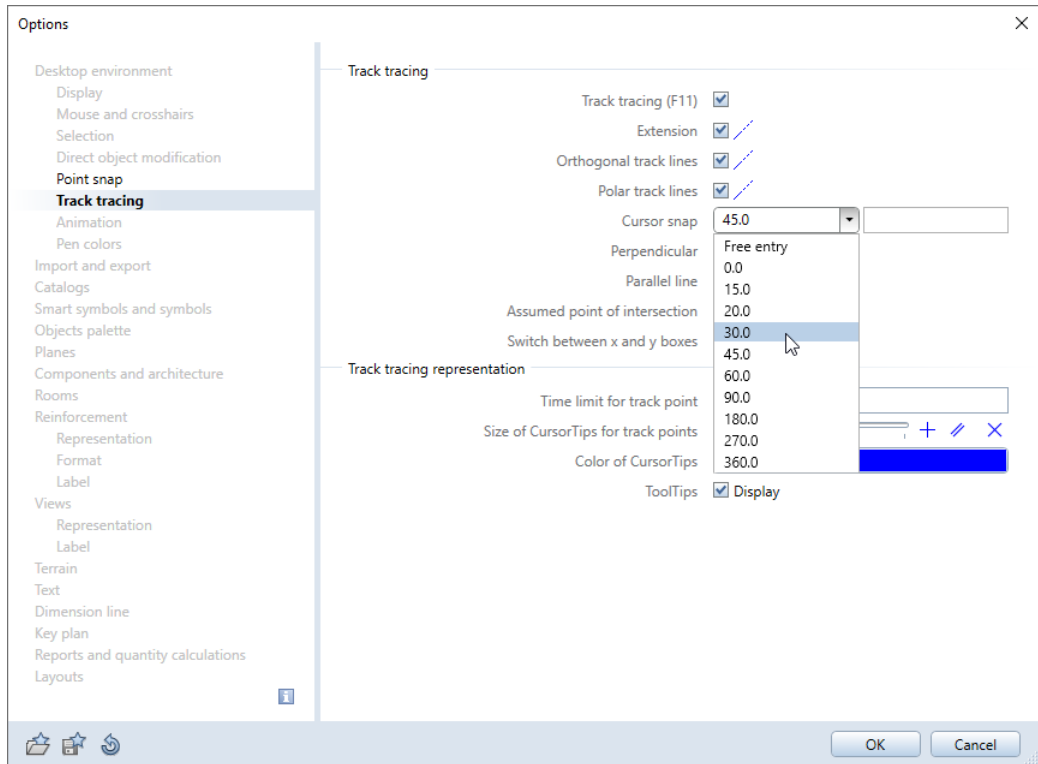


A = Starting point
B = End point

- 4 The construction line is parallel to the rafter. Open the shortcut menu by right-clicking and select  **Track tracing options**.

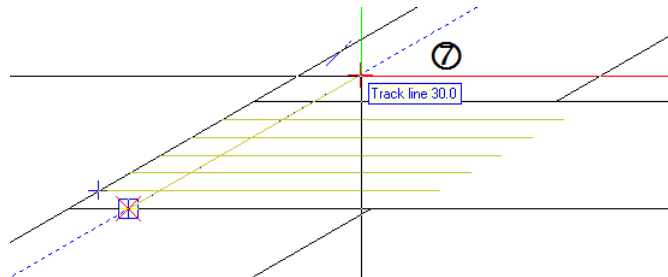


- 5 The **Options** dialog box opens. Change the **Cursor snap** angle to **30°**.

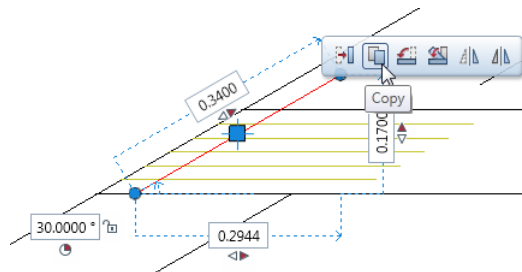


- 6 Click **OK** to close the **Options** dialog box.

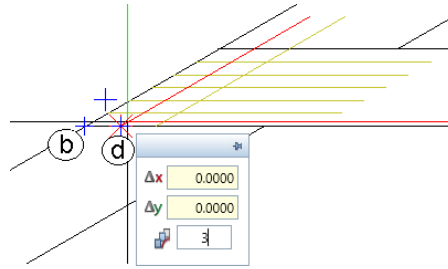
- 7 Move the crosshairs along the **30.0**-degree track line. Use the mouse to define the length of the construction line. The exact length is not important. However, make sure that the line projects beyond the horizontal line at the top.



- 8 Select ESC to close the  **Line** tool.
- 9 Create three copies of the construction line and place them to the right:
 - a) Click the construction line and select the  **Copy** tool on the context toolbar.



- b) *From point or enter offset*: Click the point where the outer edge of the rafter intersects the bottom edge of the collar beam (see illustration).
- c) *To point or enter offset*: Enter the  **Number** of copies in the coordinate dialog box: **3**.
- d) Click the point where the outer edge of the rafter intersects the sloping construction line (see illustration).



Now you have created the temporary grid which helps you place the nails.

- 10 Select ESC to close the **Copy** tool.
- 11 To turn construction-line mode off again, clear the **Construction line** check box in the **Properties** palette.

Nails

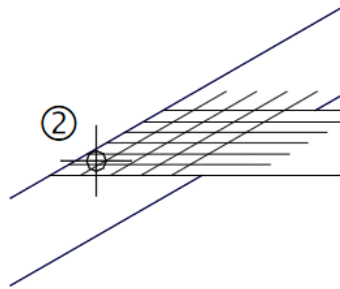
You will place the nails based on the temporary grid consisting of construction lines. First draw a nail as a circle. Then copy this circle to the points where the construction lines intersect. Finally, mirror and copy the complete design onto the opposite side.

Tip: Before placing the copies of the circle, check that you have selected the **Point of Intersection** option in the **Point snap** area of the **Point snap options** (shortcut menu). Allplan must not emit an acoustic signal when you place the copies of the circle.

To place nails

- 1 To draw a nail as a circle, click **Circle** (2D Objects task area). The **Circle** dialog box opens. Select **Circle based on center** and **Enter full circle**.
- 2 To define the center of the circle, click the point where the horizontal construction line at the bottom intersects the vertical construction line on the left.

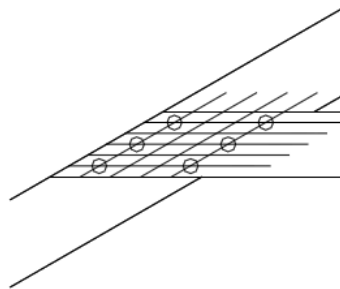
- 3 Enter the radius in the dialog line: **0.01**



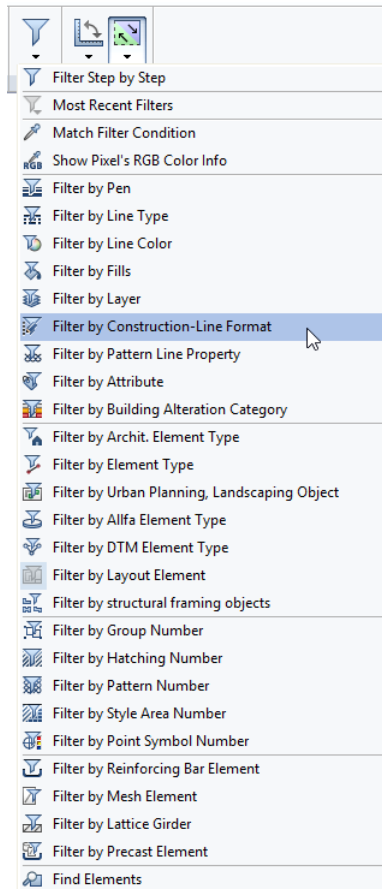
- 4 To create the other nails, click  **Copy** on the shortcut menu of the circle.

From point: Select the center of the circle as the reference point.

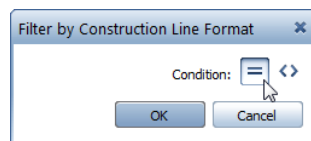
To point: Copy the circle to the points where the construction lines intersect (as shown in the illustration).



- 5 Delete the temporary grid consisting of construction lines by using an element filter. Right-click in the workspace and select **Delete** on the shortcut menu.
- 6 Point to the icon in the **Filter** task area and click within the blue part at the bottom of this area. Click  **Filter by Construction-Line Format** in the drop-down list.



- 7 Select the condition (=) in the dialog box and click **OK** to confirm.

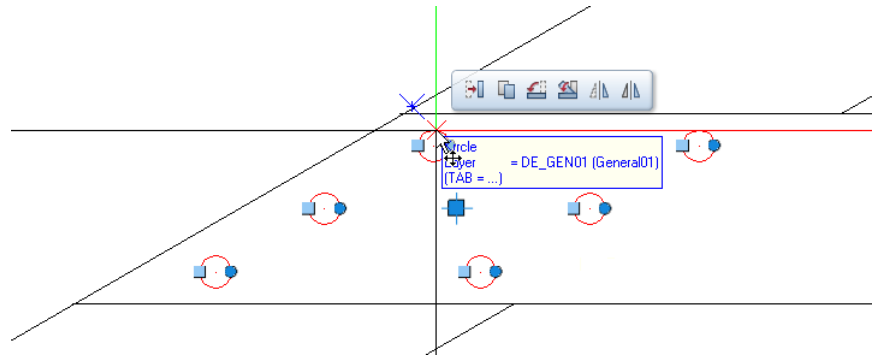


- 8 Enclose the temporary grid in a selection rectangle.
Due to the filter, Allplan deletes only the construction lines (not the nails).
- 9 Select ESC to close the  **Delete** tool.

To finish, you will mirror the nails onto the right rafter.

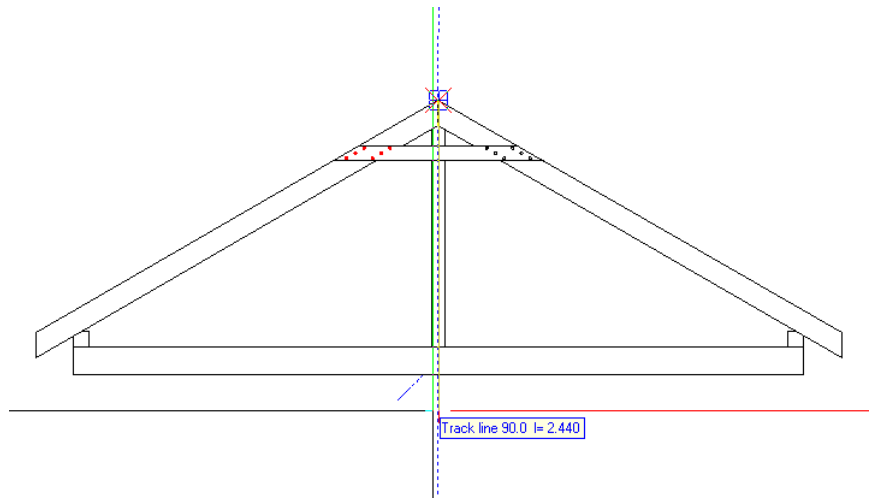
To mirror the nails

- 1 Enclose all nails in a selection rectangle.
- 2 Point to a circle.
Make sure that you do *not point to a handle*!



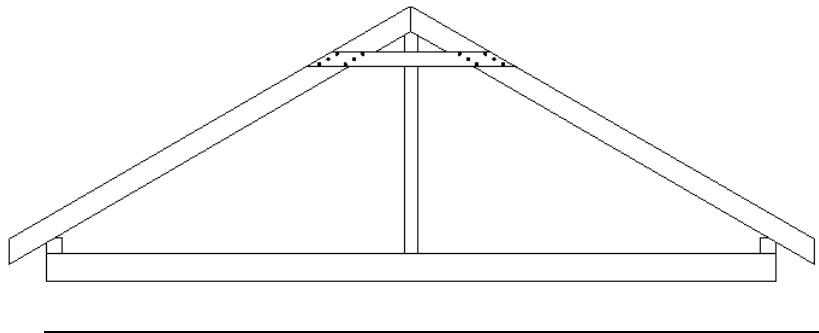
- 3 Click  **Copy and Mirror** on the context toolbar.
- 4 To obtain a mirror axis that is exactly vertical:
 - a) Click the gable peak.
 - b) Move the crosshairs vertically downward along the 90.0-degree track line.
 - c) Click in the workspace below the design.

Tip: Instead of defining the vertical mirror axis by using the 90.0-degree track line, you can click the vertical line between the rafters.



5 Select ESC to finish.

The design should now look like this:



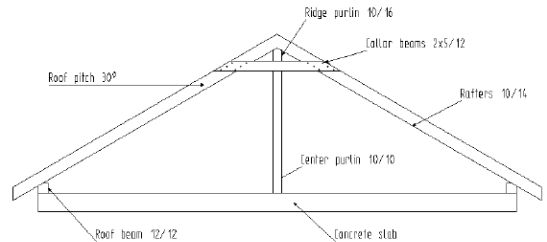
Task 2: labeling the purlin roof

In this section, you will label the purlin roof.

Tools:

Objective:

A Horizontal Text



Labeling

To label the purlin roof

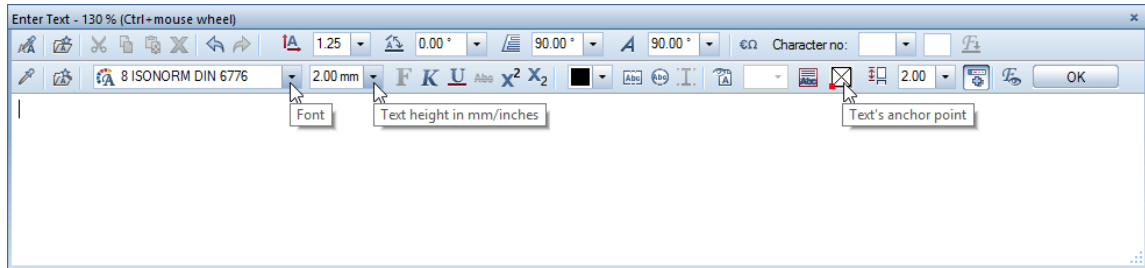
- 1 Maximize the **Quick Access** task area on the **Actionbar**.
- 2 Click **A** **Horizontal Text** (**Quick Access** task area) and define the position of the text by clicking in the workspace (see illustration). Enter the text and define its parameters in the dialog box.



See also: You can find detailed information about entering and editing text in the Allplan Help.

- 3 You do not need the track tracing feature to create labels. Therefore, turn off track tracing by selecting the F11 key.
- 4 To change the text height to **2.0** mm, click in the **Text height** box and enter **2.0**. The text width automatically changes with the aspect ratio.
- 5 Click **Font** and select front number **8 ISONORM DIN 6776** in the drop-down list.

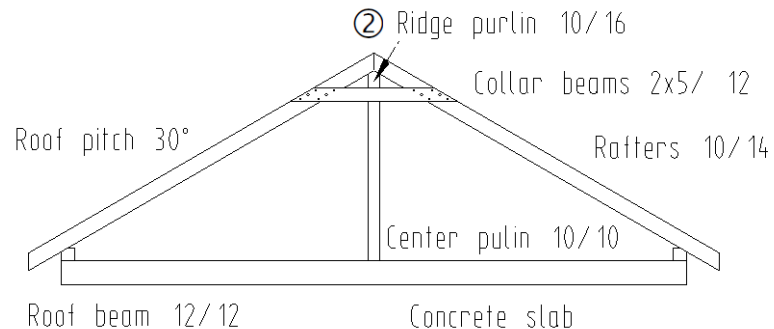
- 6 Click the lower-left corner to define the  **Text's** anchor point.



Tip: To place text, you can also use CTRL+ENTER instead of clicking **OK**.

Tip: As long as you have not placed the text, you can change the drop-in point by clicking anywhere in the workspace.

- 7 Type **Ridge purlin 10/16** for the text and click **OK** to confirm.
This places the text in the workspace.
The  **Horizontal Text** tool is still open.



- 8 Click to define the starting point for the next line of text and label the drawing as shown in the illustration at the beginning of this exercise.
- 9 When you have entered all texts, select ESC to close the  **Horizontal Text** tool.

Creating leaders

A leader connects the text with the design. A leader is always at a defined distance from the text. Allplan creates a leader as a simple line with the current pen. You can apply symbols to the start and end of this line.

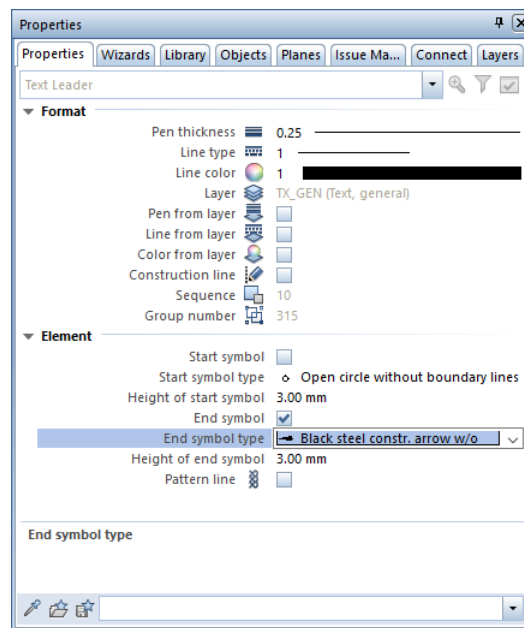
The starting point of the leader is always a defined point of the text. Every text has eight points where the leader can start:



When you move the text, the leader "sticks" to the starting point defined.

To create leaders

- 1 Click  **Text Leader** in the **Quick Access** task area.
- 2 Click  **Individual lines** on the **Text Leader** context toolbar.
- 3 Select the **End symbol** option in the **Properties** palette and choose **Black steel construction arrow without boundaries** in the drop-down list.
- 4 The **Symbol height** is **3.00** mm. Do not change this setting.



- 5 *Leader to text:* Click the text that is to get the leader. Make sure that the point you click is the starting point of the leader (lower-left corner).

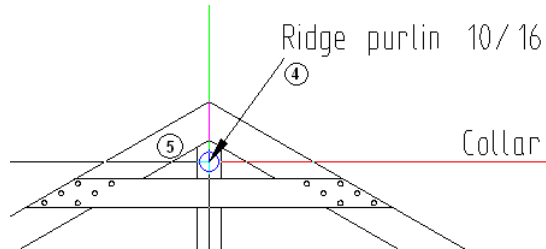
The text appears in the selection color.

Tip: If the leader is not in the correct position, click

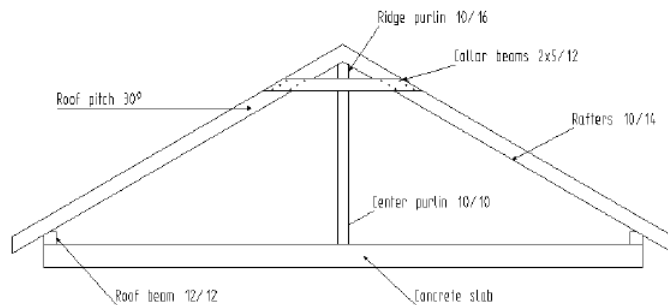
← **Back** on the **Text**

Leader context toolbar and place the leader again.

- 6 *To point:* Click the endpoint of the leader.



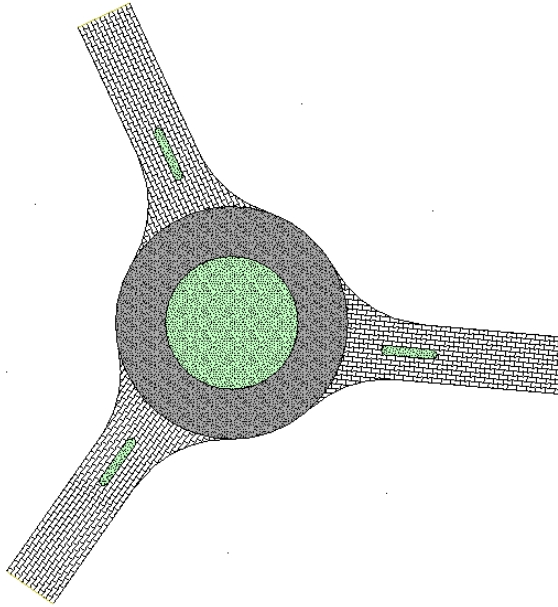
The  **Text Leader** tool is still open. To add a leader to the next line of text, repeat steps 4 and 5. Create more leaders as shown in the illustration.



- 7 Select ESC to close the  **Text Leader** tool.

Exercise 4: rotary with three roads

In this exercise, you will design a rotary with three roads.



You will use the tools in the **2D Objects**, **2D Areas**, and **Change** task areas.

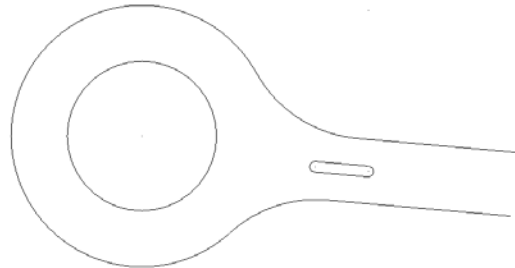
Task 1: designing a rotary with one road

In the first part of this exercise, you will design a rotary with one road leading up to it. In addition, you will create a traffic island in the middle of the road. You will start by drawing a rough outline consisting of a circle, lines, and parallel lines. You will learn about the **Circle** and **Fillet** tools. The **Fillet** tool will produce the final outline.

Tools:

-  **Circle**
-  **Parallel to Element**
-  **Polar coordinates**
-  **Auto-Delete Segment**
-  **Fillet**

Objective:



Rotary with one road

In the first part of this exercise, you will design the rotary and one of the three roads leading up to it.

To draw the rotary

- 1 Click  **Open on a Project-Specific Basis** and open an empty drawing file. Name it **Rotary** and close all the other drawing files.
- 2 Click  **Circle** in the **2D Objects** task area.
- 3 The **Circle** context toolbar opens. Click  **Circle based on center** and  **Enter full circle**.
- 4 Click in the workspace to define the center of the circle.
- 5 Define the **radius** by entering **12.25 m** in the dialog line.
- 6 Select ENTER to confirm.
- 7 Switch to the  **Parallel to Element** tool.

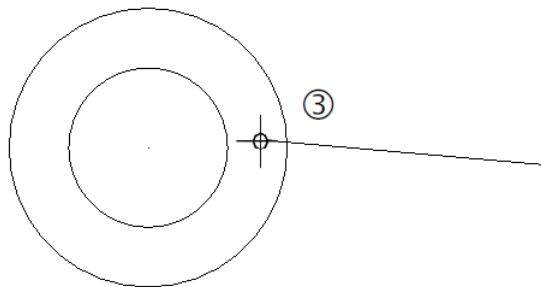
- 8 Click the circle.
- 9 Define the **offset** by entering **5.25** m; select ENTER to confirm.
- 10 Click within the circle to specify the side where you want to create the parallel line.
- 11 Enter **1** for the **Number** and select ENTER to confirm.

This produces a lane width of 5.25 m in the rotary.

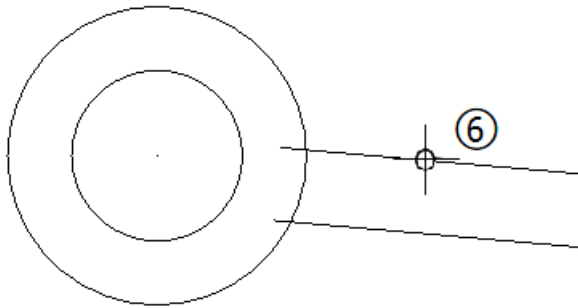
Draw the first road as a line. The road is at a specific angle. Therefore, you will use **polar coordinates**.

To design the first road

- 1 To draw the top edge of the road, click  **Line** (2D Objects task area).
- 2 Click  **Individual lines** on the **Line** context toolbar.
- 3 Click within the rotary to define the starting point of the line (as shown in the illustration).
- 4 To draw the road at a defined angle, click  **Polar coordinates** in the dialog line.
<Line> To point:
The angle between the road and the rotary is 5°. Positive angles work in a counterclockwise direction. Enter **355**.
Select the TAB key; enter **25** for the *length*.
Select ENTER to confirm.

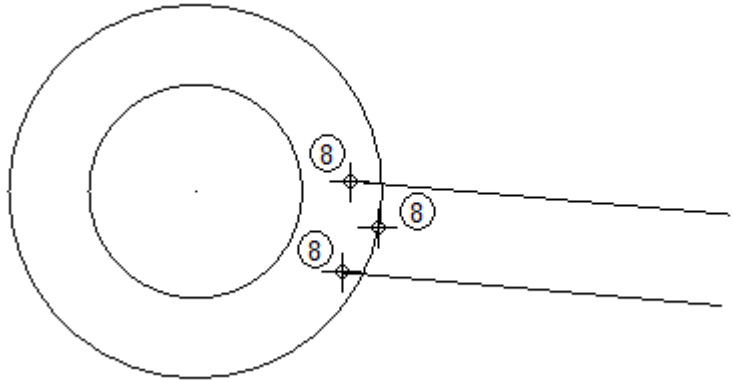


- 5 To draw the bottom edge of the road, click  **Parallel to Element** in the **2D Objects** task area. The  **Line** tool closes automatically.
- 6 Click the line you just created. Enter the following values in the dialog line:
Offset: **6.00**
Which side? Click below the line.
Number: **1**



- 7 You can now delete the redundant line segments protruding into the rotary.
Right-click a line that you want to delete.

- 8 Choose  **Auto-Delete Segment** on the shortcut menu and click the line segments that you want to delete.
Allplan deletes the lines as far as the point where they intersect the rotary.



- 9 Select ESC to close the  **Auto-Delete Segment** tool.

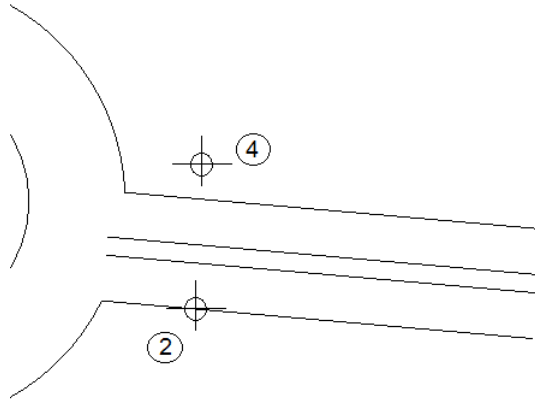
Traffic island for road

The next step is to design the traffic island, which consists of lines that are parallel to the road you just created. You will then use lines to connect the parallel lines.

To design the traffic island

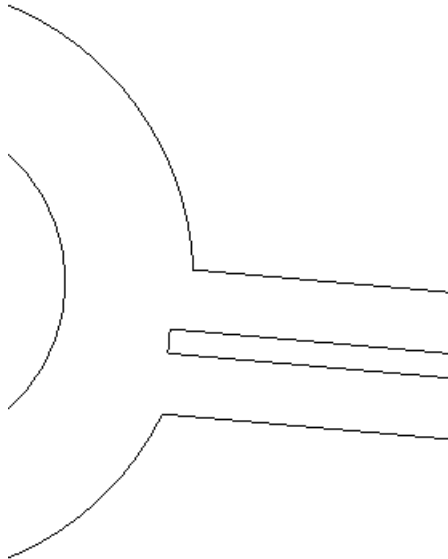
- 1 Click  **Parallel to Element** (2D Objects task area).
- 2 Click the bottom line of the road to use it as the reference element for the traffic island (see illustration).
- 3 To create the bottom edge of the traffic island, enter **2.50** for the offset.
- 4 Click above the reference element to define the side and enter the *Number* in the dialog line: **1**.
This creates the first parallel line; the  **Parallel to Element** tool is still open.

- 5 The dialog line prompts you again to enter the offset. The offset you enter now applies to the parallel line you just created. Enter the offset between the bottom and top of the traffic island: **1.00**



- 6 Click  **Line** in the **2D Objects** task area. The  **Parallel to Element** tool closes automatically.
Make sure that  **Individual lines** is selected on the **Line** context toolbar.
- 7 Choose  **Delta Point** in the dialog line.

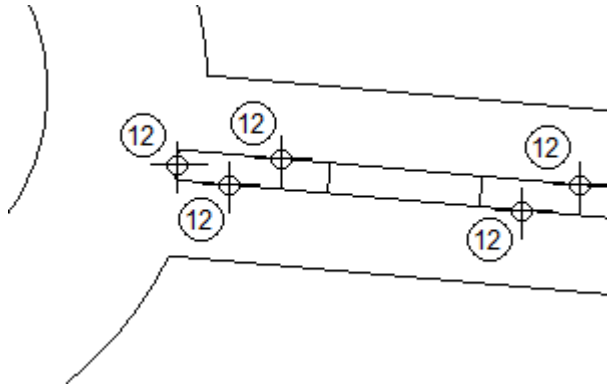
- 8 Join the end points of the two parallel lines (see illustration) and select ESC to close the tool.



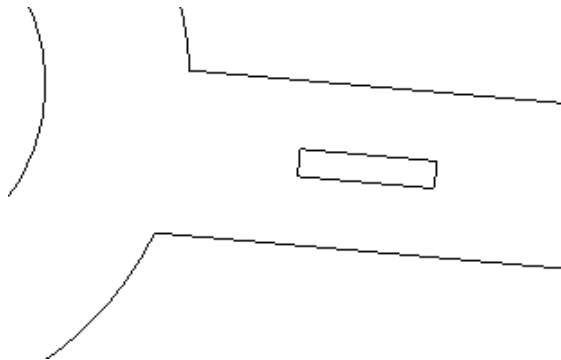
You will use this line as the reference line for the next step.

- 9 To design the missing sides of the traffic island, click  **Parallel to Element** in the **2D Objects** task area.
- 10 Click the line you just created to use it as the reference element. The dialog line prompts you to make various entries. Enter the following values:
 - *Offset:* **5.00**; *Which side?* **To the right**; *Number:* **2**
 - Select ESC to close the tool.
- 11 Right-click a line of which you want to delete redundant segments.

- 12 Choose  **Auto-Delete Segment** on the shortcut menu and click the line segments that you want to delete (see illustration).



- 13 Select ESC to close the  **Auto-Delete Segment** tool. The result should look like this:

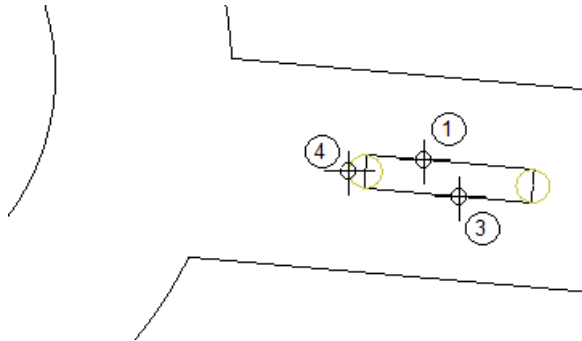


Filleting the road and traffic island

In the next exercise, you will create the final outlines of the road and the traffic island by means of the **Fillet** tool. You can use this tool to fillet corners and to join nontouching lines with arcs. After having clicked the two elements, you can choose one of the auxiliary circles presented.

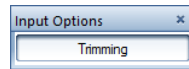
To fillet the road and the traffic island

- 1 Use the  **Fillet** tool to fillet the traffic island.
To open this tool, point to the top line of the traffic island.
- 2 Open the shortcut menu and click  **Fillet**. This selects the first element.
- 3 Click the bottom line of the traffic island.
The fillet radius is **0.5**. Select ENTER to confirm.
Two auxiliary circles appear on the screen.
- 4 Click the circle that you want to use for the fillet.



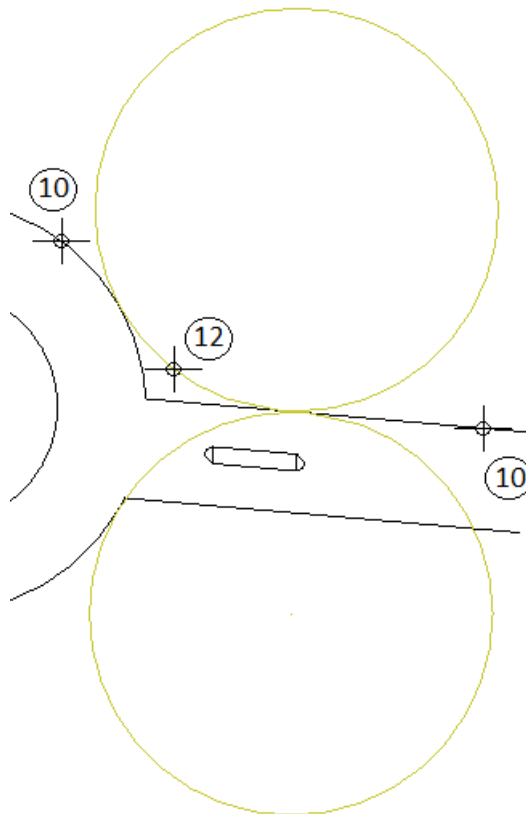
- 5 To fillet the opposite side of the traffic island, click the top and bottom lines again.
- 6 Here, too, the fillet radius is **0.5**. Select ENTER to confirm.
- 7 Click the second circle to use it for the fillet.
- 8 If you can't see the result, click F8 to update the screen contents.

- 9 The  **Fillet** tool is still open. You can see **Trimming** in the input options. When this button is selected (default setting), Allplan automatically shortens or lengthens the initial lines as far as the fillet. If this button isn't selected, click it to select trimming.

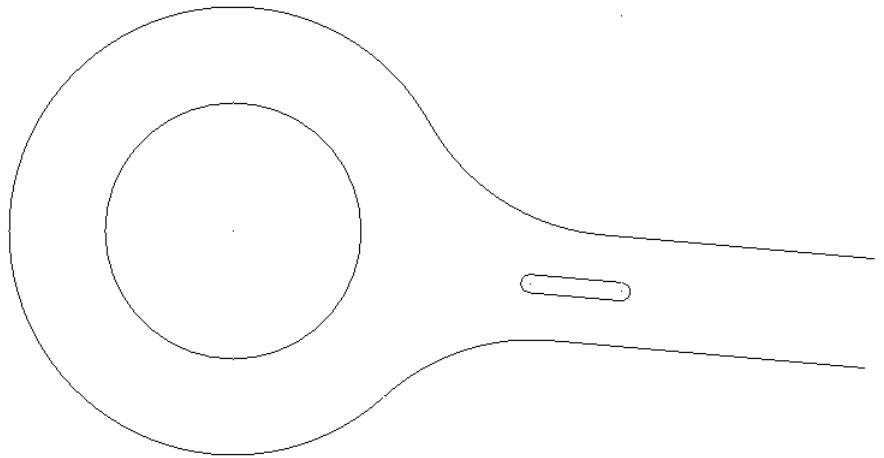


This creates the traffic island. The next step is to fillet the road leading up to the rotary.

- 10 Click the top edge of the road and the adjoining outer arc to create the fillet at the top (see illustration).
- 11 Enter the radius: **12.00**
- 12 Click the circle that you want to use for the fillet.



- 13 Click the bottom edge of the road and the adjoining outer arc to create the fillet at the bottom (see illustration). Repeat steps 7 and 8.
- 14 Delete redundant segments by selecting  **Auto-Delete Segment** (shortcut menu). Your screen should now look like this:



- 15 Select ESC to close the  **Auto-Delete Segment** tool.

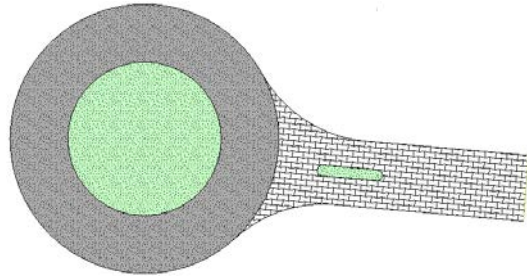
Task 2: pattern

In this task, you will apply a pattern to the road leading up to the rotary. You will learn about the **Pattern** and **Pattern definition** tools.

Tools:

-  Pattern
-  Pattern defaults
-  Area detection
-  Island detection
-  Select Pattern
-  Pattern Width
-  Pattern Height
-  Modify Format Properties
-  Convert Surface Element

Objective:

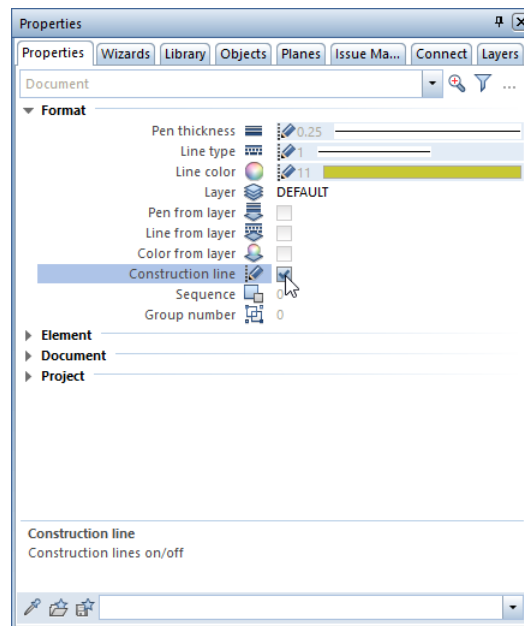


Creating closed outlines

The first step involves creating closed outlines. This is necessary if you want to use **area detection** to apply patterns to the rotary and the road.

To create a closed outline for the road

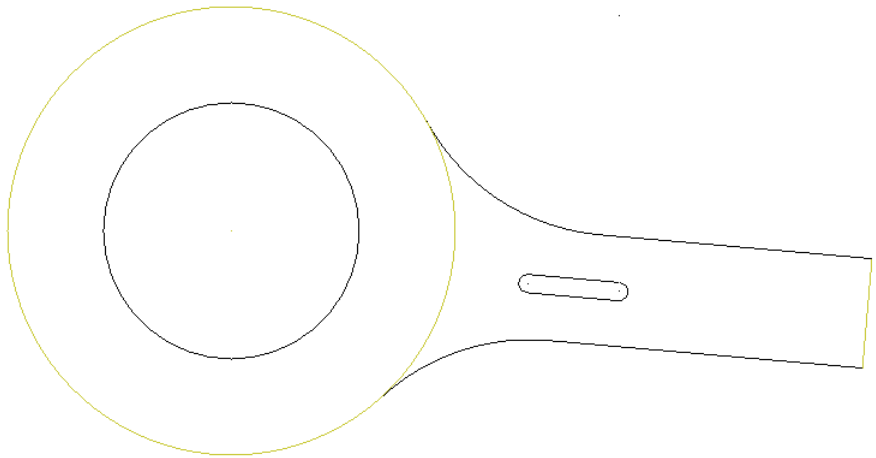
- 1 To do this, you will use construction lines.
Select the  **Construction line** check box in the **Properties** palette.



Note: Construction lines are like erasable pencil lines on conventional drawings. When you turn on construction-line mode, Allplan draws all elements with the construction-line color and line type selected in the  **Options – Desktop environment – Display page – Drawing file and NDW window** area. Elements drawn as construction lines are excluded from printouts.

- 2 Click  **Circle** in the **2D Objects** task area.
- 3 The **Circle** context toolbar opens. Click  **Circle based on center** and  **Enter full circle**.

- 4 Click the existing center of the inner circle.
- 5 Define the **radius** by entering **12.25** m in the dialog line.
- 6 Select ENTER to confirm.
- 7 Switch to the  **Line** tool (**2D Objects** task area) to join the right ends of the two lines representing the boundaries of the road.
The  **Circle** tool closes automatically.
Make sure that  **Individual lines** is selected on the **Line** context toolbar.
- 8 Choose  **Delta Point** in the dialog line.
- 9 Join the end points of the two parallel lines (see illustration) and select ESC to close the tool.
- 10 Turn off construction-line mode again.
Your drawing should look like this:



Applying a pattern to the road

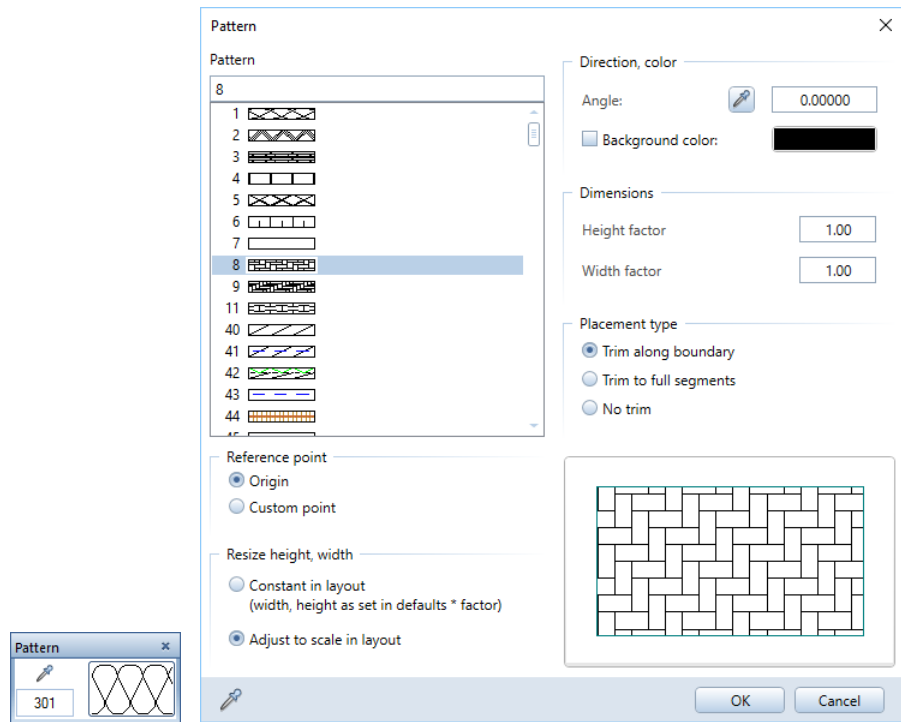
In this section, you will apply a pattern to the road leading up to the rotary. You will use area detection to define the area which gets a pattern. The traffic island will not get a pattern.

To apply a pattern to the road

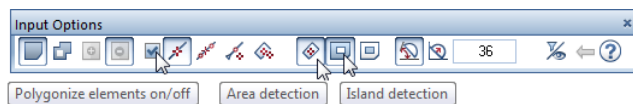
- 1 Click  **Pattern** (2D Areas task area).
- 2 Click  **Single** in the input options.



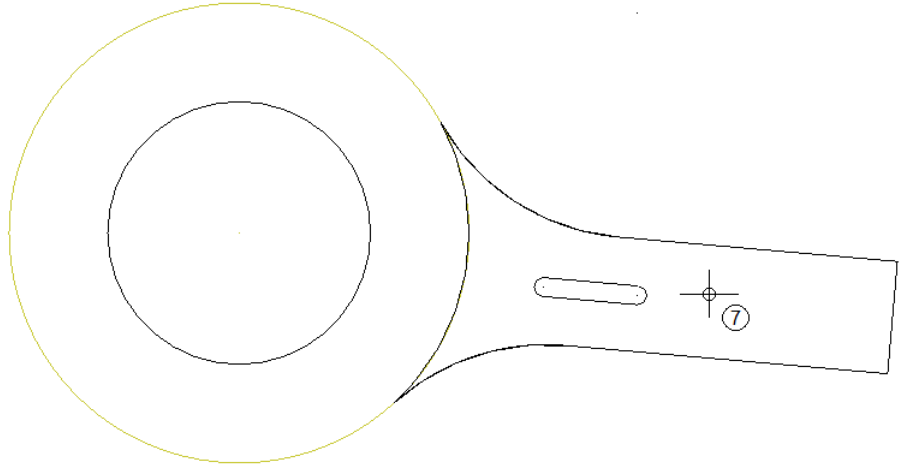
- 3 Click **Properties** on the **Pattern** context toolbar.
- 4 Select pattern **8** and define the following parameters:
 - **Reference point** area:
Origin
 - **Resize height, width** area:
Adjust to scale in layout
 - **Dimensions** area:
Height factor and width factor: 1.00
 - **Placement type** area:
Trim along boundary



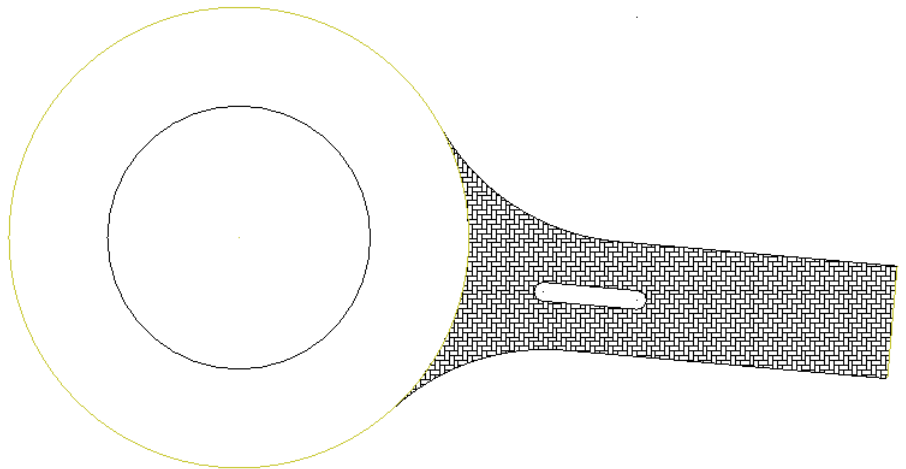
- 5 Check that the **Polygonize element** check box is selected in the input options.
- 6 Select  **Area detection** and  **Island detection** in the input options.
 **Island detection** detects closed outlines within an area, cutting them out automatically.



- 7 Click within the closed outline of the road.
Make sure that you do not click within the part that you want to cut out.



Allplan detects the outline of the road as a closed area, automatically cutting out the traffic island. This is indicated by an acoustic signal.



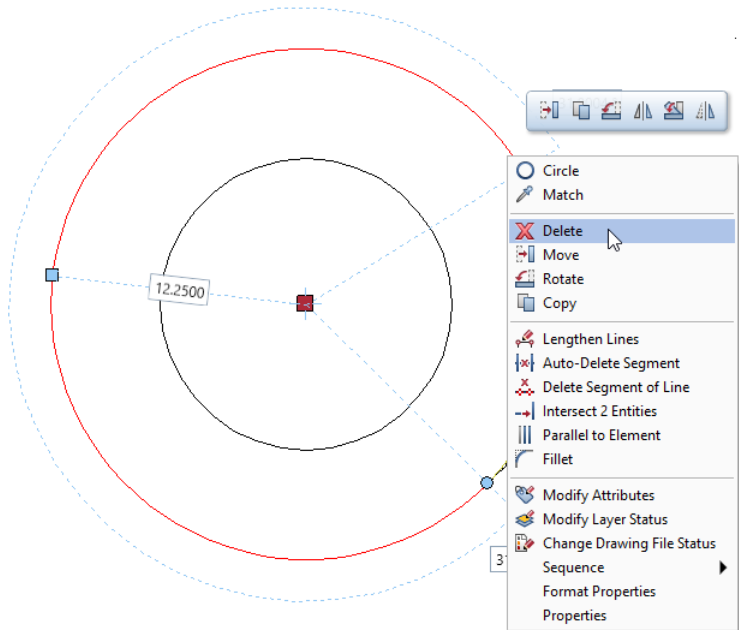
- 8 Select ESC to close the  Pattern tool.

Applying a pattern to the rotary

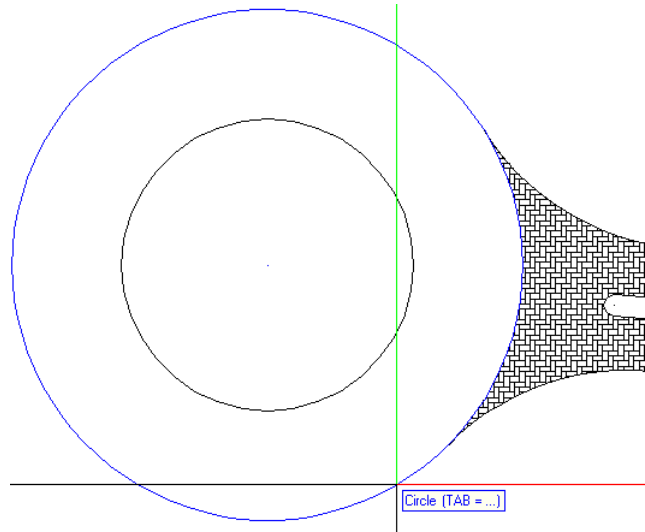
The next step is to apply two different patterns to the rotary. The procedure is basically the same as the one described in the previous step.

To apply a pattern to the rotary

- 1 Before you apply patterns to the areas, delete the arc that is directly under the circle in construction-line format.
To do this, click the arc.
- 2 Right-click to open the shortcut menu and select **Delete**.

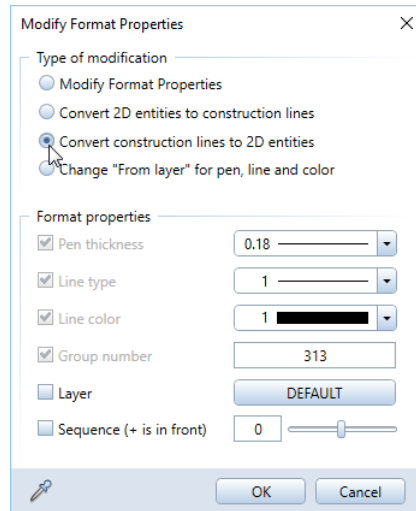


- 3 Point to the outer circle of the rotary. Make sure that you do not select the part of the circle that belongs to the road. The arc appears in the selection preview color.



- 4 Click the arc.
- 5 Convert the circle in construction-line format to a design entity. Select the  **Modify Format Properties** tool in the **Change** task area.

- 6 Select **Convert construction lines to 2D entities** and click **OK**.



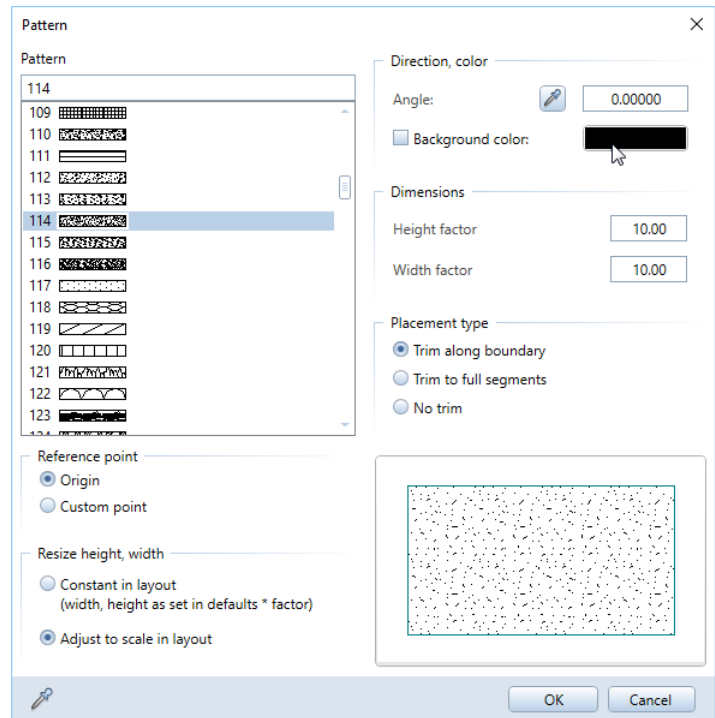
- 7 Select ESC to finish.
- 8 Click  **Pattern** (2D Areas task area).
- 9 Click  **Single** in the input options.



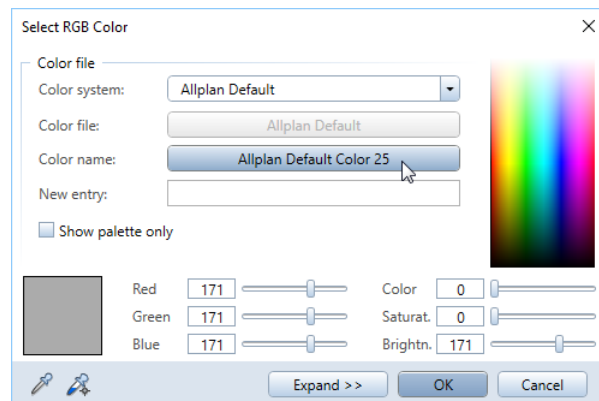
- 10 Click **Properties** on the **Pattern** context toolbar.
- 11 Select pattern **114**, enter **10** for the **Height factor** and the **Width factor**, and define the following parameters:
- **Reference point area:**
Origin
 - **Resize height, width area:**
Adjust to scale in layout
 - **Placement type area:**
Trim along boundary

12 The pattern gets a background color.

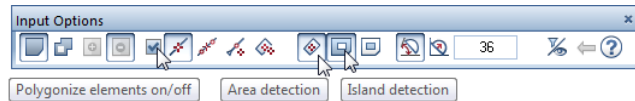
Select the **Background color** option and click in the box to select a color.



13 Click the **Color name** button and select **Default Allplan Color 25**.



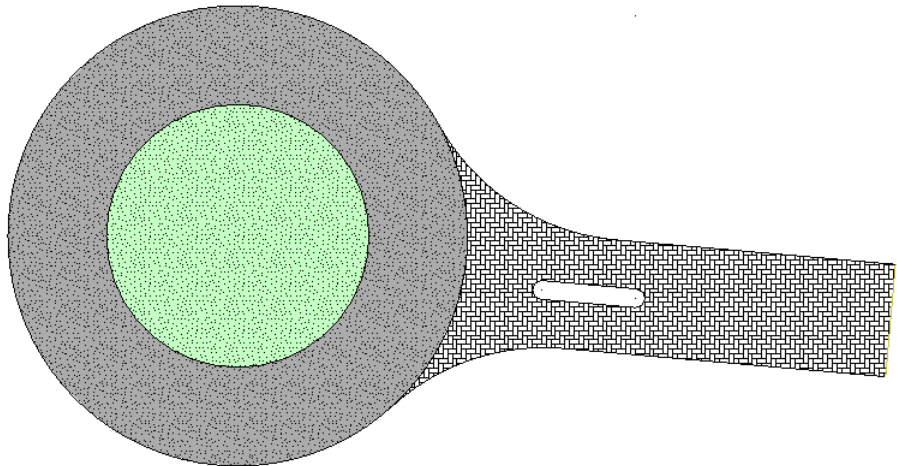
- 14 Close the **Select RGB Color** and **Pattern** dialog boxes by clicking **OK**.
- 15 Check the input options to see if the **Polygonize elements on/off** check box,  **Area detection**, and  **Island detection** are turned on.



- 16 Click the outer circle.
- 17 Select **ESC** again to close the  **Pattern** tool.

Apply pattern **105** to the inner circle yourself. Enter **10** for the **Height factor** and the **Width factor**. Select **Allplan Default Color 78** for the background color.

The result should look like this:



Applying a pattern to the traffic island

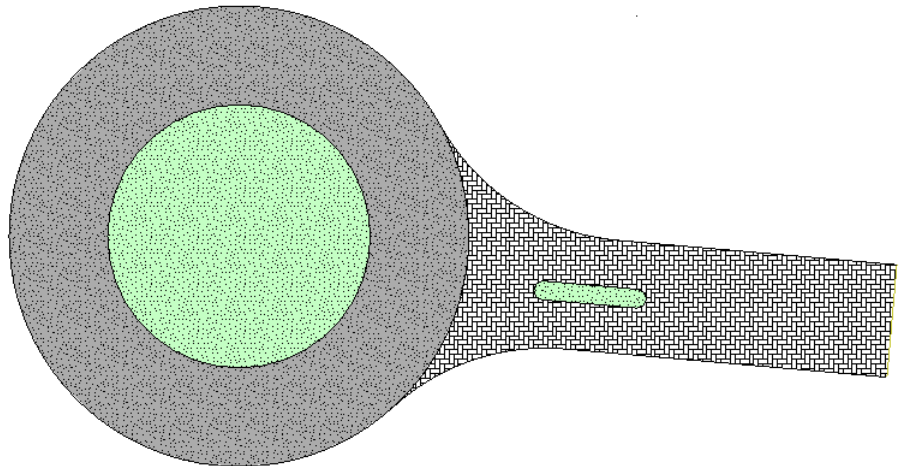
Finally, you will apply a pattern to the traffic island of the road. You will use the same pattern as for the inner circle of the rotary.

To apply a pattern to the traffic island

- 1 Click  **Pattern** (2D Areas task area).
- 2 To match a pattern you have already applied, click  **Match parameters** on the **Pattern** context toolbar.
- 3 Click the pattern in the inner circle of the rotary.
- 4 Click  **Single** in the input options.
- 5 Check that  **Area detection** is selected in the input options.
- 6 Zoom in on the area around the traffic island.
- 7 Click within the traffic island.

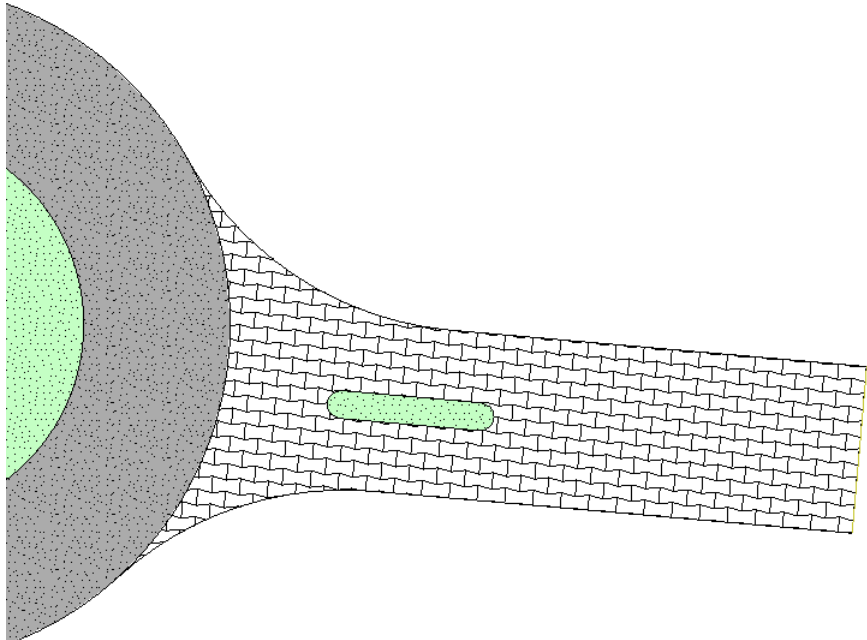
The traffic island gets the pattern.

The result should look like this:



Defining a new pattern

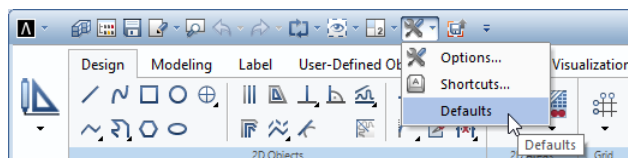
Allplan 2022 comes with various ready-made patterns (depending on the configuration). You can also modify existing patterns and define new ones. In the following exercise, you will learn how to define a new pattern and apply it to the road.



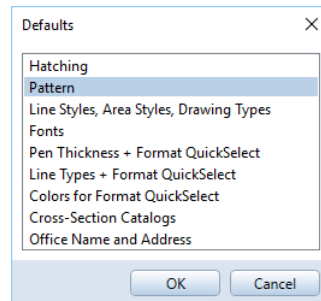
Please read the notes on defining hatching styles. They also apply to patterns.

To define a new pattern

- 1 Open the  **Default Settings** drop-down list on the Quick Access Toolbar and select **Defaults**.



- 2 Click **Pattern** on the context toolbar.

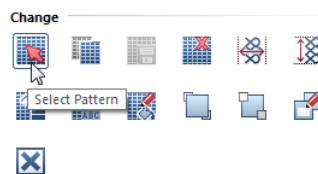


Tip: The patterns that are already defined depend on the configuration you purchased. Patterns 10 and the following numbers are usually free.

When you select a free pattern, only the editing frame and the temporary crosses are visible on the screen.

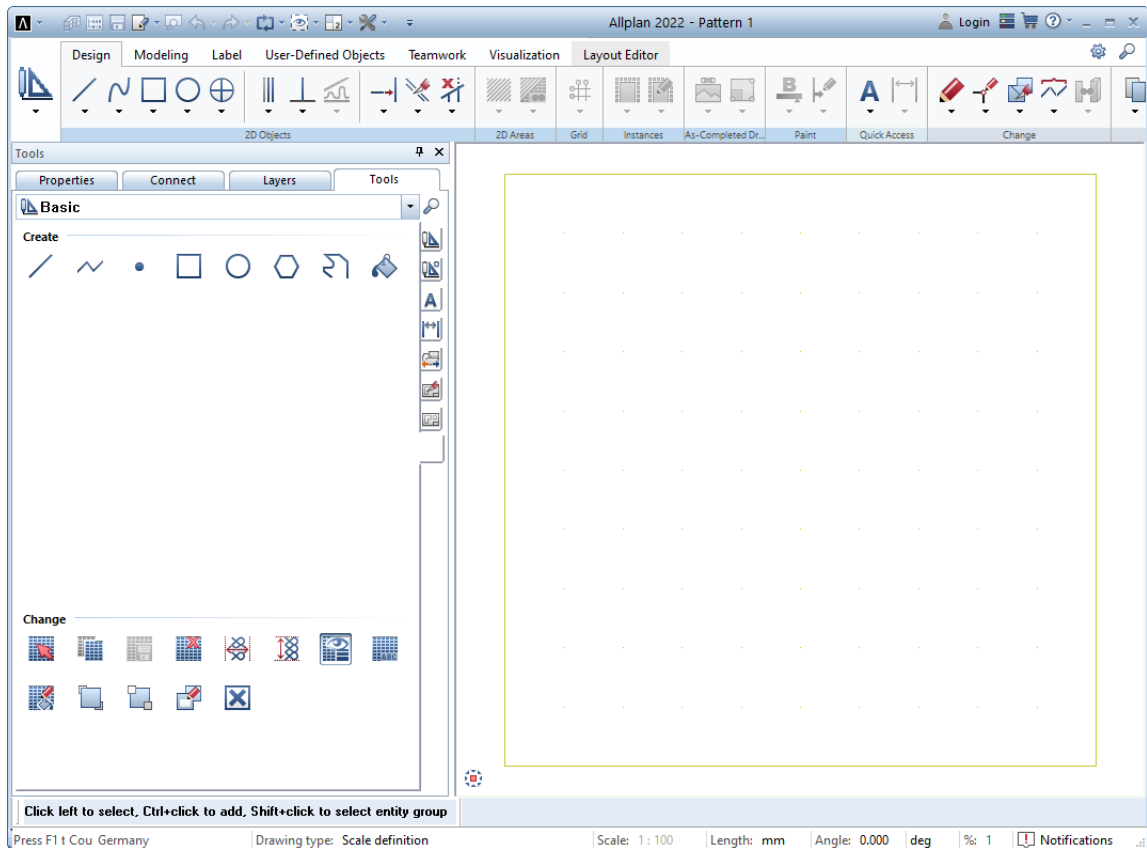
Begin by selecting an unassigned pattern.

- 3 The **Tools** palette opens. It helps you create your own patterns. Go to the **Change** area and click  **Select Pattern**.



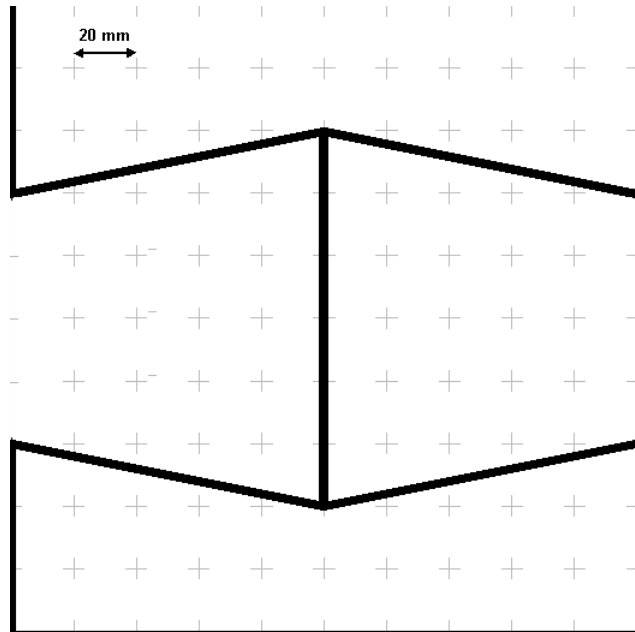
- 4 Select a free number (for example, 11) in the **Select Pattern** dialog box.

Allplan displays an editing frame on the screen, making it easier for you to define patterns. The frame contains a grid of dots to help you draw the pattern.



- 5 Click  **Pattern Width** in the **Tools** palette (**Change** area) and enter the *width* of the pattern in mm in the dialog line: **200**. Select ENTER to confirm.
- 6 Click  **Pattern Height** in the **Tools** palette (**Change** area) and enter the *height* of the pattern in mm in the dialog line: **200**. Select ENTER to confirm.

- 7 Click  **Line** in the **Tools** palette (**Create** area).
Click  **Polyline** on the **Line** context toolbar.
Draw the pattern as shown in the illustration.



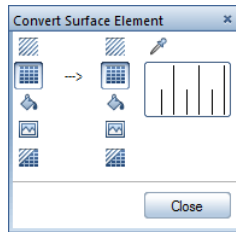
- 8 Select ESC three times to close the  **Line** tool and to finish defining the pattern.
- 9 Click **Yes** when you see the **Would you like to save the pattern definition?** prompt.
- The **Tools** palette closes automatically.

Applying the new pattern

The pattern is defined. Now you can apply the new pattern to the road. You need to make some settings in the **Pattern** dialog box to adjust the pattern to the road.

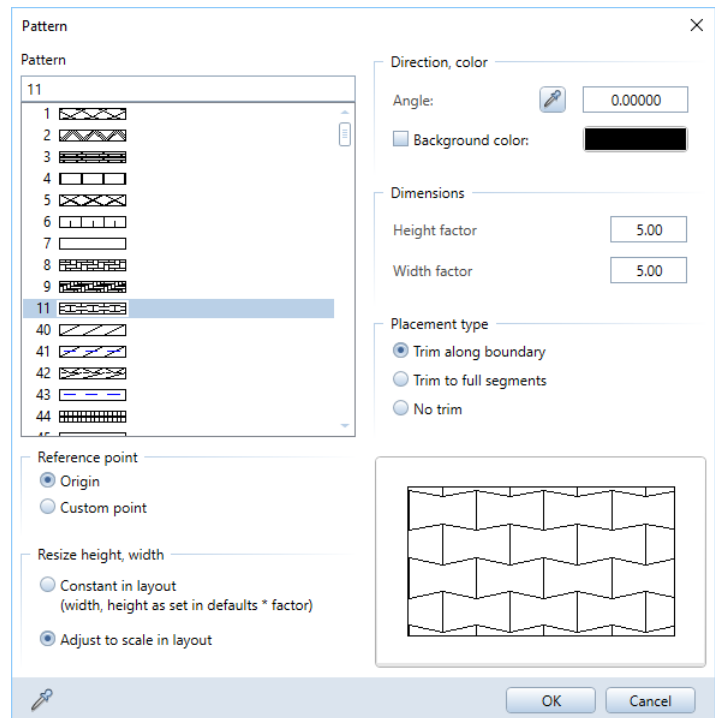
To apply the new pattern

- 1 Click  **Convert Surface Element** (2D Areas task area).
- 2 On the context toolbar, select **Modify, Convert Pattern to Pattern** and click **Properties**.



The **Pattern** dialog box opens.

- 3 Select pattern **11** and enter **5** for the **Width factor** and **Height factor** in the **Size** area. In addition, define the following parameters:
 - **Reference point area:**
Origin
 - **Resize height, width area:**
Adjust to scale in layout
 - **Placement type area:**
Trim along boundary



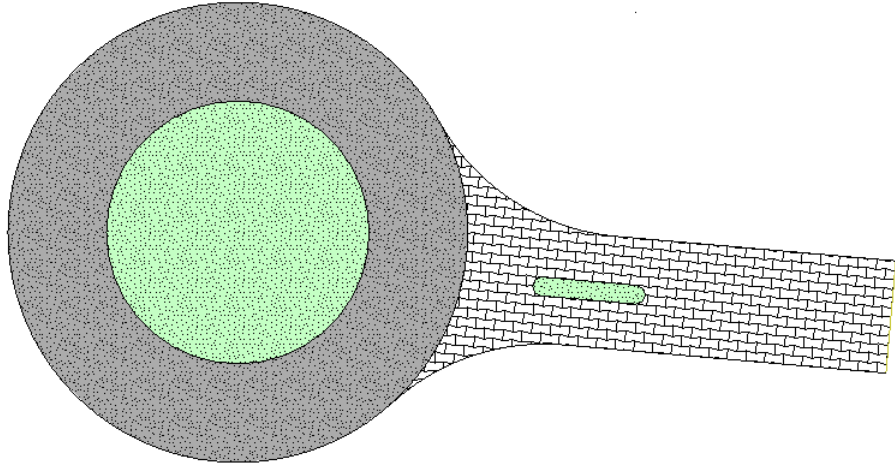
Place the pattern at an angle of 5°. You can match the angle directly from the drawing.

- 4 Go to the **Direction, color** area and click  next to **Angle**. The dialog box closes; you are back in drafting mode.
- 5 Click the top line of the road. The dialog box opens again, displaying an angle of -5°.
- 6 You want to generate the pattern from the lower-right point; this is the reference point. Therefore, to define the reference point yourself, click **Custom point** in the **Reference point** area. The dialog box closes temporarily.
- 7 Click the lower-right point to define it as the reference point.
- 8 Click **OK** to confirm the settings.
- 9 Click the pattern. The pattern changes accordingly.

Tip: Allplan generates the pattern from the reference point.

To define a new reference point, switch the reference point to **Origin** and then to **Custom point**.

10 Select the ESC key to close the tool.



Task 3: completing the design

In this task, you will add the two missing roads to the rotary. You will learn about the **Copy and Rotate** tool.

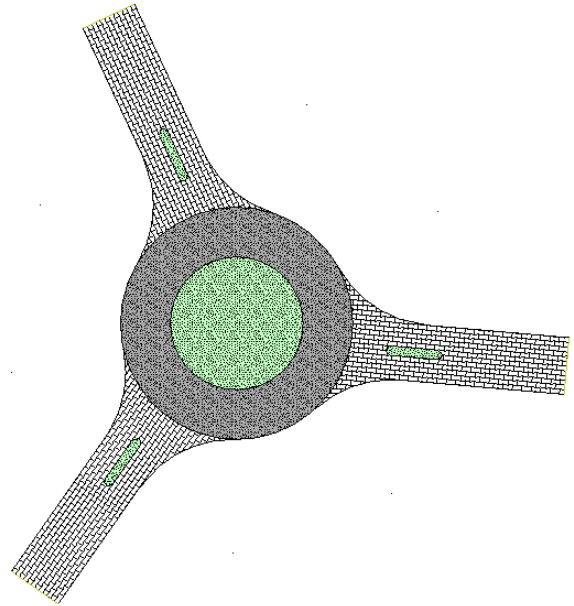
Tools:

Objective:



Copy and
Rotate

Pattern
Line



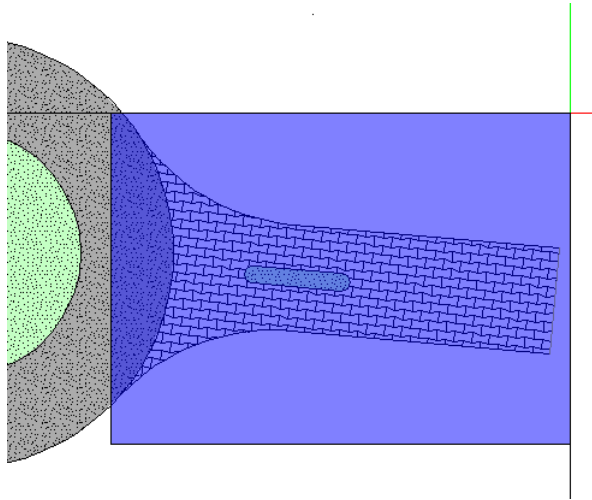
Creating the missing roads

You will complete the rotary so that three roads lead up to it.

To add two more roads

- 1 Enclose the entire road in a selection rectangle that you open from the lower left to the upper right (positive x-direction)..

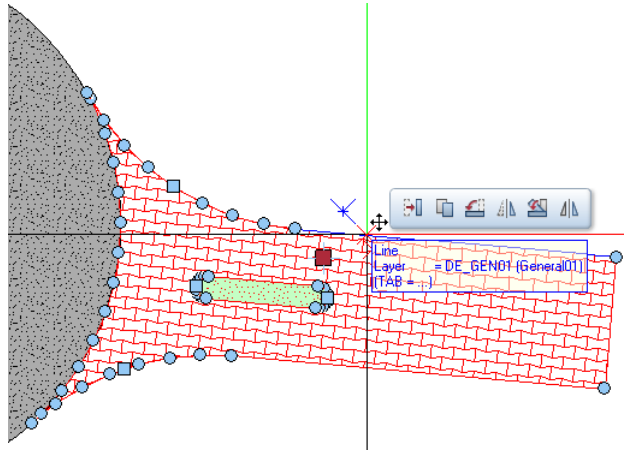
Tip: You can select elements by enclosing them in a selection rectangle. The default setting is  **Select Elements Based on Direction** (Actionbar – Work Environment task area): By opening the selection rectangle in the positive x-direction, you select only the elements that are fully bounded by the selection rectangle. By opening the selection rectangle in the negative x-direction, you select all elements that are fully bounded or partially bounded by the selection rectangle.



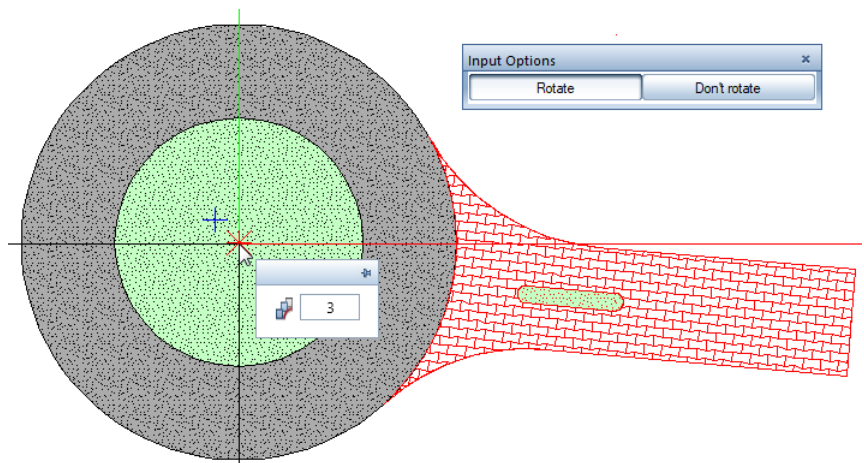
The road, pattern, and traffic island appear in the selection color. In addition, you can see handles.

- 2 Point to the upper boundary line of the road. Make sure that you do not point to a handle!

- 3 Click  **Copy and Rotate** on the context toolbar.

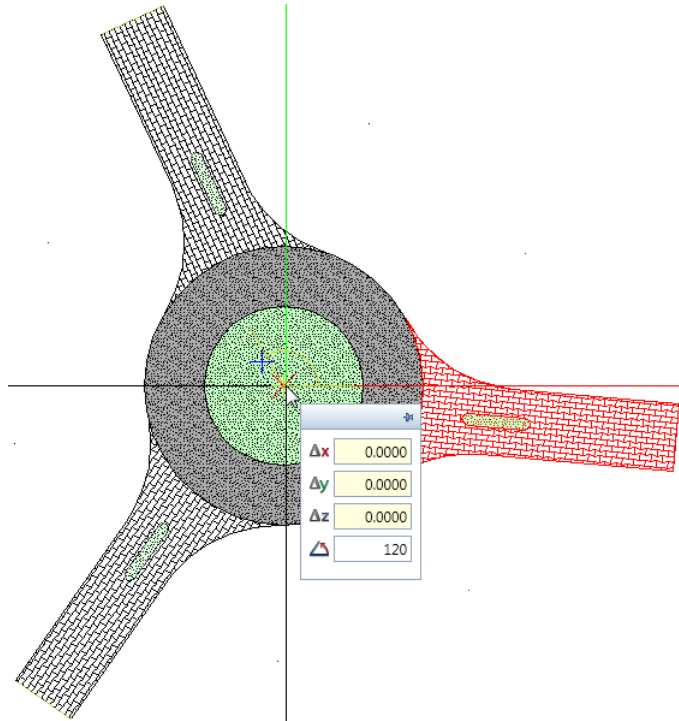


- 4 *Click base point of rotation*
Click the center of the rotary.
- 5 Select **Rotate** in the input options to rotate the elements at the same time.
- 6 *How many times?*
Enter  **3** in the coordinate dialog box; select ENTER to confirm.

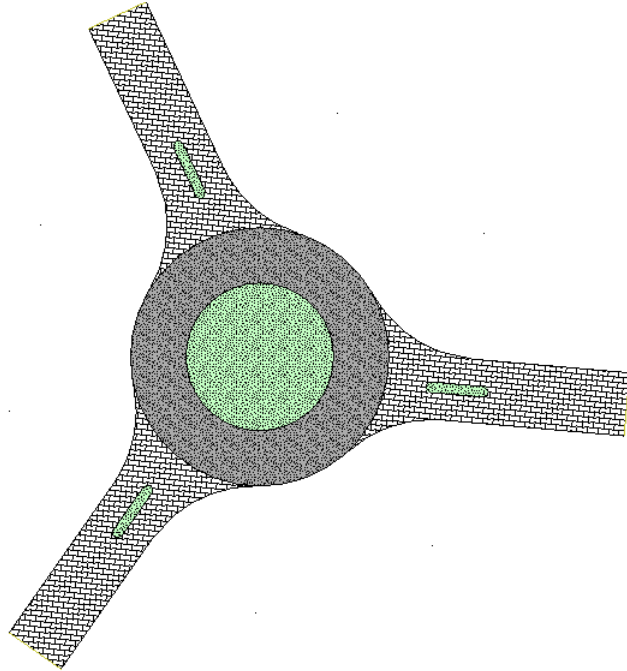


7 *Starting point, reference line, or rotation angle*

Enter **120°** for the angle of rotation in the coordinate dialog box;
select ENTER to confirm.



8 Select ESC to close the tool.



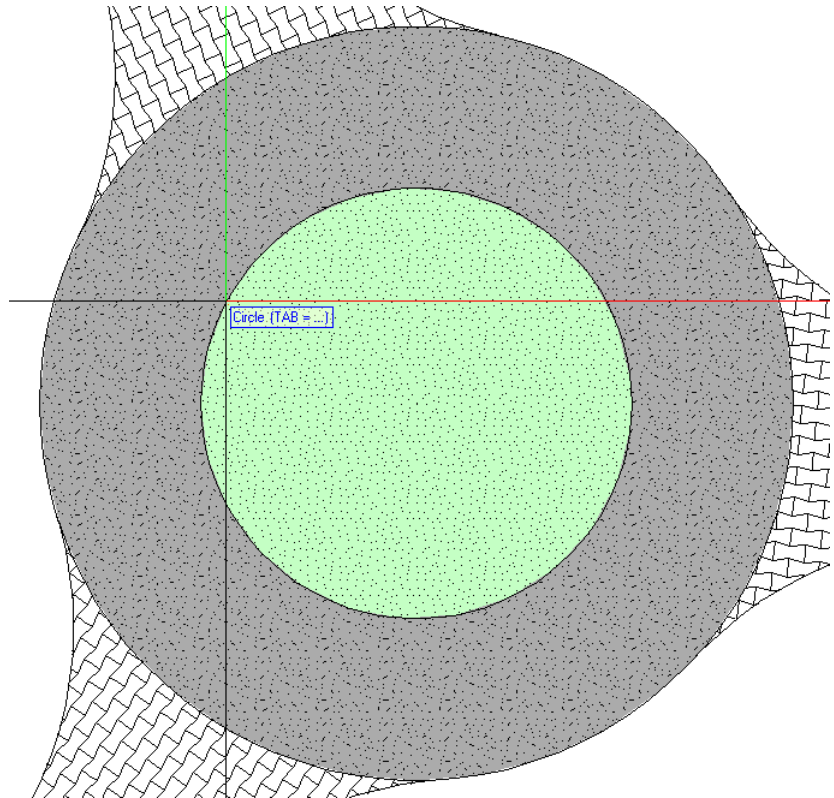
You can now adjust the pattern in the two new roads to the road angle (see "Applying the new pattern (on page 153)").

Pattern line

Finally, you will add a row of large paving stones to the edge of the inner circle of the rotary. To do this, you will use the **Pattern Line** tool.

To add a pattern line

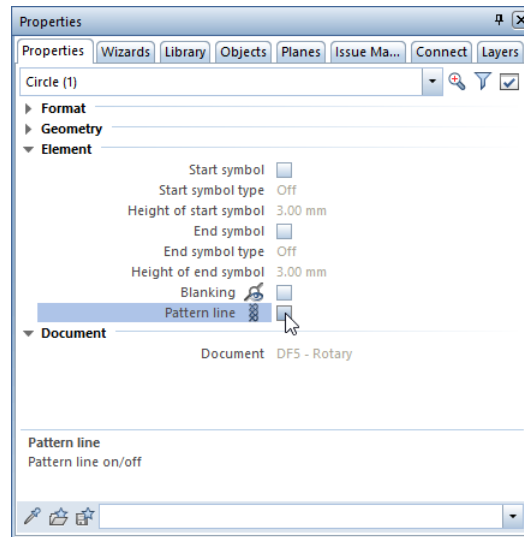
- 1 Zoom in on the inner circle of the rotary.
- 2 Point to the inner circle.
Pay attention to **element info**, which must display **circle**.



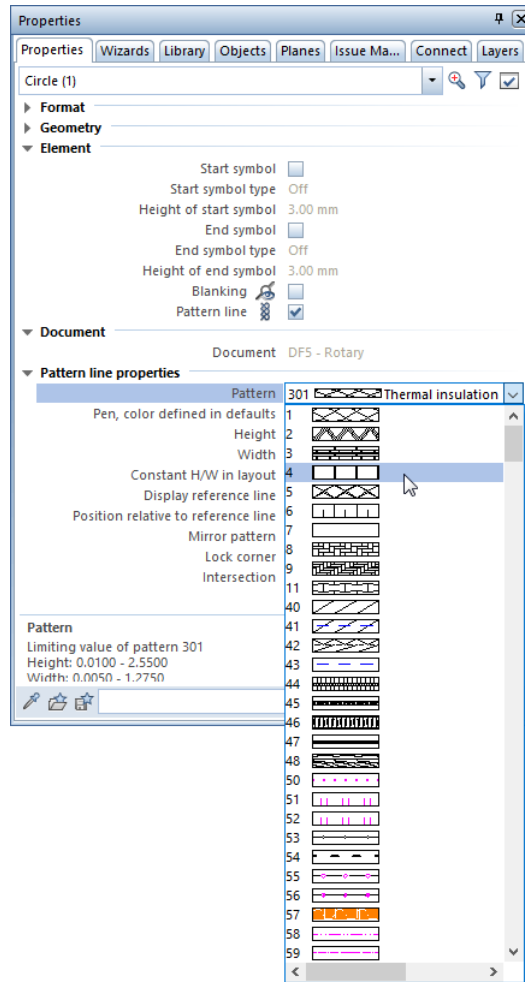
- 3 Double-click the inner circle.

The circle appears in the selection color. The **Properties** palette opens. You can see the properties of the selected circle.

- 4 Select the **Pattern line** option in the **Properties** palette.

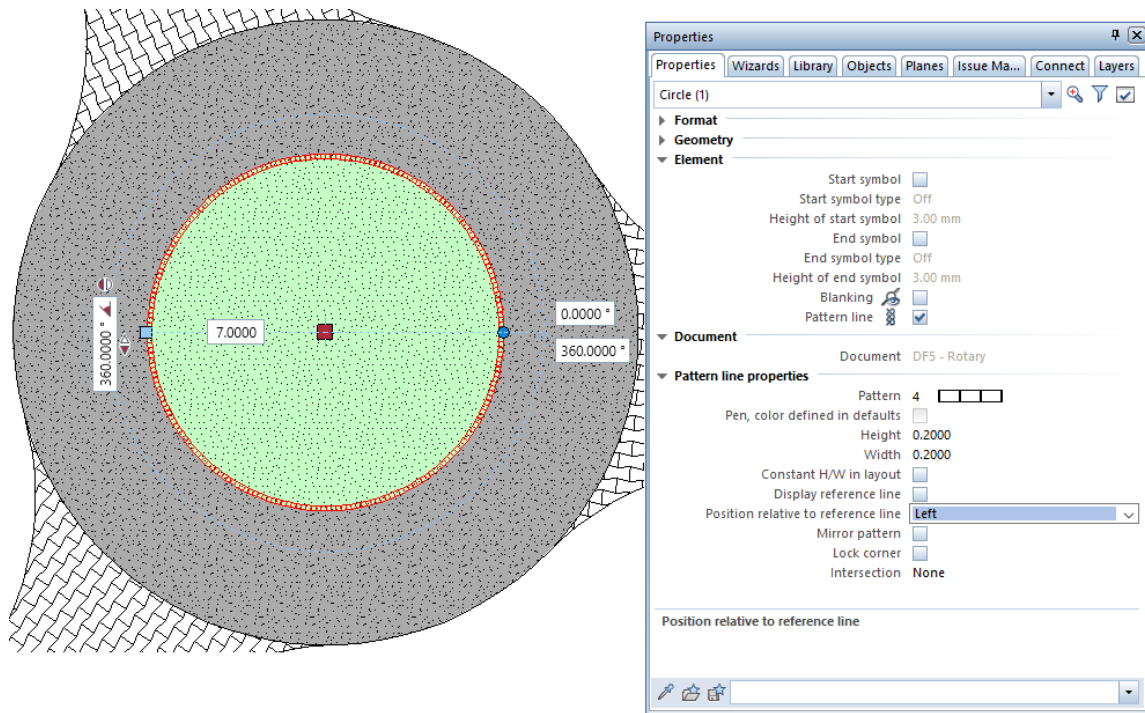


- 5 The pattern line properties are highlighted in yellow.
Open the list of patterns and select pattern 4.



- 6 Adjust the **Height** and **Width** of the pattern. Select **0.20 m** for both values.

7 Position relative to reference line: Left



8 Select ESC.

Exercise 5: title block

In this exercise, you will create and label a title block.

Index	Type of Change	Date / Name
Contents		
Precast Balcony, Type 12		
Project		
New Condominium With Underground Parking		
Client	Client Street, Munich	Date XX.XX.20XX
		Drawn by: Name
Architect	Architects Street, Munich	Checked by: Name
		Scale: 1:50/25
Engineer	Consulting Engineers Street, Munich	Layout number: XXX

You will use the tools in the **2D Objects** and **Quick Access** task areas.

Note: In addition to the method described in this exercise, Allplan provides other options for creating and labeling title blocks:

The task **Layout** contains the tool  **Label layout border** in the task area **Layout Editor** in which you can select from several plan headers. These title blocks, which are saved as label styles, transfer specific details and project attributes (date, project name, edited by, and so on) directly from the system to the title block. For an example of creating a title block as a label style, see the Architecture tutorial, Lesson 8: Layouts.

In addition, you can use the  **Legend, Title Block** tool (**Layout Editor** task – **Layout Editor** task area). These legends analyze the current layout attributes and project attributes. The program creates the legends as associative legends, thus automatically updating the attributes to reflect changes. After having placed a legend, you can explode it into its design entities.

Task 1: designing the title block

In the first part of this exercise, you will draw the layout of the title block with the tools in the **2D Objects** task area (**Design** task).

Tools	Objective																		
 Reference Scale	<table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>																		
 Rectangle																			
 Parallel to Element																			
 Auto-Delete Segment																			
 Delete																			
 Modify Format Properties																			

Each task on the **Actionbar** has its own set of basic settings known as  **Options** ( **Default Settings** drop-down list on the Quick Access Toolbar). The options contain defaults that affect the way the tools function.
So, you can configure the program to suit your own preferences.

Selecting the unit and reference scale

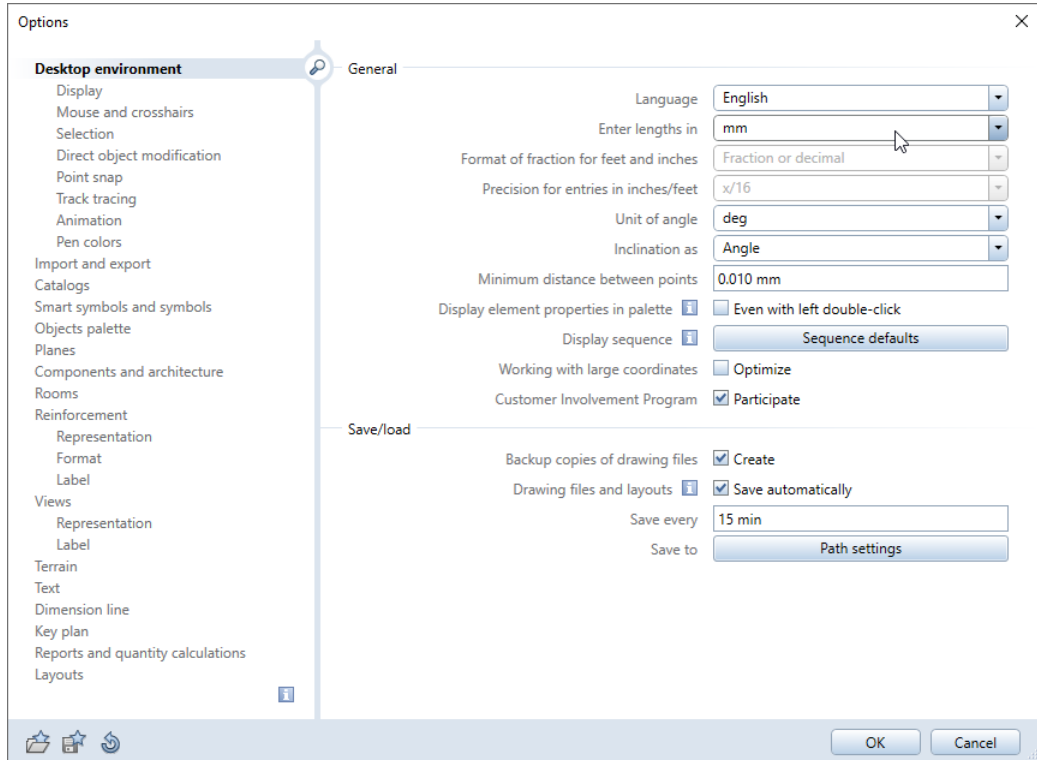
Start by specifying the unit of length for the values that you enter in this exercise. You will use **mm**.
Before you do this, select an empty drawing file so that the new unit of length and reference scale apply only to the drawing file in which you create the title block.

To select a new drawing file

- Click  **Open on a Project-Specific Basis** and open an empty drawing file. Name it **Title block** and close all the other drawing files.
-

To select units

- 1 Open the  **Default Settings** drop-down list on the Quick Access Toolbar and click  **Options**. You can see the **Options** dialog box. Select the **Desktop environment** page.
- 2 Go to the **Enter lengths in** box and click **mm**.



Tip: Alternatively, select the unit on the status bar: Click in the box next to length and select **mm**.

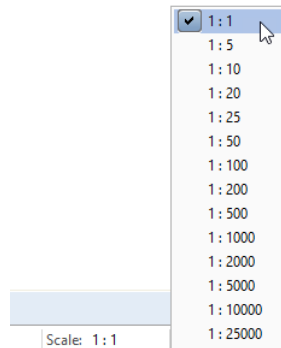
- 3 Click **OK** to confirm the settings.

Change the reference scale too. Until now you have worked at a scale of 1:100.

You will draw the title block at a scale of 1:1.

To select the reference scale

- 1 Go to the status bar, click in the box next to **Scale** and select 1:1.



Border of title block

Start by drawing the outer border of the title block.

To draw the outer border as a rectangle

- Drawing file **Title block** is current; all the other drawing files are closed.

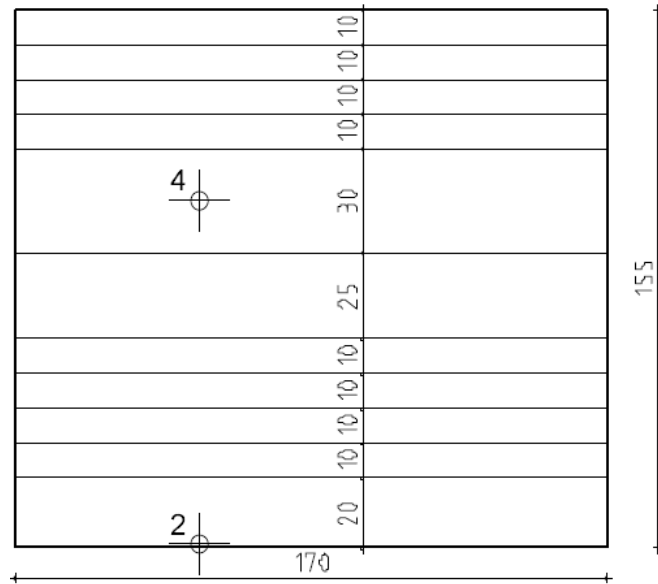
- 1 Click **Rectangle** (2D Objects task area).
- 2 The **Rectangle** context toolbar opens. Select **Based on diagonal line**.
- 3 Place the first point in the workspace.
- 4 *Diagonal point*
Enter a length of **170**. Select the TAB key and enter **155** to define the width. Select ENTER to confirm.
Allplan draws the rectangle.
- 5 Click **Zoom All** on the viewport toolbar.

Tip: Did you make an incorrect entry? Click **Undo** (Quick Access Toolbar). You can undo all steps back to the last save.

Create the inner lines as lines parallel to the border.

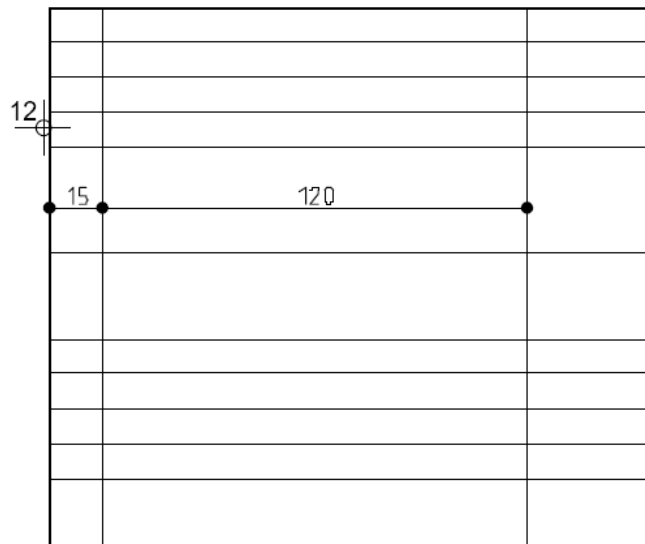
To draw the inner lines

- 1 Click  **Parallel to Element** (2D Objects task area).



- 2 *Click element*
Click the bottom line of the border.
- 3 *Through point or offset*
Enter **20** for the offset.
Select ENTER to confirm.
- 4 *Which side?*
Click inside the rectangle.
- 5 *Number: 1.*
- 6 *Through point or offset: 10.*
Number: 4.
- 7 *Through point or offset: 25.*
Number: 1.
- 8 *Through point or offset: 30.*
Number: 1.

- 9 *Through point or offset:* **10**.
Number: **3**.
- 10 Select ESC to close the tool.
- 11 To draw the parallel lines that are vertical, click  **Parallel to Element** again.
- 12 Click the outer line on the left and create two parallel lines – one at an offset of **15** and the other at an offset of **120**.

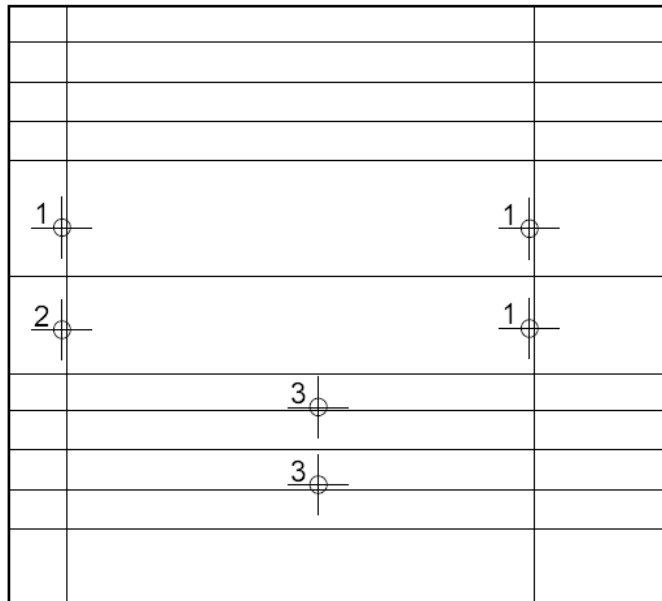


Deleting lines

Finally, delete the lines that you do not need.

To delete lines and segments of lines

- 1 Open the shortcut menu of a line that you want to delete and select the  **Auto-Delete Segment** tool. Click all line segments that you want to delete. Select ESC to close the tool.
- 2 Open the shortcut menu of the vertical line on the lower-left side and click  **Delete**. Select ESC to close the tool.
- 3 Click the  **Repeat** drop-down list on the Quick Access Toolbar and select the  **Auto-Delete Segment** tool. Delete the horizontal lines that you no longer need. Then select ESC to close the tool.

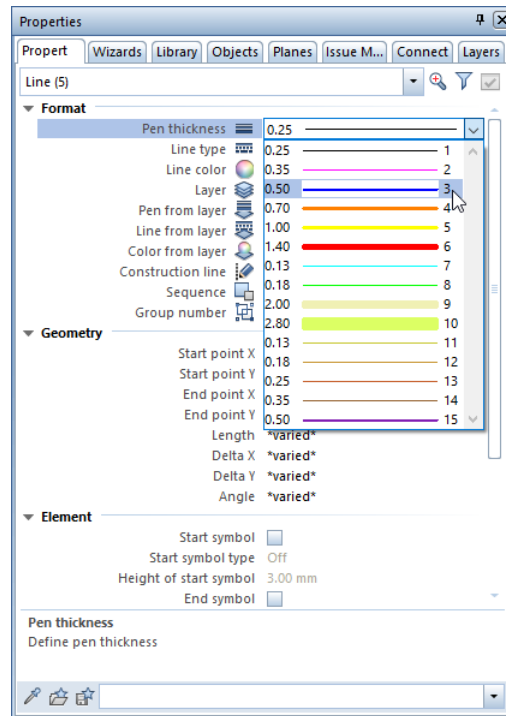


Modifying the pen thickness

The border of the title block needs to stand out.

To modify the pen thickness

- 1 To select the border of the title block, select and hold the SHIFT key and click a line of the border. This selects all lines with the same group number.
- 2 The **Properties** palette shows the format properties of the selected lines.
Click the **Pen thickness** drop-down list and select pen 3 0.50.



- 3 To confirm, click in the workspace.

Tip: The  **Modify Format Properties** tool (shortcut menu or **Change** task area) produces the same result.

Your drawing should now look like this:



Task 2: labeling the title block

In the following part of the exercise, you will label the title block with the tools in the **Quick Access** task area (**Design** task).

Tools

x-coordinate
(delta point)

y-coordinate
(delta point)

Copy

Edit Text

Explode
Paragraph

Change Text
Parameters

Objective

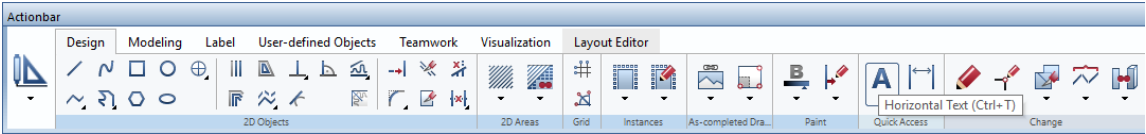
Index	Type of change	Date / Name
Contents	Precast Balcony, Type 12	
Project	New Condominium With Underground Parking	
Client	Client Street, Munich	Date XX.XX.20XX Drawn by: Name
Architect	Architects Street, Munich	Checked by: Name Scale: 1:50/2:5
Engineer	Consulting Engineers Street, Munich	Layout number XXX

Entering centered text

Start by entering the name of the layout contents in the title block.

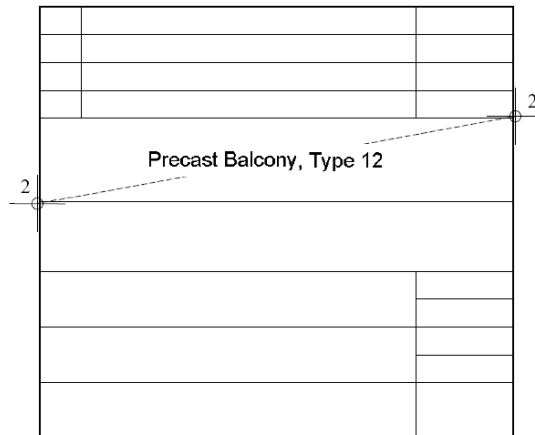
To enter centered text

- 1
- Click **Horizontal Text** in the **Quick Access** task area.

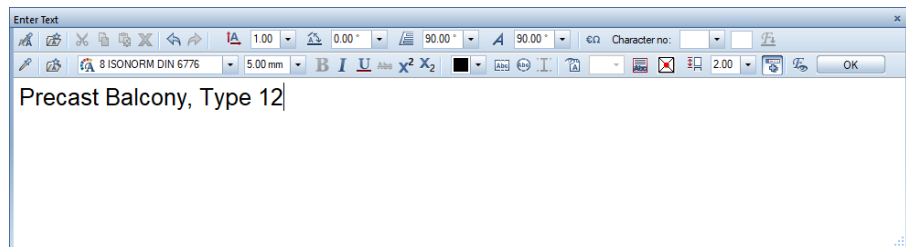


The anchor point of the text is exactly in the middle of the rectangle – in other words, the midpoint of an imaginary diagonal line.

- 2 Right-click in the workspace to open the shortcut menu (**Point Assistant**). Select  **Midpoint** and click the two diagonally opposite points of the rectangle.



- 3 Click  to expand the dialog box so that all the parameters are visible. Enter the text parameters:
 -  **Text's anchor point:** Centered
 - **Text height:** 5.00 mm
The **Text width** automatically adapts to the **Aspect ratio** (in this example: 1.00).
 - **Line spacing:** 2.00
 - **Font:** 8 ISONORM DIN 6776



- 4 Type the text: **Precast Balcony, Type 12**
- 5 Click **OK** or select CTRL+ENTER.
The  **Horizontal Text** tool is still open.

Paragraph text

Enter the name of the construction project in the next box. The lines are left-aligned. Allplan creates multiline text that you write in a single operation as paragraph text.

Use   to delete individual lines from a paragraph.

  resolves a paragraph into lines.

Line spacing:

The spacing between lines results from **line spacing multiplied by text height**. Select Enter to go to the next line.

Point snap and offset entry:

To exactly position text, use the point snap feature and enter an offset. This is very useful when you want to place a point relative to an existing point.

Tip: You can use  to save combinations of text parameters as favorites:

Use  to enter a name in the list and specify the parameters.

To enter paragraph text

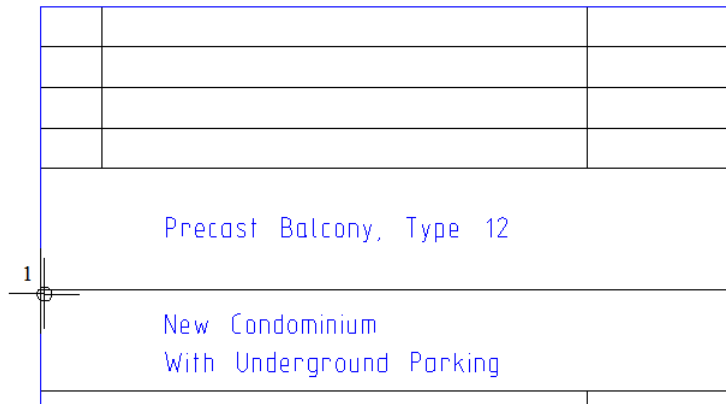
➡ The  **Horizontal Text** tool is still open.

Define the text's anchor point anchor by **snapping a point and entering the offset**.

- 1 Point to the point in the title block as shown in the illustration. Do not click the point!

By pointing to this point, you snap the reference point to which the values that you enter will apply. The point is marked with a cross.

The values that you enter for  and  are based on this reference point (the point snapped). To indicate this, the  **x-coordinate** and  **y-coordinate** boxes in the dialog line turn yellow.



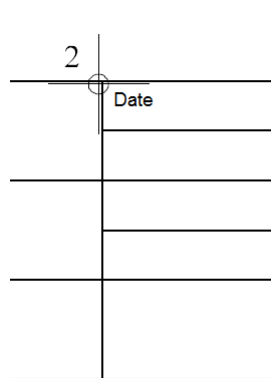
- 2 Enter **30** for **dx**, select the Tab key, enter **-5** for **dy**, and select ENTER to confirm.
- 3 Change the position of the text's anchor point to **upper left**.
- 4 Enter the following text:
New condominium [ENTER]
with underground parking
- 5 Select CTRL+ENTER or click **OK** to finish entering text.
- 6 Select ESC to close the tool.

Horizontal text

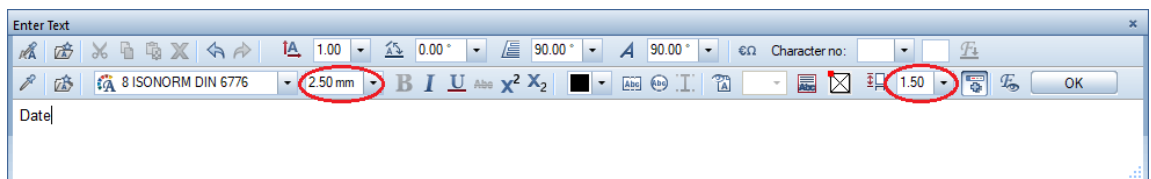
Enter a line of text on the right side of the title block and copy it to the boxes below.

To enter and copy text

- 1 Click  **Horizontal Text** in the **Quick Access** task area.
- 2 Use point snap and offset entry to define the text's anchor point:
 - A) Point to the upper-left corner of the box (see illustration)
 - b)  **dx = 2**; TAB key
 - c)  **dy = -2**
 - d) Select ENTER to place the point

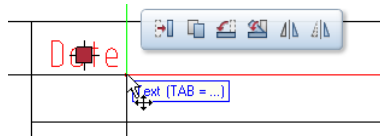


- 3 Switch the text's anchor point to  **upper left**. Change the **Text height** to **2.5 mm** and the **Line spacing** to **1.5**. Enter the following text: **Date**.

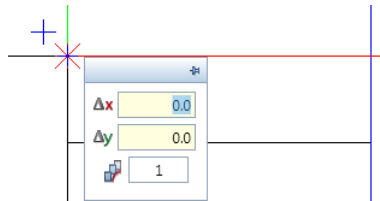


- 4 Click **OK** to confirm.
- 5 Select ESC to close the tool.

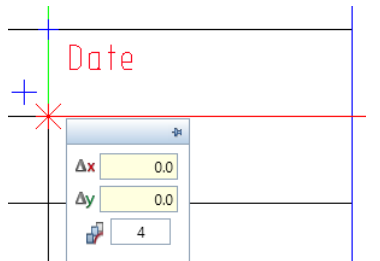
- 6 Click the **Date** text.
- 7 Point to a part of the text that is not marked with a handle.
- 8 Click  **Copy** on the context toolbar.



- 9 *From point or enter offset*
Click the upper-left corner of the box.



- 10 *To point or enter offset*
Enter **4** for the  **Number** of copies in the coordinate dialog box and click the upper-left corner of the box below.



- 11 Select ESC to close the tool.

Editing text

Modify the text by means of the  **Edit Text** tool.

Text modification:

You can edit text at any time. This makes it much easier for you to enter text, such as boilerplate, which must be updated.

To edit text

- 1 Right-click the **Date** text at the top and select  **Edit Text** on the shortcut menu.

The dialog box opens; you can make changes.

- 2 Select ENTER at the end of the **Date** text to go to the next line. Enter a placeholder for the date in the next line: **XX.XX.20XX**
- 3 Click **OK** to confirm.
- 4 Change the entries below as described in steps 2 and 3; use the placeholders in the following illustration.

Date XX.XX.20XX
Drawn by: Name
Checked by: Name
Scale: 1:50/25
Layout number: XXX

Changing text parameters

Change the appearance of the placeholder for the layout number. Assign different text parameters: Use a character height of 5 mm and a character width of 6 mm. Start by exploding the paragraph.

Defining text parameters:

You can choose from 20 Allplan fonts and from all TrueType fonts installed.

The **text height** and **text width** parameters are absolute values. This means that Allplan uses the selected values to print the text, regardless of the selected reference scale.

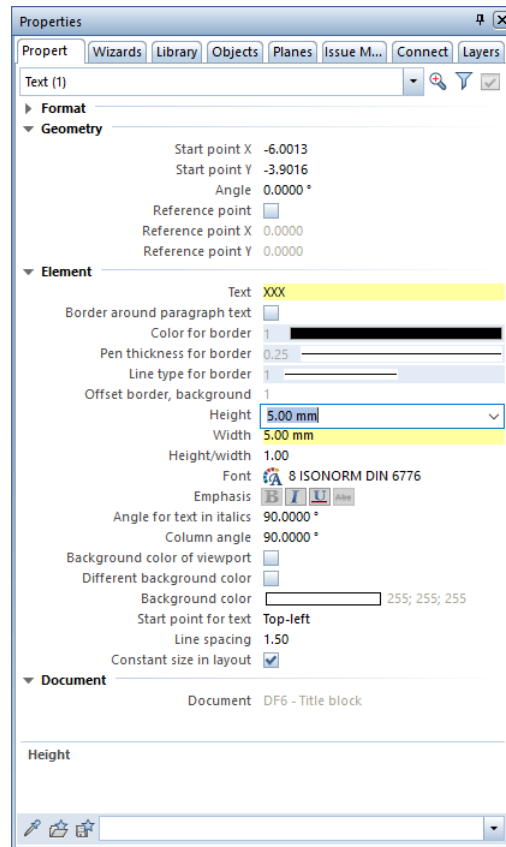
To change text parameters

- 1 Right-click the **Layout number...** paragraph. The shortcut menu opens. Select  **Explode Paragraph**.

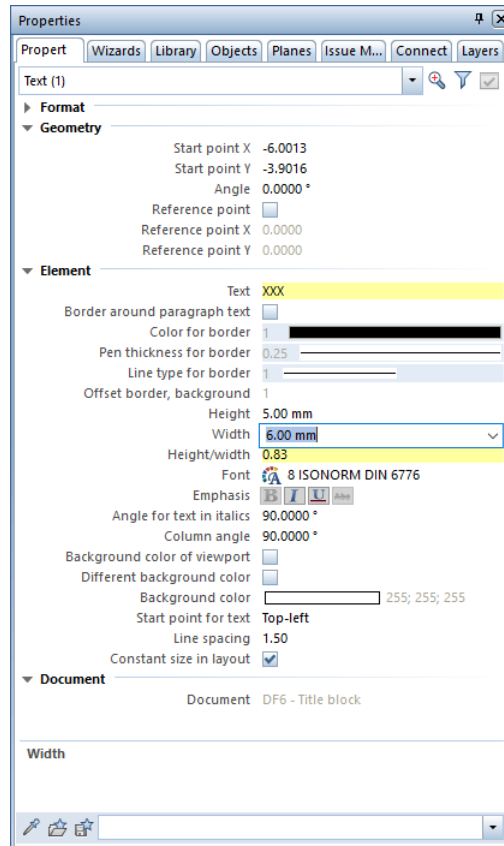
This explodes the paragraph; you can now modify each line separately.

- 2 Select the Esc key to close the  **Explode Paragraph** tool.
- 3 Click the placeholder XXX.

- 4 The **Properties** palette shows the parameters of the selected text.
Click in the box next to **Height** and select **5.00 mm**.



- 5 Click in the box next to **Width**, enter **6.00**, and select the Enter key to confirm.



- 6 Click in the workspace.

Note: You can also use the  **Change Text Parameters** tool (shortcut menu of the text that you want to modify).



7 Add the missing labels as shown in the illustration.

Index	Type of Change	Date / Name
Contents		
Precast Balcony, Type 12		
Project		
New Condominium With Underground Parking		
Client	Client Street, Munich	Date XX XX.20XX
		Drawn by: Name
Architect	Architects Street, Munich	Checked by: Name
		Scale: 1:50/25
Engineer	Consulting Engineers Street, Munich	Layout number XXX

Task 3: saving the title block as a symbol in the library and retrieving the title block

In the last part of this exercise, you will save the title block as a symbol in a folder of the library. Finally, you will retrieve the title block and place it in an empty drawing file.

Tools

Library palette



New group



Insert element



Insert symbol



Find

Symbols

Symbols are design entities that you can use whenever you need. Symbols automatically adapt to the scale of the drawing file. You can select (click the middle mouse button and then left mouse button) and modify a symbol as a single entity. After having modified a symbol, you can save it under another name.

Symbols are often used for drawings and other common components. In time, you will develop and use your own extensive libraries for title blocks, fixtures, connecting elements, and so on.

Symbols in the Library palette

Symbols and other library elements (smart symbols, SmartParts, PythonParts) can be stored and managed in any folder in the library. You can create your own folders in the following data paths:

Office: The folders in this path belong to the office standard; they are available to the entire office:

- Standalone computer: for all local projects
- Computer on the network: for all users and projects on the network; only the administrator can store and manage symbols.

Project: The folders in this path belong to a particular project; they are available in that project only.

Private: The folders in this path belong to the user that is currently signed in; they cannot be accessed by other users on the network.

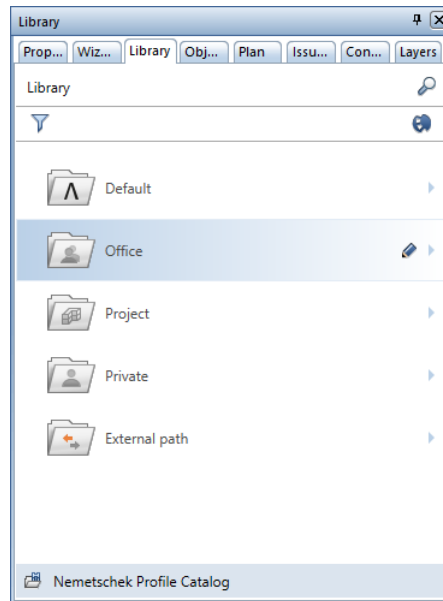
Each folder can contain numerous library elements. You can copy, delete, and rename symbols in the **Library** palette or in File Explorer.

Saving a symbol in the library

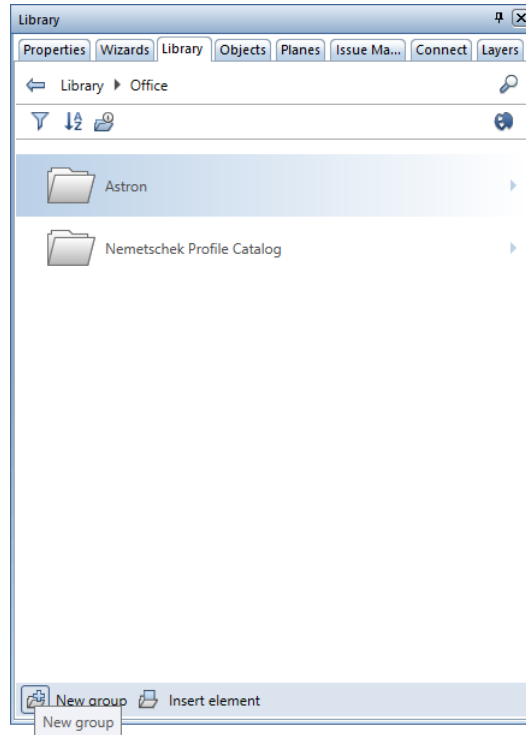
Add the title block as a symbol to a folder of the library.

To add a symbol to the library

- 1 Open the **Library** palette.
- 2 The title block is to be made available to the entire office. Therefore, open the **Office** folder.



- 3 Click  **New group** at the bottom of the **Library** palette.

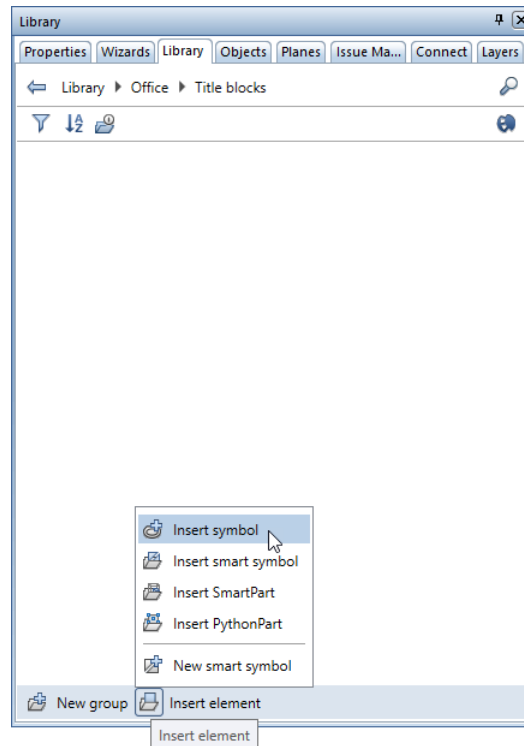


- 4 Enter **Symbols** for the name of the new group and select ENTER to confirm.



- 5 Open the new **Symbols** folder.
- 6 Click  **New group** at the bottom of the **Library** palette again.
- 7 Enter **Title blocks** for the name of the new group and select ENTER to confirm.
- 8 Open the new **Title blocks** group.

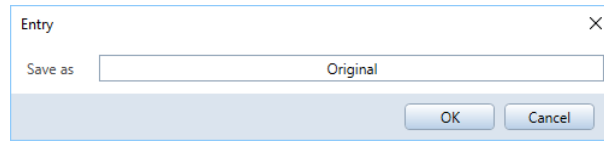
- 9 Click  **Insert element** and then  **Insert symbol** at the bottom of the **Library** palette.



- 10 *Select elements you want to save as a symbol file*
Open a selection rectangle around the title block. It appears in the selection color.
- 11 *Set the symbol's base point*
Click the lower-right corner. This is the point at which the symbol will be attached to the crosshairs when you retrieve it.
- 12 A dialog box opens. Choose **Dumb symbol without Snoop functionality** and click **OK** to confirm.

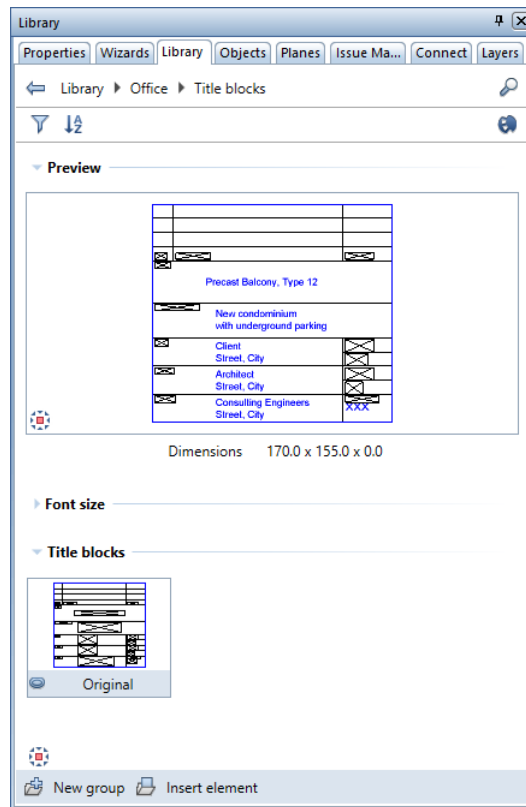
Tip: You can change the symbol's base point when you retrieve the symbol.

- 13 Enter **Original** for the name of the new symbol and click **OK** to confirm.



- 14 This saves the new **Original** symbol to the **Title blocks** folder.

Tip: You cannot see the symbol in the preview? Check that the **Wireframe** view type and **Plan** view are selected.



Retrieving a symbol from the library

Now retrieve the symbol and place it in an empty drawing file.

In practice, the new drawing file might be a drawing file for another construction project. All you would need to do then is change the project-specific information and save the title block again as a symbol – this time in the **Project** folder.

In addition to the symbols that you create and save yourself, you can use this approach to retrieve symbols from the Allplan library.

To retrieve a symbol from the library

- 1 Click  **Open on a Project-Specific Basis** (Quick Access Toolbar), open an empty drawing file, and close the drawing file with the **Title block**.

Click the scale on the status bar and select **1:1**.

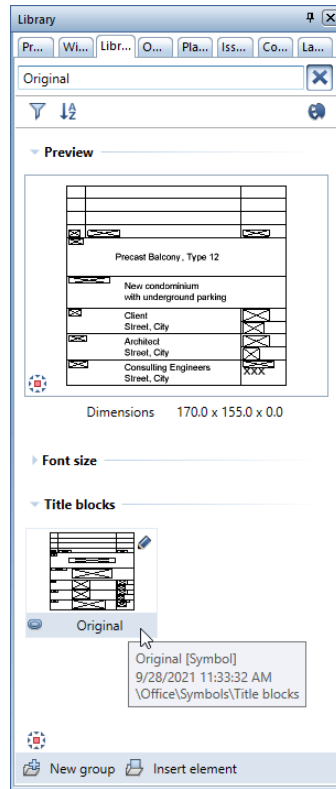
- 2 As you have just saved the **Original** symbol, this folder is still open in the **Library** palette.

If it isn't, click  to move through the **Library** palette or click  to find the required file.

- 3 If you want to  **Find** the file, enter its name in the line at the top of the **Library** palette. In this example, enter **Original**; select ENTER to confirm.

The **Library** palette displays the required file.

Point to the image of the file in the lower area to display a ToolTip with information about the file type (symbol in this example), date saved, and folder (**\Office\Symbols\Title blocks** in this example).



- 4 Double-click the **Original** symbol or drag it into the workspace.
The symbol's base point is attached to the crosshairs.
- 5 Click in the workspace to place the symbol.
- 6 If the title block appears too small, click  **Zoom All** on the viewport toolbar.

Tip: You can use the **Input options** to change the symbol's base point and define a cursor snap angle. Use the tools on the shortcut menu to exactly define the symbol's drop-in point.

When placing the title block, you can use the input options to change the size of the title block by resizing it in the x-direction and y-direction. Define the fixed point and enter the resizing factors.

Additional tools for placing elements

The Input Options provide a number of tools that help you place elements:



Use this to position the drop-in point.

Df Pnt

You can use the default drop-in point. This is the point you chose when you defined the smart symbol or symbol.

Ang=

Use this to switch between a fixed ('0') and freely definable ('?') drop-in angle. When you have selected ?, you can enter the angle after placing the element, or you can click a line and match its angle.

Note: Using the + and - keys, you can rotate the element displayed attached to the crosshairs by a cursor snap angle in a clockwise or counter-clockwise direction. You can specify this angle using **Cursor snap angle** (see below).

Num=

Use this to define the number of the elements to be placed. **Num=1** places one element; **Num=?** places any number of elements.

Resize

You can use this to resize the element in the x-direction, y-direction and z-direction by first defining a fixed point and then specifying the resizing factors.

Tip: When you activate **Ang=?** before you click **Scale**, you can define an angle and then resize the (smart) symbol you are about to place.

Snoop

You can use this to rotate the element so that it is parallel to existing architectural lines (e.g. walls). This is done automatically when you position the cursor over the architectural line. In the case of (smart) symbols defined as **Intelligent symbol/smart symbol with snoop functionality**, the distance to the architectural line is based on the setting made when you defined the relevant (smart) symbol. For more information, see Intelligent (smart) symbols. 'Normal' (smart) symbols or elements from the Clipboard are placed using an offset value of 0.

Drop

You can use this to define a custom drop-in point by clicking it in the dialog box.

Attribute

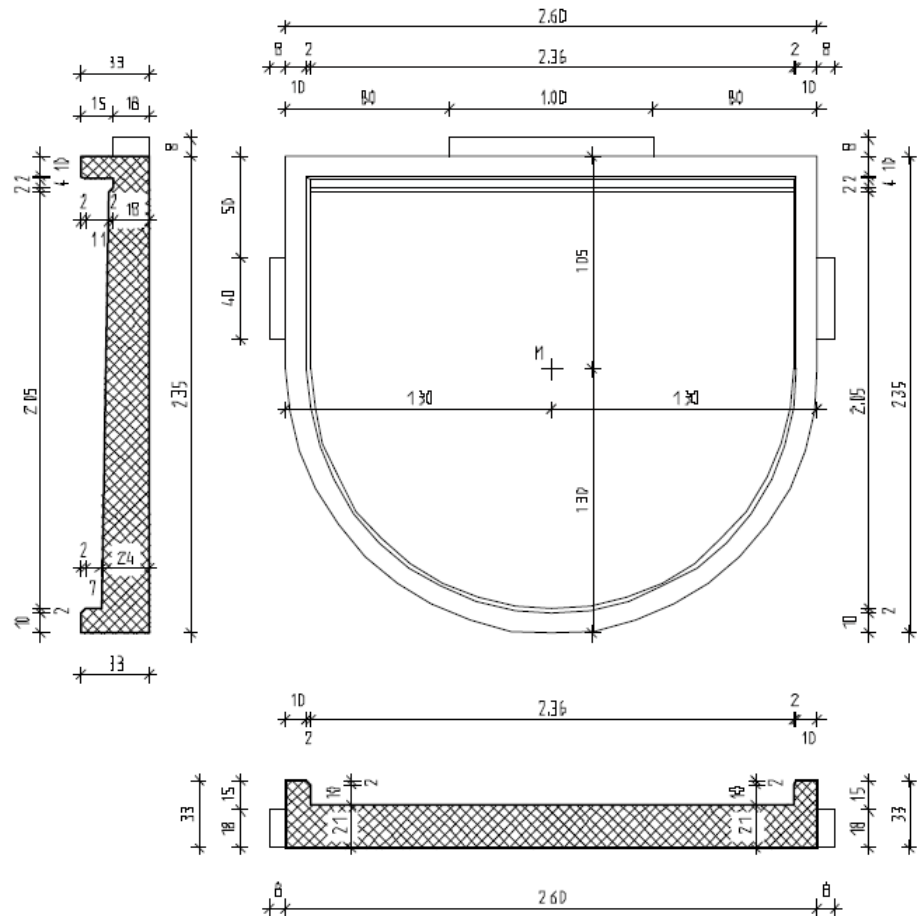
You can use this to assign attributes to (smart) symbols.

Cursor snap angle

You can use this to set the angle by which the element is rotated when you click the + or - keys.

Exercise 6: precast balcony

In this exercise, you will create a general arrangement drawing for a precast balcony with dimensions and hatching. Finally, you will print out the precast element.



You will design the precast element with the tools in the **2D Objects** and **Quick Access** task areas. To print the result, you will use the  **Print Preview** tool (Quick Access Toolbar).

Task 1: designing the precast balcony

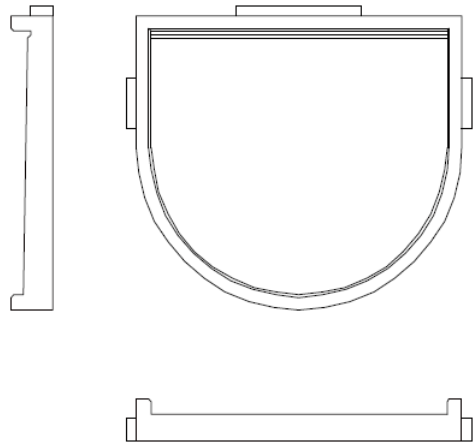
In the first part of this exercise, you will draw the floor plan and two sections for a precast balcony.

You will use the tools in the **2D Objects** task area (**Design** task) and **Change** task area.

Tools:

-  Rectangle
-  Fillet
-  Parallel to Element
-  Auto-Delete Segment
-  Line
-  x-coordinate (delta point)
-  y-coordinate (delta point)
-  Copy and Mirror
-  Midpoint
-  Stretch Entities
-  Fold Line

Objective:



Default settings

Start by defining the default settings.

To select a new drawing file and define the options

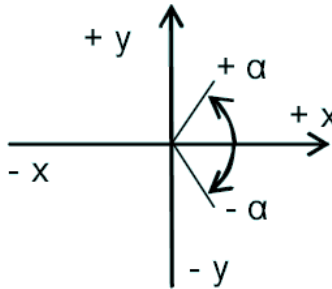
- 1 Click  **Open on a Project-Specific Basis** and open an empty drawing file. Enter **Precast balcony** for its name and close all the other drawing files.
- 2 Click the **Length** on the status bar and select **m**.
- 3 Click the **Scale** on the status bar and select **1:25**.

Tip: If you want to change the scale by means of the  **Reference Scale** tool on the **View** menu, you must use the Alt key to temporarily show the hidden menu bar. When you define a scale by using the  **Reference Scale** tool and the  icon, the status bar also presents this scale for selection.

Note: You can also show the menu bar all the time. To do this, open the drop-down list on the Quick Access Toolbar (title bar) and click **Show menu bar**.

You will start by drawing the outline.

Pay attention to the coordinate system and the angles' direction of rotation!



To display the required section, use the tools on the viewport toolbar:



Zoom All



Zoom Section

You can also use the mouse to define the section that is visible on the screen:

- Open the  **Options - Desktop environment - Mouse and crosshairs: Keeping the right mouse button held down - Zoom function** is selected by default. As a result, you can zoom with the right mouse button.
- You can use the wheel to dynamically enlarge or reduce the section on the screen.
- Double-click the middle mouse button to display the screen contents in their entirety.
- You can pan by selecting and holding the middle mouse button; a hand appears at the crosshairs.
If you work with a two-button mouse (for example, Mac computers), you can select the **Ctrl+Shift+left mouse button held down - Pan function (for two-button mouse)** option ( **Options - Desktop environment - Mouse and crosshairs**).

Drawing the outline

To draw the outline in plan

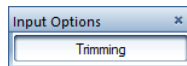
- 1 Click  **Rectangle** (2D Objects task area).
- 2 The **Rectangle** context toolbar opens. Select  **Based on diagonal line**.

Note:  Make sure that **Create rectangle as a polyline** is not selected in the input options because you will edit some lines of the rectangle later.

- 3 Click in the workspace to define the starting point of the rectangle.

When you move the mouse, a preview of the diagonally opposite point is attached to the crosshairs ("rubber-band").

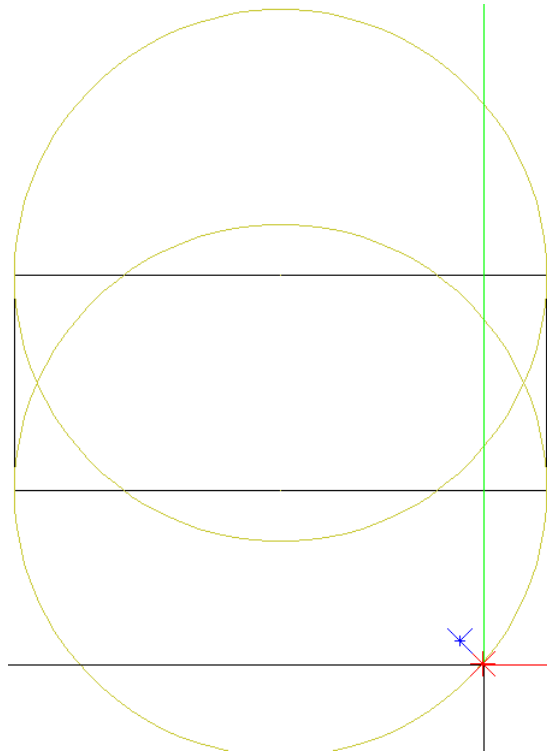
- 4 *Diagonal point*
Enter a length of  **2.60**. Select the TAB key and enter **1.05** for the  width.
Select ENTER to confirm.
- 5 Click  **Zoom All** on the viewport toolbar.
- 6 Use ESC to close the  **Rectangle** tool.
- 7 The next step is to fillet the vertical lines. To do this, open the shortcut menu of the left or right side of the rectangle and select the  **Fillet** tool.
- 8 You can see **Trimming** in the input options. When this button is selected (default setting), Allplan automatically shortens or lengthens the initial lines as far as the fillet. If this button isn't selected, click it to select trimming.



- 9 Click the other vertical side of the rectangle and confirm the fillet radius proposed.

Tip: Did you make an incorrect entry?
Click  **Undo** (Quick Access Toolbar).

You can choose from four options:
Select the semicircle at the bottom.



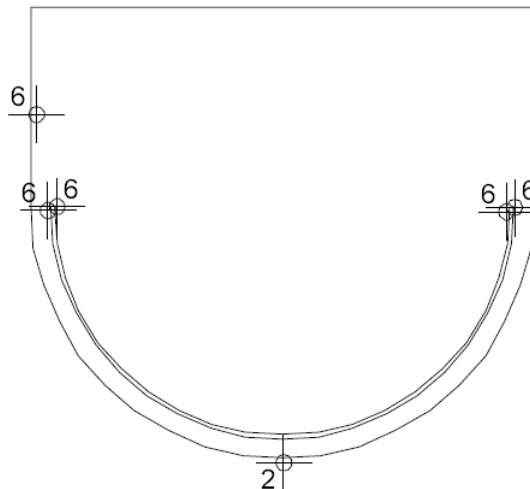
- 10 Click  **Zoom All** again.
 - 11 Select ESC to close the  **Fillet** tool.
 - 12 Delete the bottom line of the rectangle.
Right-click this line and select  **Delete** on the shortcut menu.
(Alternatively, you can select  in the **Edit** task area and then click the line).
 - 13 Select ESC to close the tool.
-

Creating inner parallel lines

Draw the inner lines of the precast balcony.

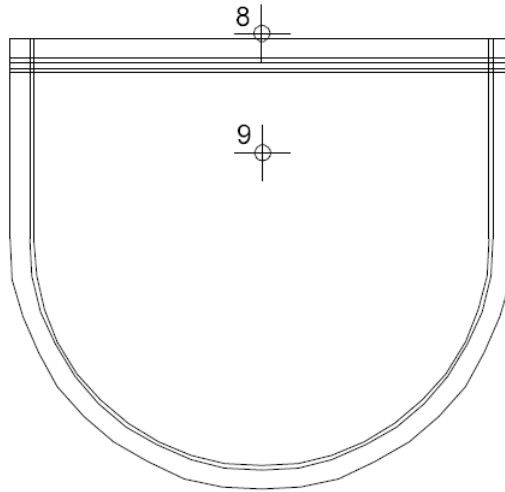
To create inner parallel lines

- 1 Click  **Parallel to Element** (2D Objects task area).
- 2 Click the semicircle you have just created.
- 3 Enter **0.1** to define the offset.
Which side?
Click within the outline to copy the circle inward.
Select ENTER to confirm the number (1).
- 4 For the next offset, enter **0.02**.
Which side? Click the inside again.
Confirm the number (1).
- 5 Click  again to create the lines that are parallel to the lateral lines.
- 6 Click the left line and then the endpoints of the semicircle one after the other (see illustration).



- 7 Click  again to create the lines that are parallel to the line at the top.

- 8 Click the line at the top and enter **0.1** for the offset.



- 9 Click below the line to specify the side where you want to create the parallel lines. Confirm the number (**1**).
- 10 Create three more parallel lines. Enter **0.02** for the first offset, **0.04** for the second offset, and **0.02** for the third offset. Confirm the number (**1**) each time.
- 11 Select ESC to close the tool.

Deleting redundant line segments and drawing fillets

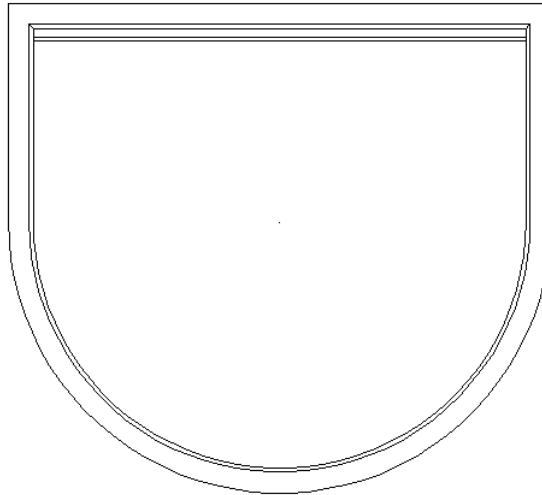
Delete the redundant line segments in the corners and complete your design by adding fillets.

Tip: If you inadvertently deleted elements, you can restore them by immediately right-clicking in the workspace twice (undoing the last action) or by using  **Undo**. Here, you can go back (undo) as many steps as you want (as far back as the last save).

To delete redundant line segments and to add fillets

- 1 Right-click one of the lines that you want to delete and select  **Auto-Delete Segment** on the shortcut menu.
- 2 Click all line segments that you want to delete.
Use  **Zoom Section** (on the viewport toolbar) to select a suitable view. This toolbar appears when you point to the lower border of the viewport.
- 3 Click  **Line** in the **2D Objects** task area.
- 4 Draw the two fillets as shown in the illustration.

Your drawing should now look like this:



-
- 5 Select ESC to close the tool.

Drawing Isokörbe

Add Isokörbe (special type of reinforcement cage) to the top and the sides.

To draw Isokörbe in plan

- 1 Click  **Line** in the **2D Objects** task area.
- 2 Click  **Polyline** on the **Line** context toolbar.
- 3 Point to the upper-left corner.

By pointing to this point, you snap the reference point to which the values that you enter will apply. The point is marked with a cross.

The values that you enter for  and  are based on this reference point (the point snapped). To indicate this, the  **x-coordinate** and  **y-coordinate** boxes in the dialog line turn yellow.

- 4 Select the Tab key to go to the  **y-coordinate** box. For **dy**, enter **-0.50**. Select ENTER to confirm.

This defines the starting point of the line.

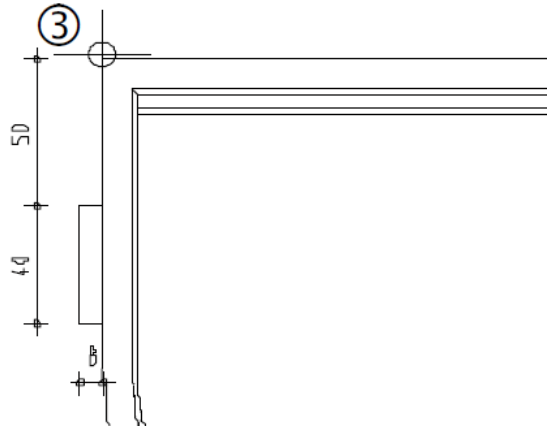
- 5 Enter the following values in the dialog line:

 **dx** = -0.08

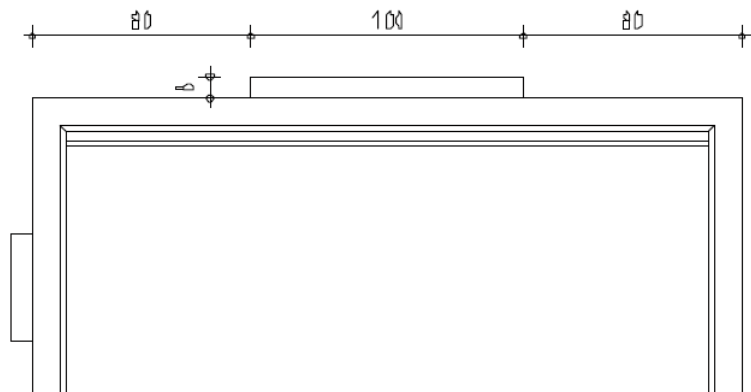
 **dy** = -0.40

 **dx** = 0.08

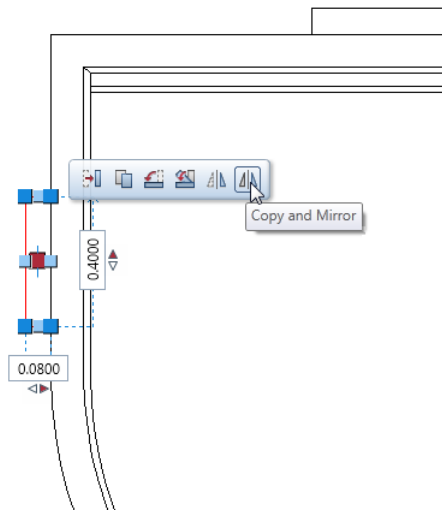
- 6 Select ESC to close the tool.



- 7 Draw the Isokorb at the top yourself; use the procedure described. Careful with the direction and the sign!



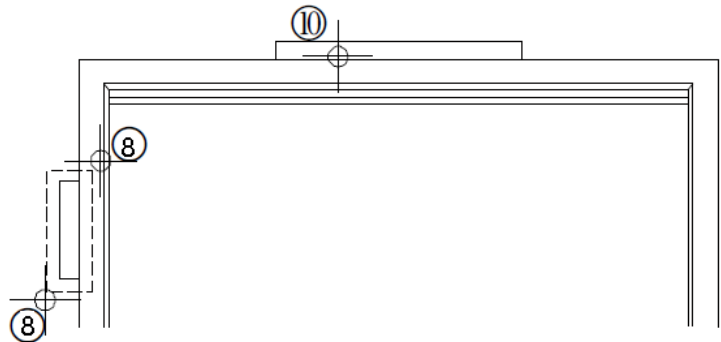
- 8 Create the Isokorb on the right side by mirroring the one on the left side. Select the Isokorb on the left side by enclosing it in a selection rectangle that you open from left to right.
- 9 Click  **Copy and Mirror**.



- 10 To define the first point of the mirror axis, right-click the line at the top and select  **Midpoint** on the shortcut menu.

Make sure that you click neither the midpoint of the line nor any other existing point.

This defines the first point of the mirror axis.



- 11 To obtain a mirror axis that is exactly vertical, select the TAB key to switch to the  **y-coordinate** box in the dialog line.
Enter any **dy** value (not equal to 0).
This creates the Isokorb on the right side, completing the design.
- 12 Select ESC to close the tool.

Outline of longitudinal section

Draw the outline of the longitudinal section and place it below the floor plan.

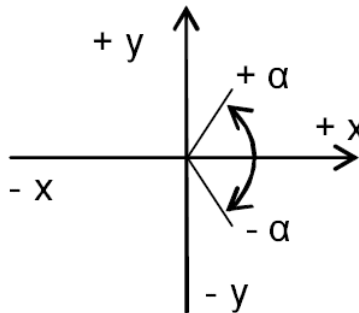
To draw the outline of the longitudinal section

- 1 Select  **Line** in the **2D Objects** task area.
- 2 Click  **Polyline** on the **Line** context toolbar.
- 3 Switch to the **Properties** palette, open the  **Pen thickness** drop-down list, and select **0.50 mm**.

Note: While you are drawing, you can change the pen thickness in the **Properties** palette at any time.

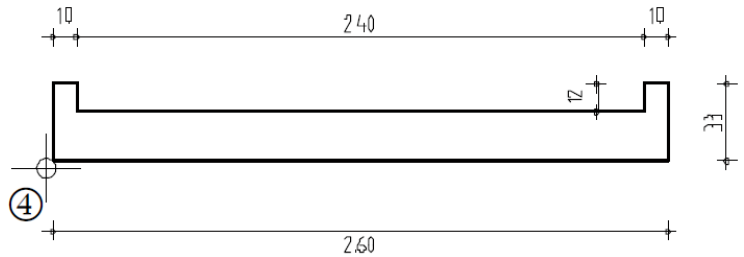
- 4 Place the first point below the plan.
Create the lines by entering the following values one after the other in the dialog line:

Tip: If the value entered is not correct, simply click  on the **Line** context toolbar. This deletes the last entry; you can resume your work at the endpoint of the previous line.



Keep the coordinate system in mind!

 Δx = 2.60,  Δy = 0.33,  Δx = -0.10,  Δy = -0.12,
 Δx = -2.40,  Δy = 0.12,  Δx = -0.10,  Δy = -0.33,



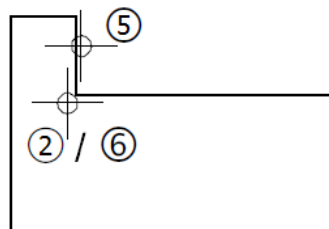
5 Select ESC to close the tool.

Modifying the outline

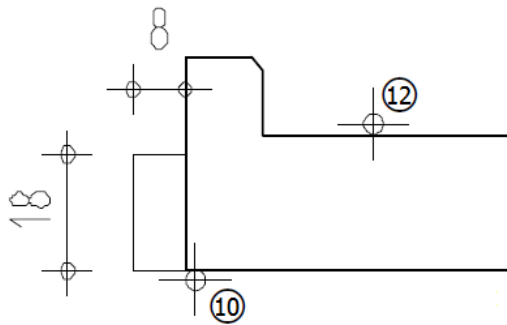
The next step involves modifying the outline.

To modify the outline of the longitudinal section

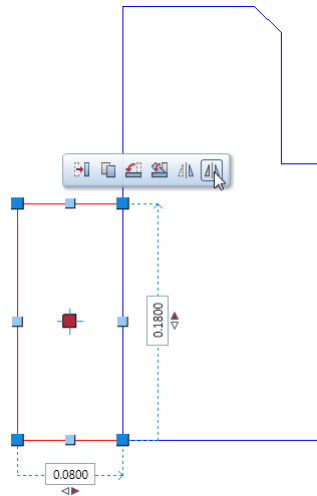
- 1 Select the  **Stretch Entities** tool in the **Change** task area.
- 2 *Select the points to modify*
Click the lower-right point of the left upstand (see illustration).
- 3 *Place a point (from point) or enter dx:*
Enter **0.02** for **dx** in the dialog line; confirm **dy** and **dz (0)**. Select ESC to finish.
- 4 Click  **Fold Line** in the **Change** task area.
- 5 Click the right line of the upstand.
- 6 *Endpoint:*
Point to the lower-right point; this is the point you have just modified. Allplan snaps this point, marking it with a cross.
- 7 Select the Tab key to go to  **y-coordinate**, enter **0.10** for **dy**, and select ENTER to confirm.



- 8 Select ESC to close the tool.
- 9 You can now modify the right upstand yourself. When finished, select ESC to close the  tool.
- 10 Switch to the **Properties** palette and select pen thickness **0.25**. Then draw the Isokorb on the left side. Use  **Line** (2D Objects task area). The starting point is the lower-left corner.
 $\Delta x = -0.08$
 $\Delta y = 0.18$
 $\Delta x = 0.08$

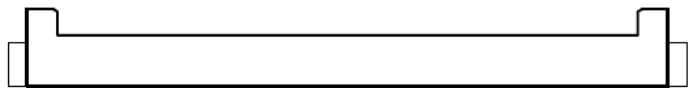


- 11 Select ESC to close the  **Line** tool.
- 12 Create the Isokorb on the right side of the longitudinal section by using  **Copy and Mirror**:
 - Enclose the Isokorb in a selection rectangle that you open from left to right.



- Point to a line and click  **Copy and Mirror** on the context toolbar.
- Right-click the line at the top and select  **Midpoint** on the shortcut menu.
- Select the Tab key to switch to the  **y-coordinate** in the dialog line and enter any value for **dy**.

Your design should now look like this:



13 Select ESC to close direct object modification.

Drawing the cross-section

Finally, you will draw the entire cross-section in a single operation.

To draw the cross-section

- 1 Select pen thickness **0.50 mm** in the **Properties** palette and click  **Line** (2D Objects task area).
- 2 Click  **Polyline** on the **Line** context toolbar.

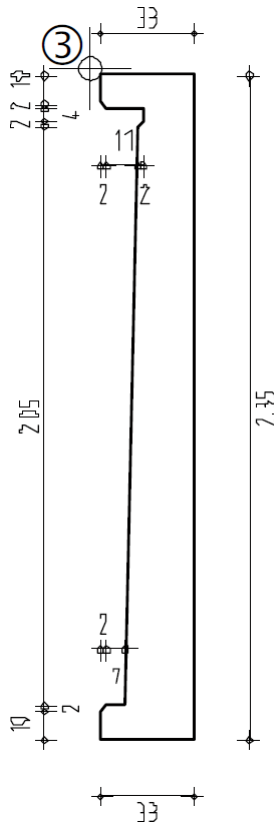
Tip: *Skip a coordinate.*

Select the TAB key to go to the next box.

Enter relative coordinates.

Enter values for Δx , Δy , and Δz in the dialog line (use the Tab key to switch between the boxes) until you have reached the drop-in point. Select the Enter key to place the point.

- 3 Place the starting point in the upper-left area so that the cross-section is to the left of the floor plan.
- 4 Enter the following values one after the other in the dialog line. The illustration on the left side helps you do so.



Δx dx = 0.33 ENTER

Δy dy = -2.35 ENTER

Δx dx = -0.33 ENTER

Δy dy = 0.10 ENTER

Δx dx = 0.02 TAB

Δy dy = 0.02 ENTER

Δx dx = 0.07 ENTER

(select the TAB key to switch to Δx)

Δx dx = 0.04 TAB

Δy dy = 2.05 ENTER

Δx dx = 0.02 TAB

Δy dy = 0.02 ENTER

(select the TAB key to switch to Δy)

Δy dy = 0.04 ENTER

Δx dx = -0.13 ENTER

(select the TAB key to switch to Δx)

Δx dx = -0.02 TAB

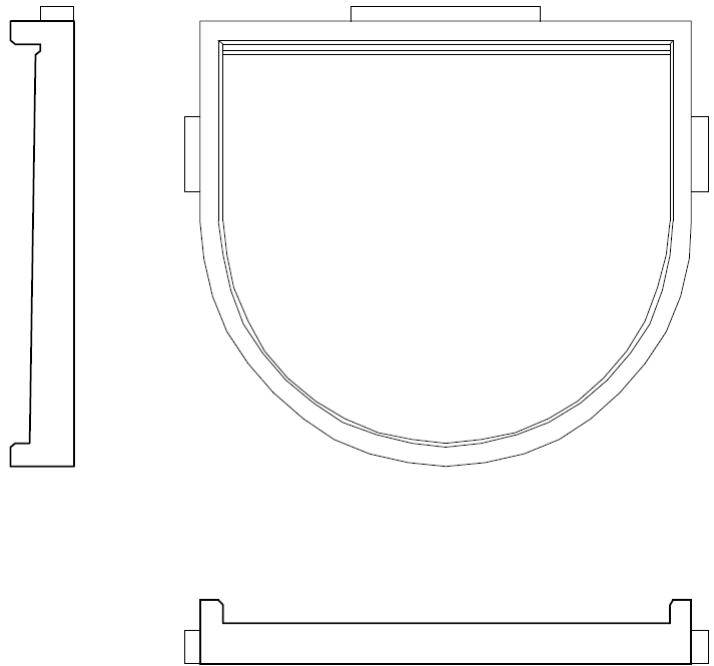
Δy dy = 0.02 ENTER

(select the TAB key to switch to Δy)

Δy dy = 0.10

- 5 Select ESC to close the tool.
- 6 Select pen thickness **0.25 mm** (**Properties** palette) and draw the Isokorb (8/18 cm) at the top.

Your drawing should now look like this:



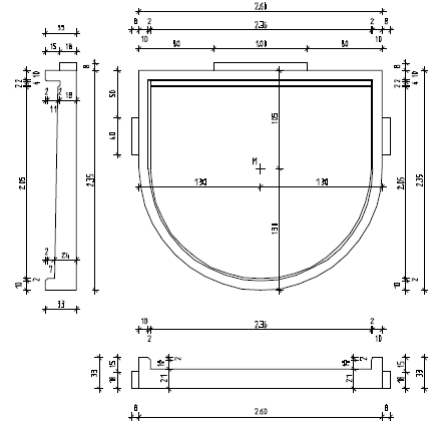
Task 2: dimensioning the precast balcony

In this section, you will dimension the precast balcony with tools in the **Quick Access** task area.

Tools:

-  Horizontal Dimension Line
-  Vertical Dimension Line
-  Auto Dimensioning
-  Add Dimension Line Point
-  Modify Dimension Line

Objective:



When you dimension a design, the first step is to define whether dimensioning is to be associative or not. After this, you define the dimension-line parameters. Dimensioning then involves three steps:

- Define the type of dimension line (vertical, horizontal, angle, or direct)
- Specify the location for the dimension line
- Click the points that you want to dimension

Tip: Open the  **Options** on the **Dimension line** page to define a tolerance for tilted (dimension) text, specify the decimal separator, and enter values for blanking.

You can modify dimension lines at any time. You can add and delete dimension-line points, move dimension lines, and modify the dimension-line parameters.

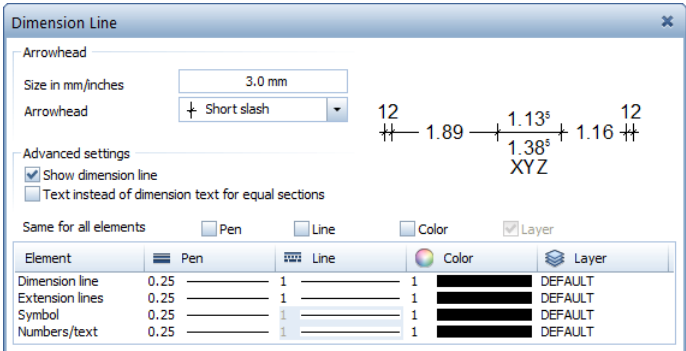
Defining dimension-line parameters

Start by making settings for the dimension-line parameters.

The most important parameters are the unit, the position of the dimension text, and the dimension-text height and width. Dimension lines are always linked with the design (dimension lines are associative; the points you click are the reference points). Dimension lines automatically adapt to changes you make to the design.

To define dimension-line parameters

- 1 Click  **Dimension Line** (**Quick Access** task area).
- Note:** The **Dimension Line** context toolbar includes the **Associative** option. Associative dimensioning means that the dimension lines and text automatically adapt to changes in the size or position of the component.
- 2 Clear the **Associative** check box in this this example.
- 3 Click  **Properties**.
- 4 Check the settings in the top (general) part of the dialog box, select an **Arrowhead** (**slash**), enter its **Size** (**3.00**), and define the **Format properties** for the components of the dimension line.

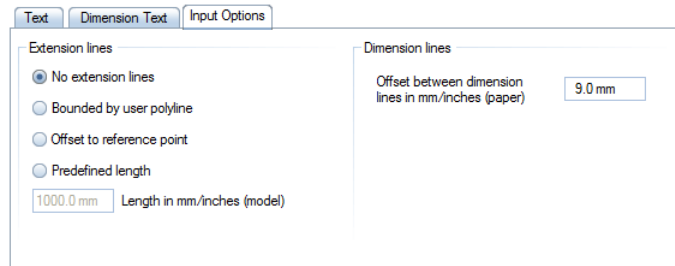


Allplan creates the dimension lines with the pen, line, color, and layer that you select here, regardless of the settings in the **Properties** palette.

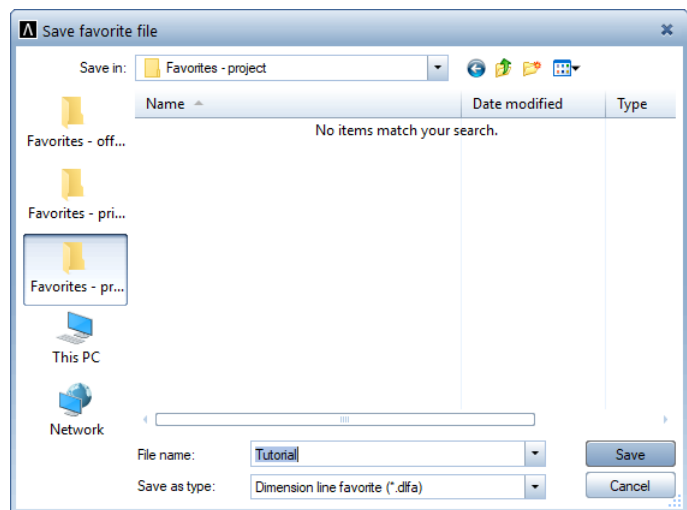
- 5 Check the settings on the **Text** tab in the bottom part of the dialog box and define the text parameters as shown in the illustration.
- 6 Select a font and define the dimension-text height and width by entering the following values:
 - **Dimension-text height: 2.50 mm**
 - **Aspect: 1.25** (this results in a text width of 2.0)
- 7 To define the **Position** of the dimension text, click the upper box in the middle.

- 8 Open the **Dimension Text** tab and check the following parameters:
 - **Dimension-text unit: m, cm**
 - **Round-off value in mm: 5**
 - **Number of decimal places: 3**
 - **Number of trailing zeros: 2**
 - **Exponent format option: selected**

- 9 Open the **Input Options** tab and select the **No extension lines** option.
Enter **9** for the **Offset between dimension lines in mm, inches (paper)**.



- 10 Click  on the lower left and save the parameters as a favorite file. For example, enter "Tutorial" for the name of the file.



- 11 Click **Save** to confirm.
12 Click **OK** to confirm the **Dimension Line** dialog box.

Creating horizontal dimension lines

The longitudinal section gets horizontal dimension lines.

To create horizontal dimension lines

➡ The  **Dimension Line** tool is still open.

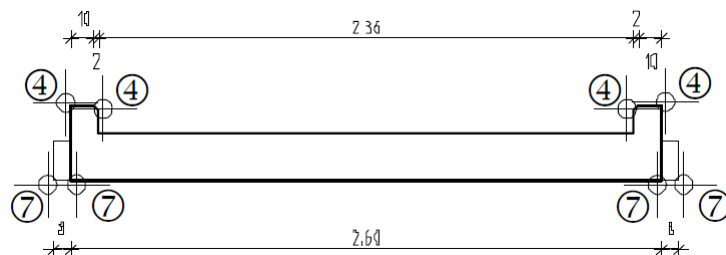
- 1 Select the section on the screen so that there is enough space at the top for the dimension line.
- 2 Click  **Horizontal** on the context toolbar.
- 3 *Through point or click dimension line*
Define the position of the dimension line by clicking above the longitudinal section. This is the point through which the dimension line will pass.
- 4 Click the six points that you want to dimension.

A preview of the dimension line appears immediately. This preview automatically changes with each new point that you add to the dimension line. You can click the points to be dimensioned in any sequence.

- 5 To finish entering points, select ESC.

 **Horizontal** is still selected so that you can create the next dimension line.

- 6 *Through point or click dimension line*
Click below the longitudinal section to define the point through which the dimension line is to pass.
- 7 Click the points to be dimensioned. Then select ESC to finish creating horizontal dimension lines.



Creating vertical dimension lines

You will continue with vertical dimension lines.

To create vertical dimension lines

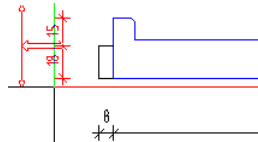
➡ The  **Dimension Line** tool is still open.

- 1 Click  **Vertical** on the context toolbar.
- 2 Click to the left of the longitudinal section to define the point through which the dimension line is to pass.
- 3 Click the corners of the Isokorb and the upstand; select ESC to close the tool.

 **Vertical** is still selected so that you can create the next dimension line.

- 4 *Through point or click dimension line*
Point to the left of the dimension line because you want to create the next dimension line to the left of the first one.

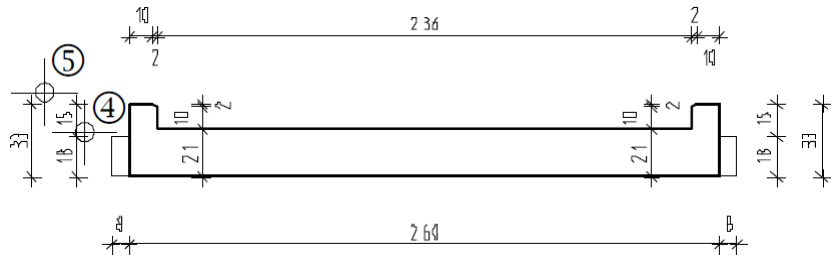
Allplan snaps this dimension line, displaying it in the selection preview color. A symbol indicates the position of the next dimension line.



- 5 If the symbol is on the correct side, click in the workspace to confirm.
- 6 Click the points to be dimensioned.

Tip: You can specify the distance between dimension lines in the  **Properties**. You can also move dimension lines later by means of the  **Move Dimension Line** tool.

- 7 Create the missing vertical dimension lines for the section yourself.

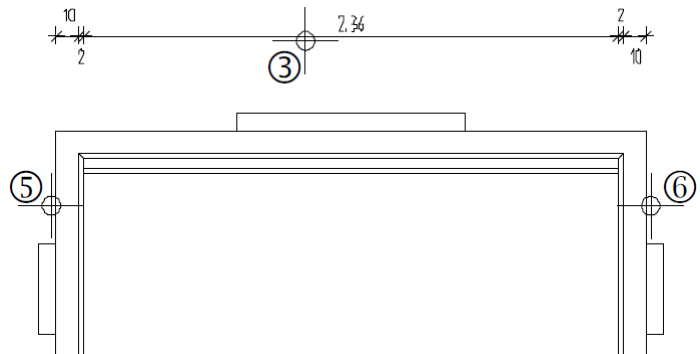


Creating dimension lines automatically

To dimension a part of the floor plan, you will use automatic dimensioning. All you need to do is draw a line through the components. Allplan will automatically dimension the points where the line intersects the components.

To apply automatic dimensioning

- 1 Click  **Auto Dimensioning (Quick Access task area)**.
- 2 Click  **Match parameters from dimension line** and click an existing dimension line.
- 3 Click above the floor plan to define the point through which the dimension line is to pass.
- 4 *Place direction point 1 or enter a direction angle or line.* Confirm the value **0.00**.
- 5 *Point 1 for clipping path.* Define the first point by clicking above the Isokorb to the left of the outer edge of the precast element.
- 6 *Next point for clipping path.* Click the equivalent point on the right side.
- 7 *Next point for clipping path.* Select ESC twice to close the tool. Allplan automatically dimensions all points where the line intersects the design.



Adding dimension line points

The dimensions of the Isokörbe are still missing. So, the next step is to add these dimension-line points to the dimension line that you have just created.

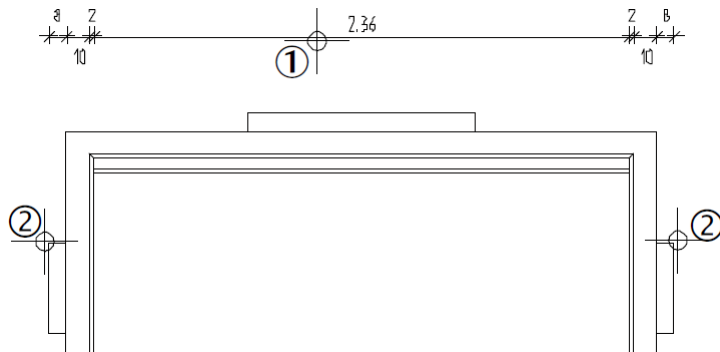
Dimension lines that were created with the **Associative** option not being selected automatically adapt in the following cases:

- When you work with drawing files: The dimension lines and the dimensioned component must be in the same drawing file, or the drawing file with the dimensions must be open in edit mode while you are making modifications.
When you work with layers: The layer containing the dimensioning must be visible.
- When modifying a dimensioned element, you must select the reference points too.

Due to changes in the design, dimension-line points are often missing or no longer required. In these cases, you do not need to create a new dimension line. You can simply add or delete dimension-line points.

To add dimension-line points

- 1 Right-click the dimension line to which you want to add dimension-line points and select  **Add Dimension Line Point** on the shortcut menu.
- 2 Click the points to be dimensioned (left and right outer edges of the Isokörbe).

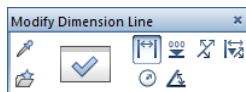


- 3 Select ESC twice to close the tool.

Tip: To remove a dimension-line point, open the shortcut menu of a dimension line and click  **Del Dimension Line Point**. Then click the point that you want to delete.

You can also change the parameters of dimension lines (except the spacing between dimension lines).

- Click  **Modify Dimension Line** (shortcut menu of a dimension line or **Label** task – **Dimension** area).

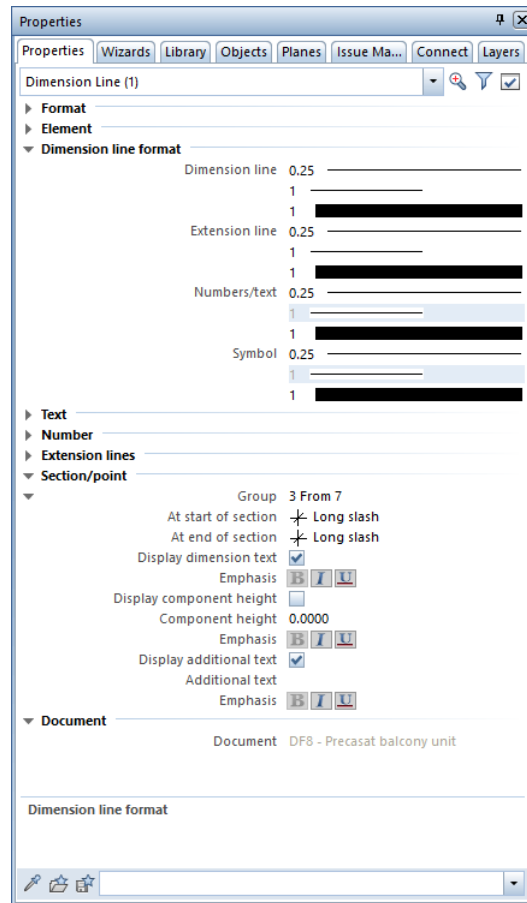


- Define the new dimension-line parameters in the  **Properties**; then click the dimension lines to apply the changes.
You can also match the parameters from existing dimension lines. Click  **Match parameters from dimension line** and click the dimension line whose settings you want to use.
- You can also use the context toolbar to select the dimension-line type to which you want to apply the changes and then enclose all dimension lines in a selection rectangle.

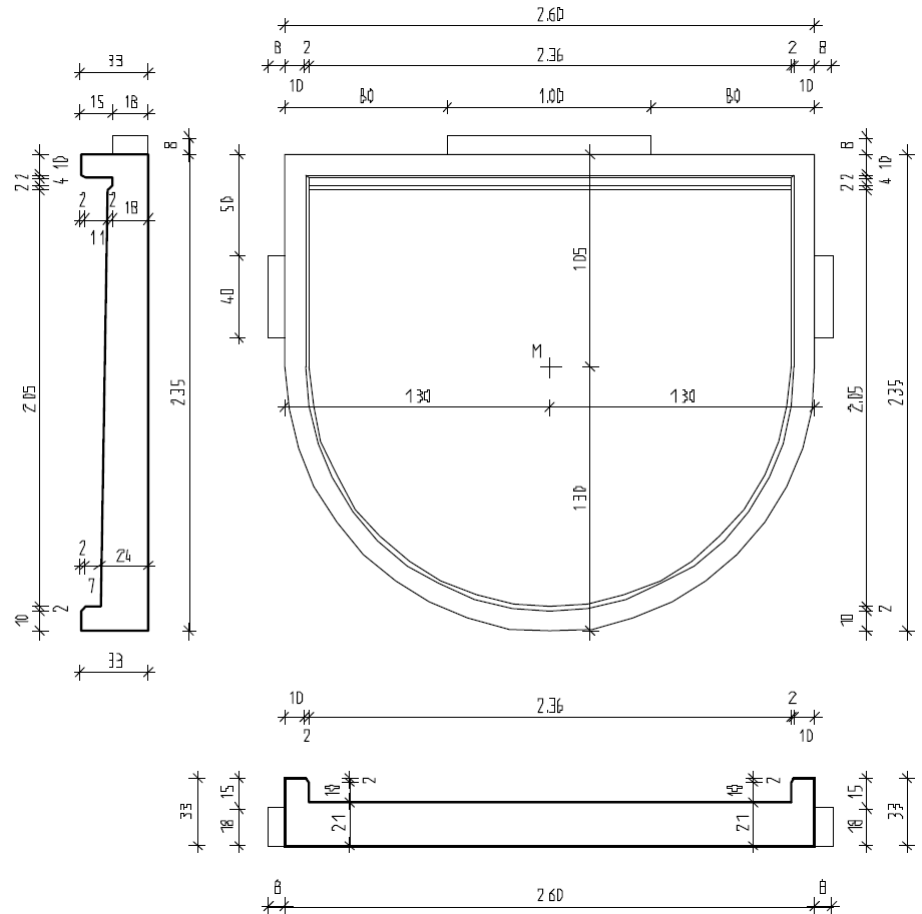
- You can also click the dimension line that you want to modify and then select and change the dimension-line parameters in the **Properties** palette.

For example, you can change the arrowheads or other parameters of the dimension-line section clicked.

Use  and  to switch between the dimension-line sections or elevation points.



Complete the dimension lines as shown in the illustration:



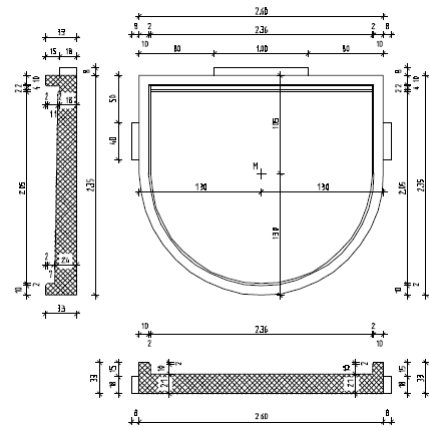
Task 3: applying hatching to the precast balcony and printing the result

In the following part of the exercise, you will apply hatching to the sections of the precast balcony. You will use the tools in the **2D Objects** and **2D Areas** task areas.

Tools:

-  Hatching
-  Area detection
-  Convert Surface Element
-  Reshape Surface Element, Architectural Area
-  Show/Hide
-  Print Preview
-  Print

Objective:



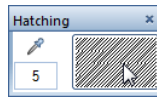
Defining and creating hatching

First, you will select a hatching style that represents reinforced concrete and apply it to the longitudinal section of the precast balcony.

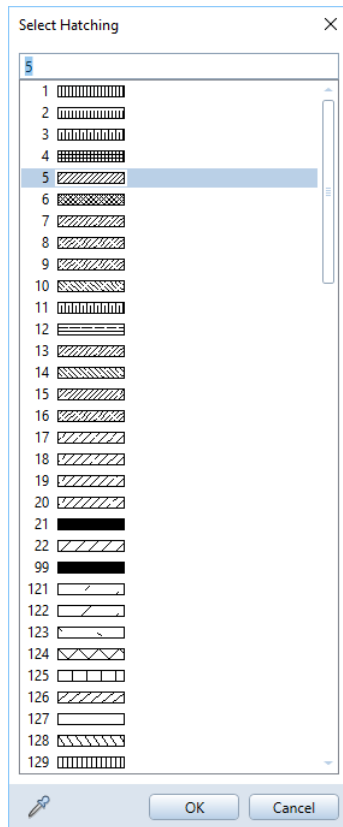
To define and create hatching

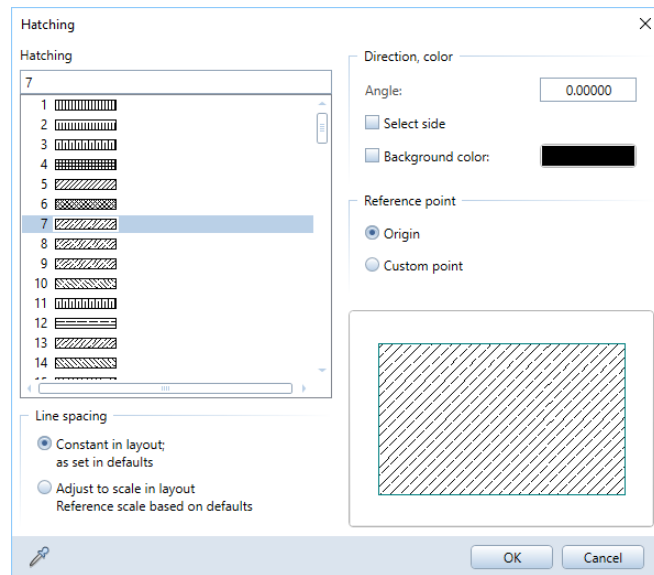
- 1 Select the  **Hatching** tool in the **2D Areas** task area.
- 2 Click the button with the hatching style on the **Hatching** context toolbar.

Tip: When you click the hatching number on the **Hatching** context toolbar, you can quickly select a hatching style in the following dialog box:



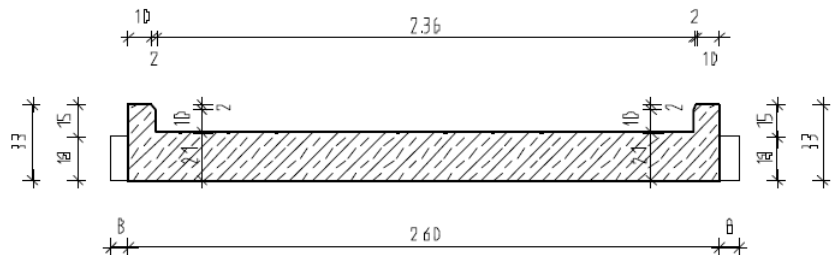
- 3 Select hatching style **7** in the **Hatching** dialog box.
- 4 Define additional hatching settings:
 - **Line spacing** area:
Constant in layout, as defined in defaults
 - **Reference point** area:
Origin





Tip: To apply hatching to rectangular areas, click two diagonally opposite points. Then select ESC.

- 5 Click **OK** to confirm the dialog box.
- 6 Click  **Area detection** (input options, icon must be pressed in).
- 7 Click within the area to which you want to apply hatching. Allplan automatically detects the boundary of the area.
- 8 Select ESC to finish defining the area.



- 9 Select ESC to close the tool.

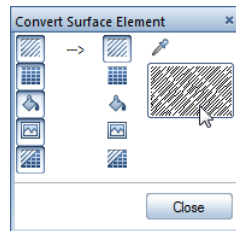
Changing the hatching style

The next step involves replacing the hatching style for reinforced concrete with a hatching style for precast elements. All you need to do is change the hatching style.

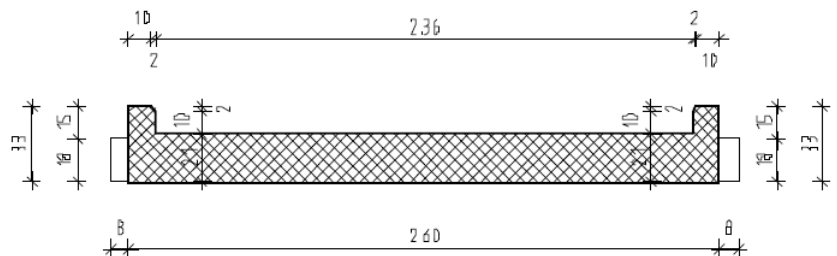
To change the hatching style

- 1 Click  **Convert Surface Element** (2D Areas task area).
- 2 You can see the **Convert Surface Element** dialog box. Make settings as shown in the illustration and click the button with the hatching style.

Tip: To change the hatching style, you can also click the hatching. The **Properties** palette opens; modify the required hatching parameters.



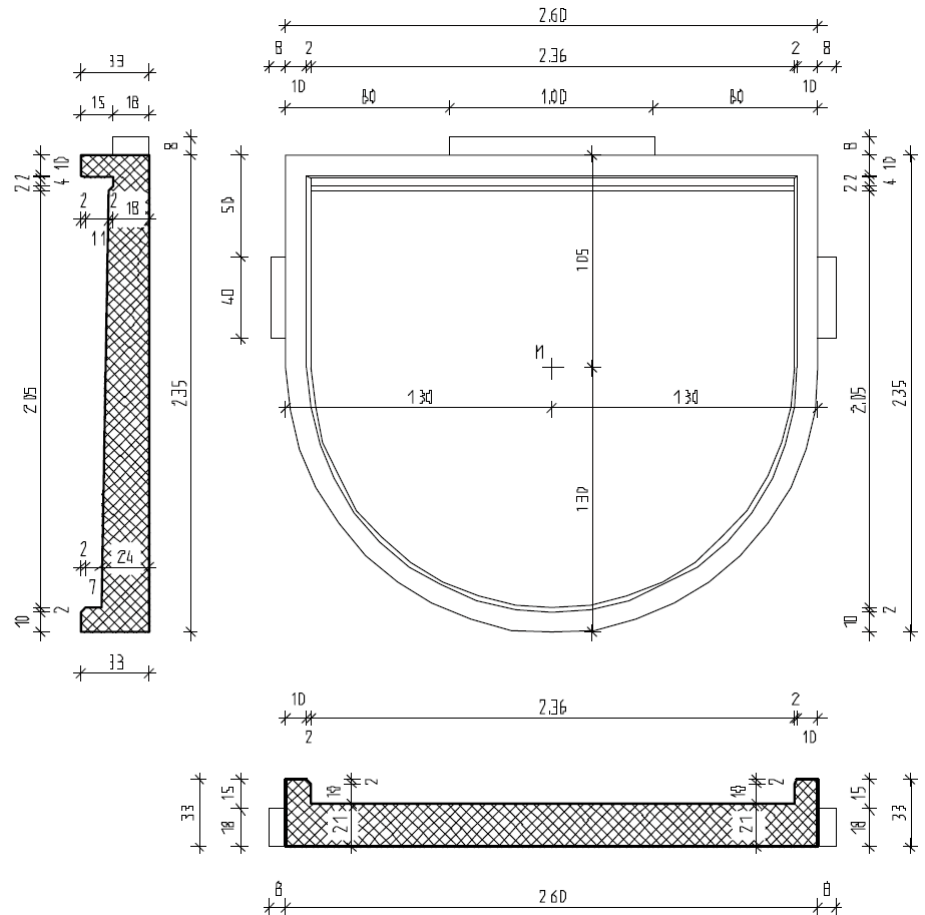
- 3 Select hatching style **6** in the **Hatching** dialog box (see "Defining and creating hatching" on page 224); click **OK** to confirm.
- 4 *Select surface elements to convert to hatching:* Click the hatching of the longitudinal section.



- 5 Select ESC to close the tool.

Applying hatching to the cross-section

Apply hatching to the cross-section as shown in the illustration:



Printing the screen contents

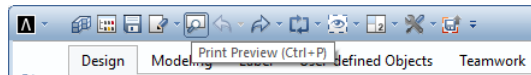
Finally, you will print out the precast element. You can directly print intermediate results without having to assemble a layout first. For example, you can quickly print details or sections of floor plans if you need the current edit status on paper.

To create quick printouts, you can use the  **Print** and  **Quick Print** tools in the drop-down list of the Allplan icon on the title bar. In addition, you can use the  **Print Preview** tool on the Quick Access Toolbar:

- The settings (for example, margins, header, footer, construction lines) in the **Print Preview** palette will be used for printing. You can also define the scale and the section that you want to print.
- Use  **Print** to print the contents of the active viewport to a printer that you select in advance.
- Use  **Quick Print** to print the current contents of the screen without defining print settings beforehand. The default printer will be used for printing.

To print the screen contents

- 1 Click  **Print Preview** (Quick Access Toolbar).

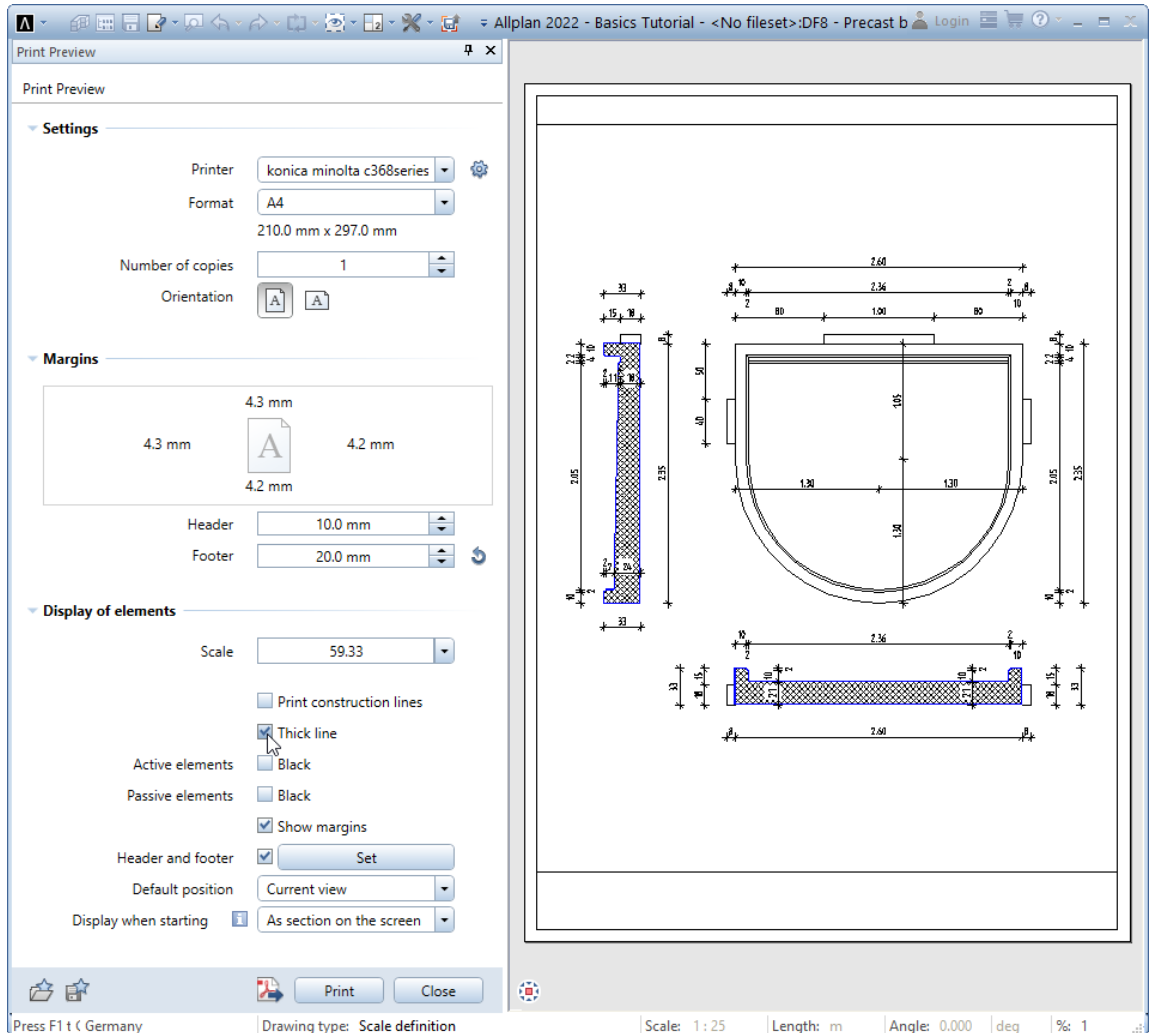


The palettes and the **Actionbar** disappear from the screen. Only the **Print Preview** palette is visible.

- 2 Define the settings for print preview in the **Settings**, **Margins**, and **Display of elements** areas.

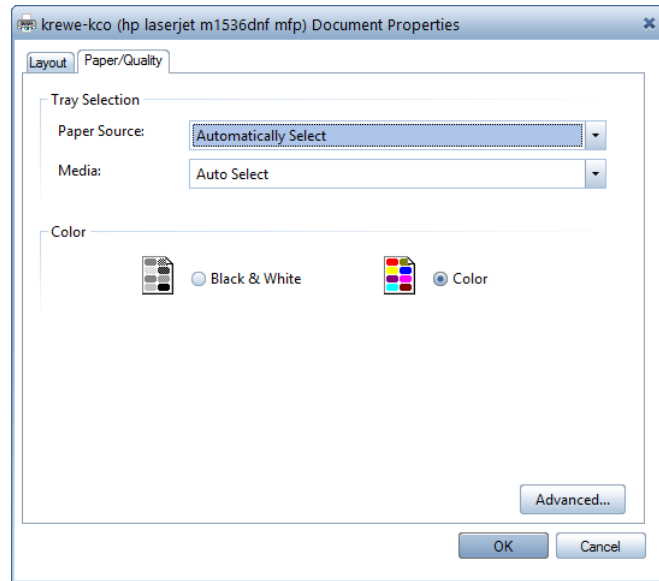
Go to the **Display of elements** area and select the **Thick line** option. As a result, different line thicknesses are visible on the screen and in printouts.

By using the **Print construction lines** option, you can choose to include construction lines in the printout. Define the other options as you want.



- 3 Define the scale in the **Display of elements** area.
- 4 Go to the **Settings** area and select the **Printer** if you do not want to use the default printer.

Note: Click the  **Settings** icon next to printer to define the properties (for example, paper options and color options) of the selected output device or raster driver.



- 5 Click **Print** at the bottom of the palette.

The program sends the document to the selected printer.

- 6 To close print preview, click **Close** at the bottom of the palette.

Note: To create more printouts, click  **Quick Print** in the drop-down list of the **Allplan icon**. This tool uses the most recent settings to print the screen contents without prompting you again.

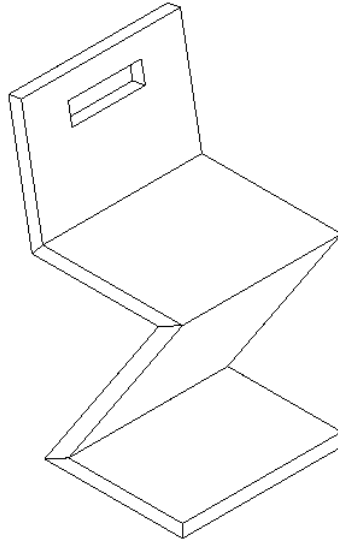
Unit 3: 3D modeling

Unit 3 introduces the **Modeling** task. You will create a chair based on the zigzag chair designed by Gerrit T. Rietveld. You will learn

- How to create the initial elements in 3D
- How to automatically create a 3D solid based on a profile and a path
- How to define a work plane so that you can draw in a sloping plane as if you were working in plan
- How to design a box and use it to create the opening in the back of the chair

Exercise 7: Rietveld chair

This exercise involves creating a chair based on the zigzag chair designed by Gerrit T. Rietveld.



You will use tools in the **Modeling** task (3D Objects, Boolean operators, and Work Environment task areas).

Task 1: designing the initial elements

You will start by drawing the profile and elevation view of the chair as 3D elements in plan. After this, you will rotate these 3D elements in space.

Tools:



3D Rectangle

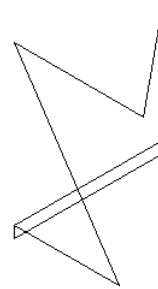


3D Line



Rotate 3D Elements

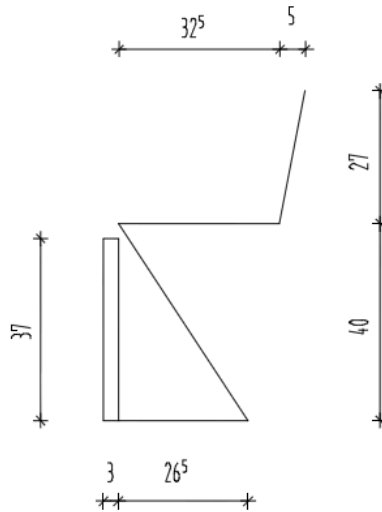
Objective:



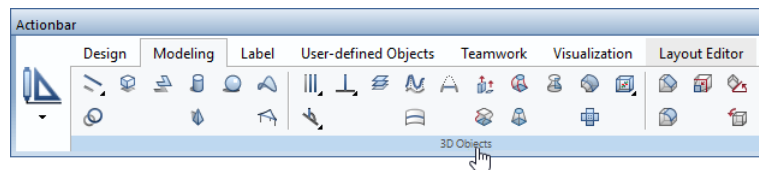
Designing the 3D elements

Start by creating the profile and elevation of the chair by using the **3D Rectangle** and **3D Line** tools.

To draw the cross-section and profile of the chair in plan

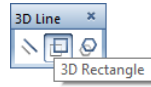


- 1 Click  **Open on a Project-Specific Basis** and open an empty drawing file. Name it **Rietveld chair** and close all the other drawing files.
- 2 Go to the **Actionbar** and switch to the **Modeling** task, which you can also find in the **Draft** role.
- 3 Expand the **3D Objects** task area by double-clicking within the name line of this task area.

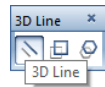


- 4 Click  **3D Line** in the **3D Objects** task area.

- 5 To draw the profile of the chair, select  **3D Rectangle** on the **3D Line** context toolbar.

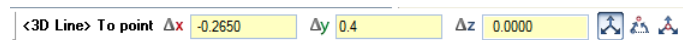


- 6 Select  **Based on diagonal line** in the input options.
- 7 Click in the workspace to define the starting point of the rectangle.
- 8 The rectangle is 0.03 m long in the x-direction. Therefore, enter  **dx = 0.03** in the dialog line. Select the Tab key to go to  **dy**.
- 9 The rectangle is 0.37 m high in the y-direction. Therefore, enter  **dy = 0.37** in the dialog line.
- 10  **dz** is **0.00**. Do not change this value. Select Enter to confirm.
- 11 Click  **Zoom All** on the viewport toolbar.
- 12 To create the elevation of the chair, select the  **3D Line** option on the **3D Line** context toolbar.



- 13 Make sure that  **Polyline** is selected in the input options.
- 14 Click the lower-right corner of the rectangle to define the starting point of the line.
- 15 To draw the first line, select  **Delta point** in the dialog line, enter  **dx = 0.265**, and select ENTER to confirm.
- 16 Enter the following values in the dialog line:
 **dx = -0.265** TAB
 **dy = 0.40** ENTER

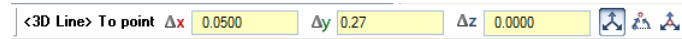
Tip: Select the Tab key to go to the next box in the dialog line.



- 17 To draw the third line, enter  **dx = 0.325** in the dialog line and select ENTER to confirm.

- 18 To draw the fourth line, enter the following values in the dialog line:

 **dx** = 0.05 TAB
 **dy** = 0.27 ENTER

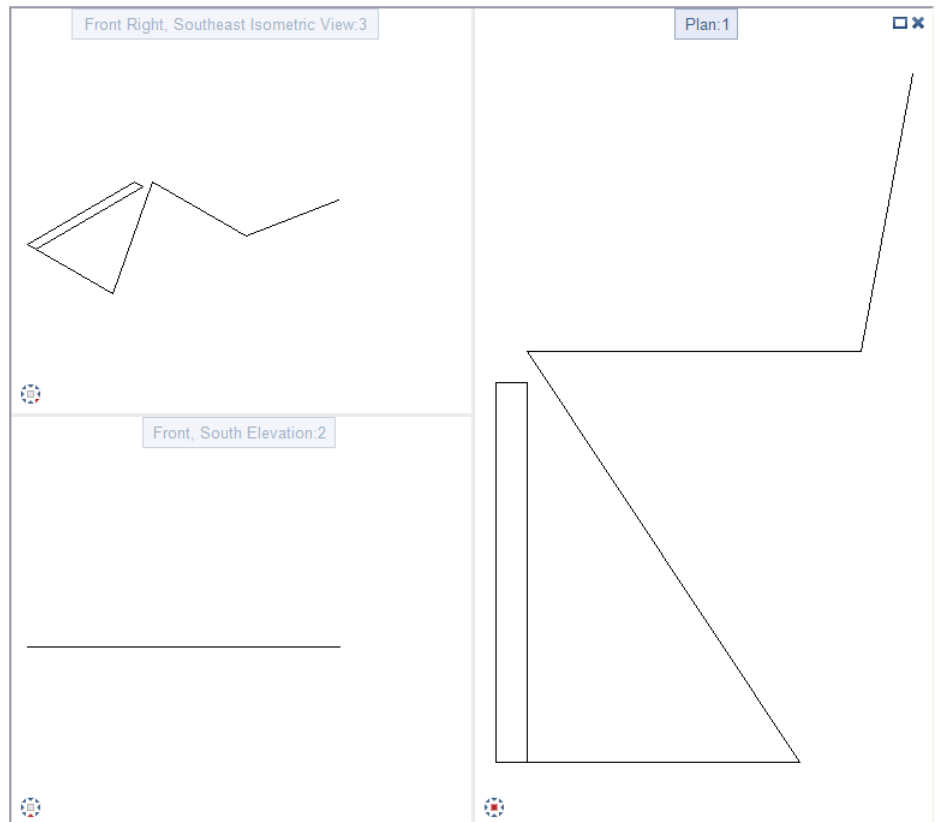


- 19 Select ESC twice to close the  **3D Line** tool.
- 20 To get an impression of how the elements look in 3D, click  **3 Viewports** ( **Window** drop-down list on the Quick Access Toolbar).

The elements are now displayed in plan view, and to the right of them in isometric and side view.

- 21 In the  **Window** drop-down list for the **viewport layout**, select the setting on the **left** and click the ALT KEY to temporarily show the menu bar.
Open the **View** menu and click **Zoom All in All Viewports**.

Your screen should now look like this:



You will continue to work in these **3 viewports**.

Rotating the 3D elements

The 3D elements are still flat on the floor (xy plane). The next step is to rotate the 3D elements in space. The difference between this and rotating elements in 2D is that you can define an axis of rotation which lies freely in space (in 2D, you can enter a point of rotation only).

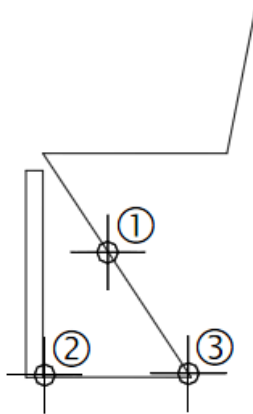
To rotate the 3D elements

➡ **Actionbar: Draft** role – **Modeling** task.

- 1 Select  **Rotate 3D Elements** (**3D Objects** task area) and click the elevation view of the chair.

The element appears in the selection color.

- 2 Define the bottom line of the chair's elevation view as the axis of rotation. First click the left point of the line. The sequence in which you enter the points is important for defining the angle later.
- 3 Click the right point of the line. This defines the axis of rotation.

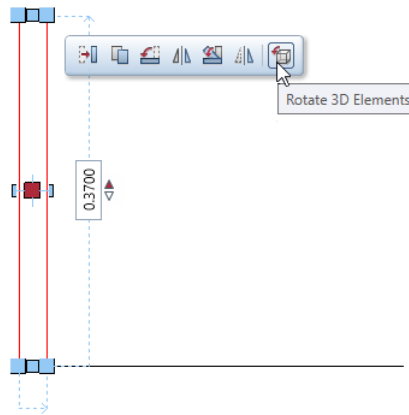


- 4 Rotate the elevation view upward by 90 degrees. Enter **90** and select ENTER to confirm.
The rotated elevation view of the chair is visible as a straight line in plan view (see illustration).
- 5 Select ESC to close the  **Rotate 3D Elements** tool.

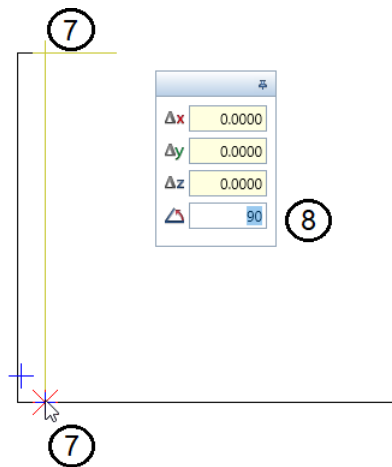
Tip: You can use the “right hand rule” to identify the positive direction of the rotation angle. Point the thumb of your right hand in the direction of the rotation axis; your fingers indicate the positive direction of rotation.

- 6 Rotate the profile of the chair downward.

You can also find the  **Rotate 3D Elements** tool on the **direct object modification** toolbar. Click the profile and select this tool.

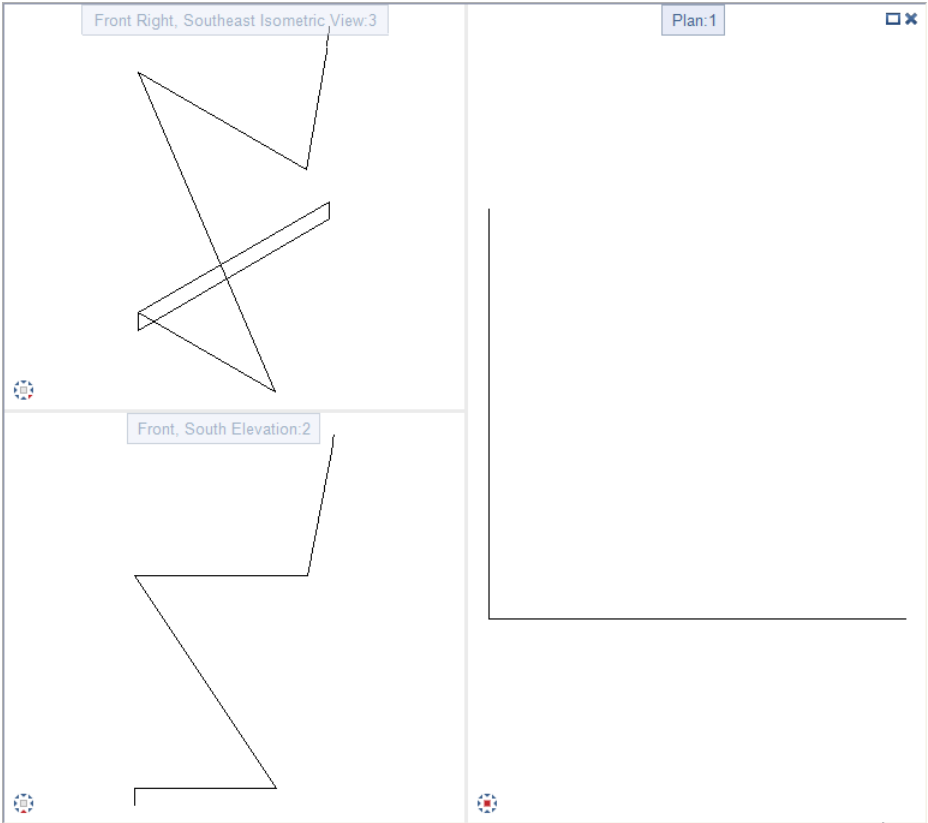


- 7 Define the axis of rotation as shown in the illustration. Here, too, click the point at the top first.
- 8 Enter the angle of rotation: **90**.



- 9 Select ESC to close **direct object modification**.
- 10 Select the ALT KEY to show the menu bar for a short time. Open the **View** menu and click **Zoom All in All Viewports**.

Now your screen should look like this:

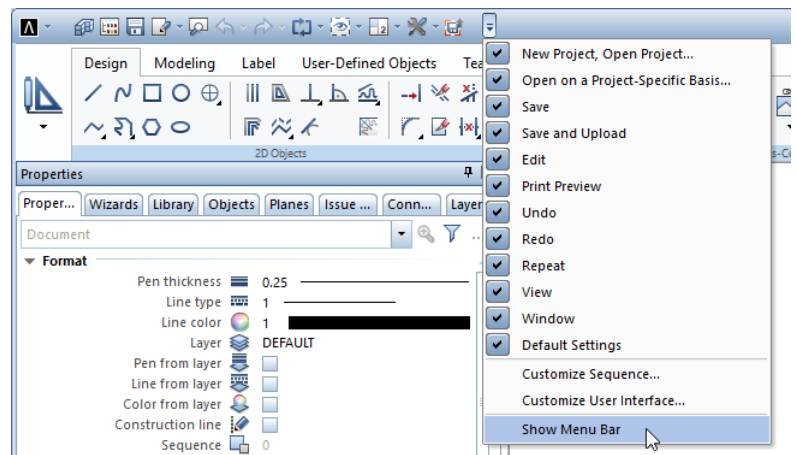


Note: When you use the Actionbar configuration, the menu bar is hidden by default. If you want, you can show the menu bar all the time.

To show the menu bar

- 1 Click the drop-down list on the Quick Access Toolbar (title bar).
- 2 Click **Show menu bar**.

Tip: As you did in the previous exercise, you can temporarily show the menu bar by selecting the Alt key. The menu bar disappears again after you have selected a tool.



The menu bar appears below the title bar.

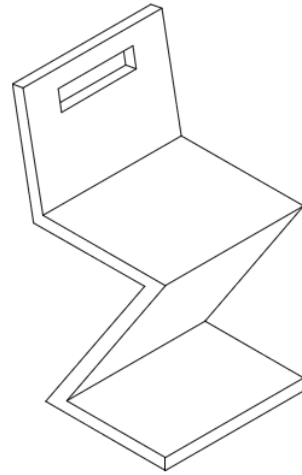
Task 2: designing the 3D solids

You will create the chair from the 3D elements. Start by designing the opening in the back of the chair as a 3D solid (box). You will then subtract this opening from the back of the chair.

Tools:

-  **Extrude Along Path**
-  **Work Plane**
-  **Box**
-  **Subtract and Remove Solid**

Objective:



Extruding along a path

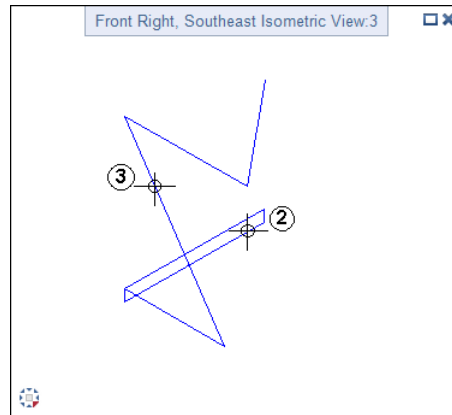
The next step is to create the chair by means of the **Extrude Along Path** tool. The profile to be extruded is the profile of the chair. The path is the elevation view of the chair.

To extrude the elevation view of the chair along a path

- 3 Viewports are still open.
- Actionbar: **Draft** role – **Modeling** task.
- 1 Click  **Extrude Along Path** (3D Objects task area).
- 2 *Select profile to extrude*
Click the profile of the chair in isometric view.

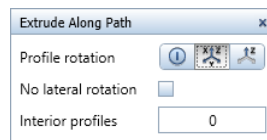
3 *Select path*

Click the elevation view of the chair in isometric view.



4 *Set parameters, press ESC to confirm*

Check the settings in the **Extrude Along Path** dialog box and select ESC to confirm.



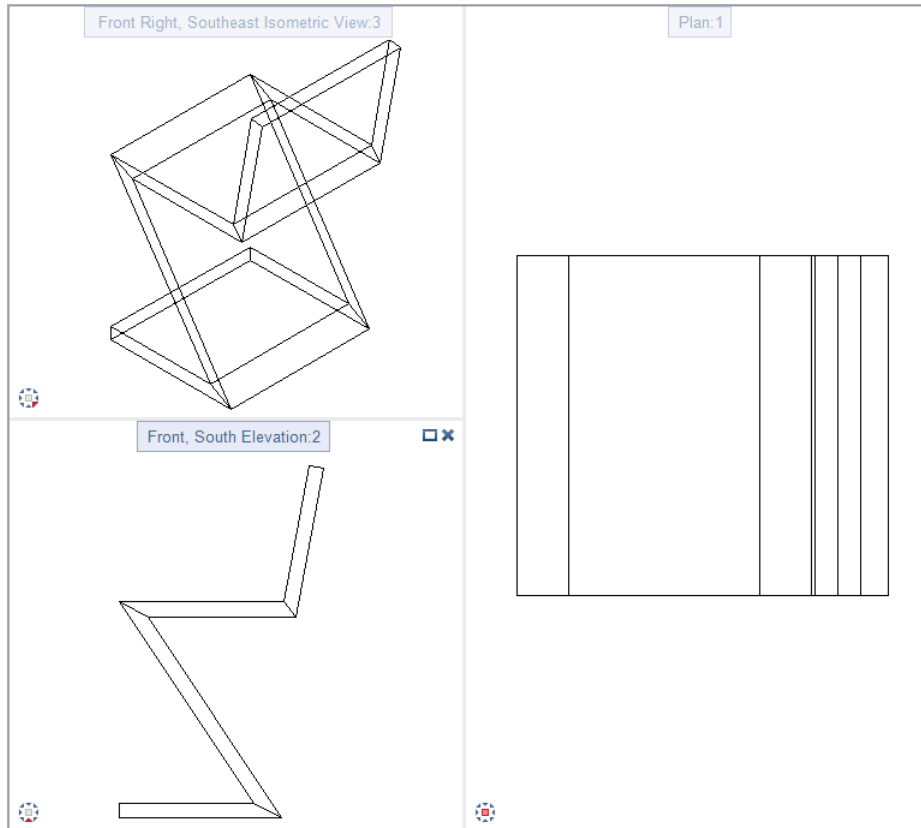
5 The menu bar is still visible (see previous exercise). If this is not so, select the ALT KEY to show the menu bar.

Open the **View** menu and click **Zoom All in All Viewports**.

Or

Select SHIFT+F5.

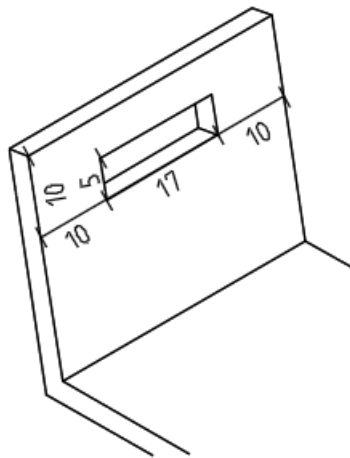
Now your screen should look like this:



6 Select ESC to close the  **Extrude Along Path** tool.

Defining a work plane and creating a 3D box

The chair is still missing the opening in its back. You will start by creating it as a box. To make it easier to position the box in the sloping back of the chair, you will define a work plane (= user-defined coordinate system) whose x-axis and y-axis are parallel to the edges of the back of the chair. Consequently, you can draw in the work plane as if you were working in plan.



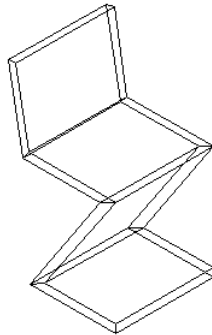
To define the work plane and create a 3D box

➔ **3 Viewports** are open.

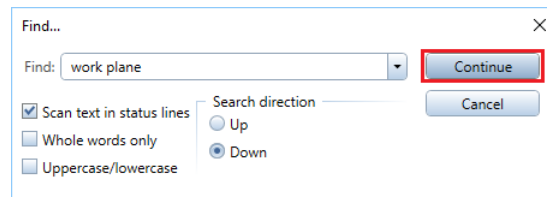
➔ **Actionbar: Draft** role – **Modeling** task.

- 1 Click in the viewport that displays the chair in plan.
- 2 To enter the work plane, display the chair in isometric view. Click  **Rear, Left Isometric View**.

The result should look like this:

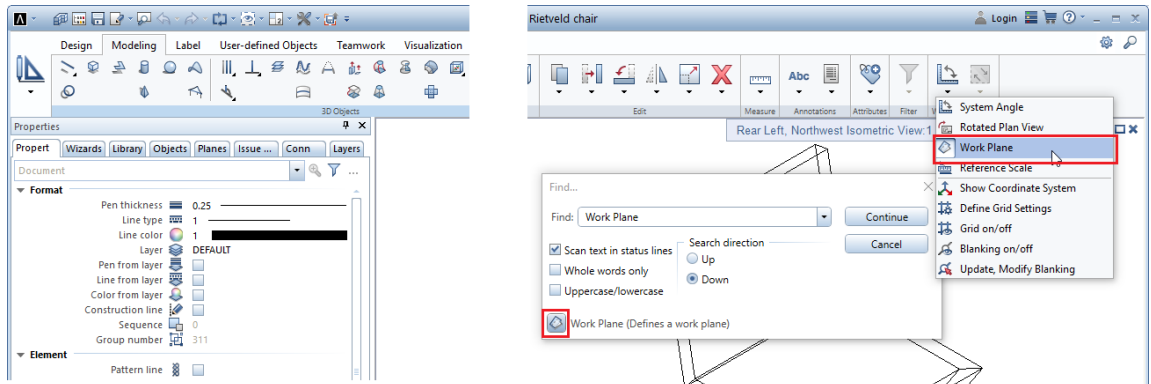


- 3 You need the  **Work Plane** tool for the following steps.
To quickly select this tool, you will use the search function of the **Actionbar**.
To do this, click  in the upper-right corner of the **Actionbar**.
- 4 Enter **work plane** in the **Find...** dialog box and click **Continue**.

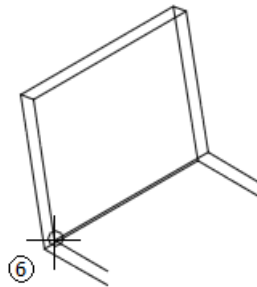


The **Work Environment** task area displays the  **Work Plane** tool.

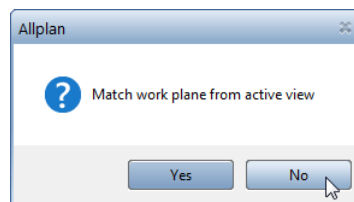
- 5 There are two ways to select the  **Work Plane** tool:
 - Click the tool in the **Work Environment** task area.Or
 - Click the  **Work Plane (Defines a work plane)** icon in the lower part of the **Find...** dialog box.



- 6 To define the origin of the work plane, click the lower-left corner of the back of the chair (see illustration).



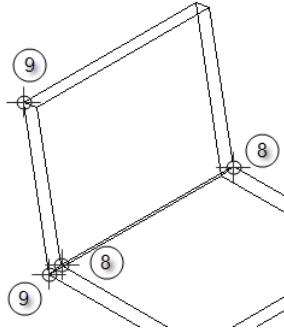
- 7 Click **No** because you do not want to use the current view as the work plane.



Define the work plane by entering four points.

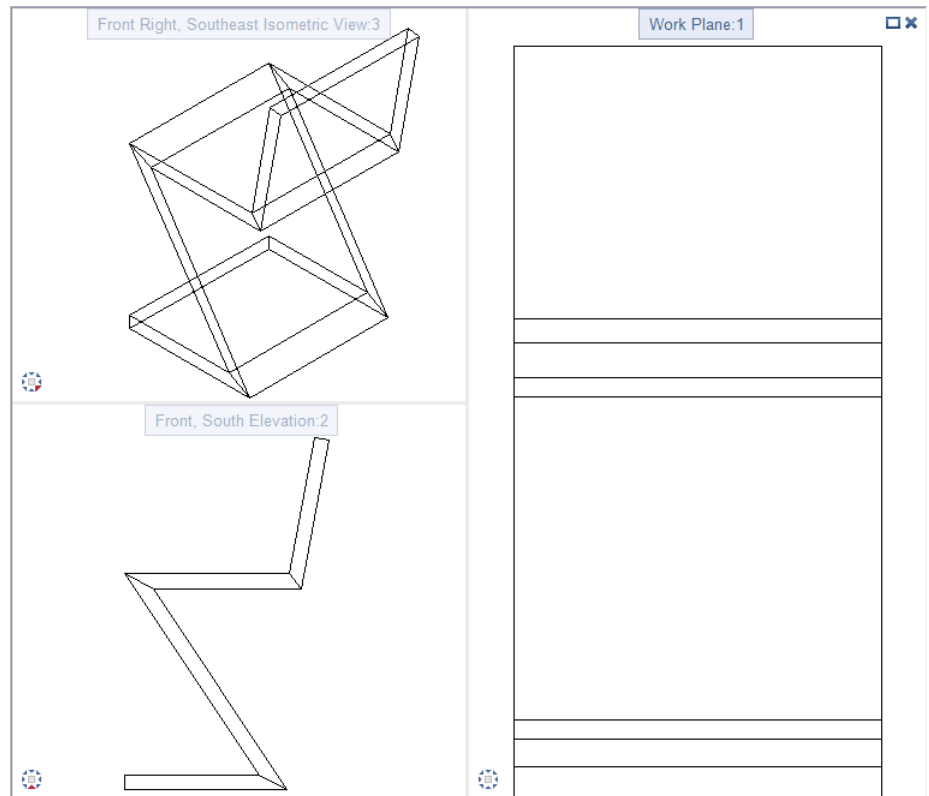
- 8 To define the x-axis, click the two end points of the bottom edge of the chair (see illustration). As the positive x-axis is to the right, click the point on the left first.

- 9 To define the y-axis, click the two end points of the rear left edge of the chair (see illustration). As the positive y-axis is upward, click the bottom point first.



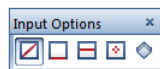
The z-axis is perpendicular to the xy plane in the origin.

The result should look like this:



Now all entries you make apply to the axes of the defined work plane: You can enter the values defining the box as if you were working in a 2D floor plan.

- 10 To draw the box, click  **Box** (3D Objects task area).
- 11 Check that the **Based on diagonal line** setting is selected in the input options. If it isn't, select it now.



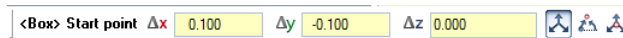
Always work in the right viewport!

- 12 Make sure that  **Delta point** is selected in the dialog line.

- 13 Point to the upper-left point to define the reference point for the corner of the box. You can see this point in the other two viewports too.



- 14  **Delta point** is selected.
Enter the offset of the corner in the dialog line:
 $\Delta x = 0.1$
 $\Delta y = -0.1$

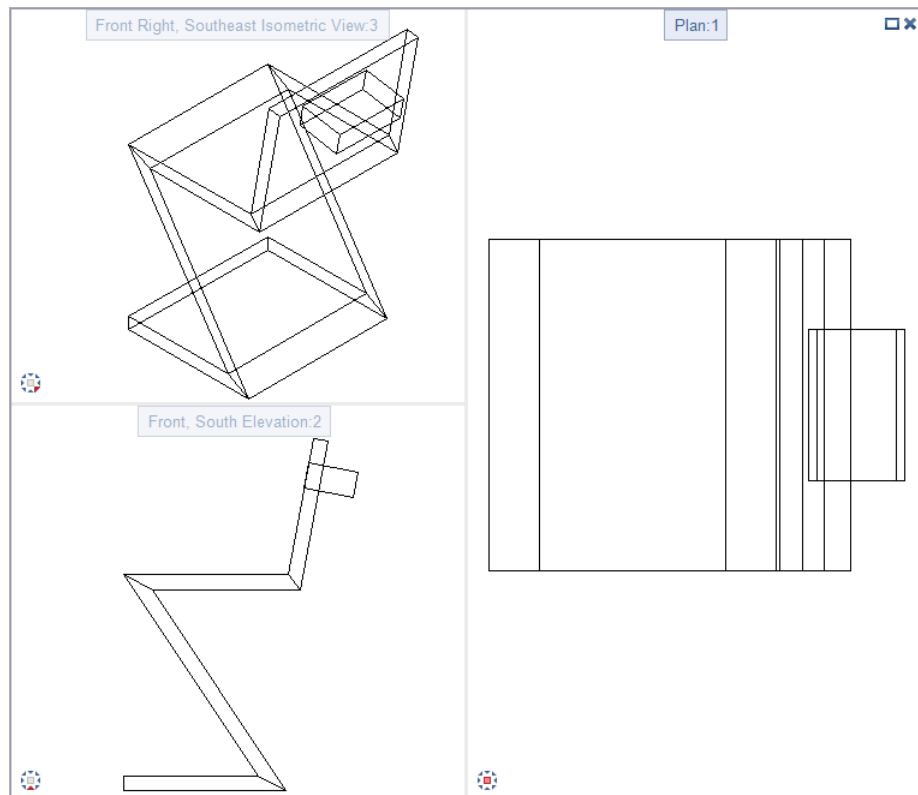


Select ENTER to confirm.

This defines the corner of the box.

- 15 Enter $\Delta x = 0.17$ for the length of the opening and $\Delta y = 0.05$ for its width. Then select ENTER to confirm.
- 16 To define the height of the box in the z-direction, enter a value that is greater than the thickness of the back of the chair. Enter -0.10 for the height. You have drawn the box.
- 17 Click  **Plan** to return to the usual work plane.

Now your screen should look like this:



18 Select ESC to close the  **Box** tool.

Creating the opening

To finish, you will remove the volume of the box from the 3D element, thus deleting the box.

To create the opening

➤ **Actionbar: Draft role – Modeling task.**

- 1 Go to the **Boolean Operators** task area and click  **Subtract and Remove Solid** to create the opening in the back of the chair and to delete the box at the same time.

Tip: Expand the **Boolean Operators** task area.

Do you still know how to do this?

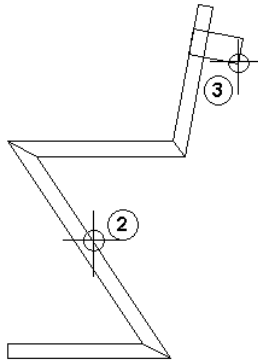
Double-click within the name line of the task area.

2 *Click 1st solid*

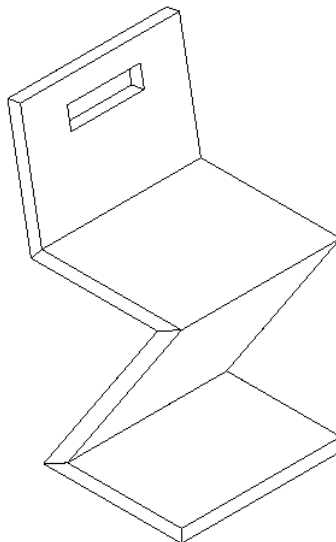
Click the chair; this is the solid in which you want to create the opening.

3 *Select solids, right-click to confirm*

Click the box; this is the (only) solid you want to subtract from the first solid. Right-click to confirm.



This creates the opening; the resulting chair might look like this (**Hidden** view type on the viewport toolbar):



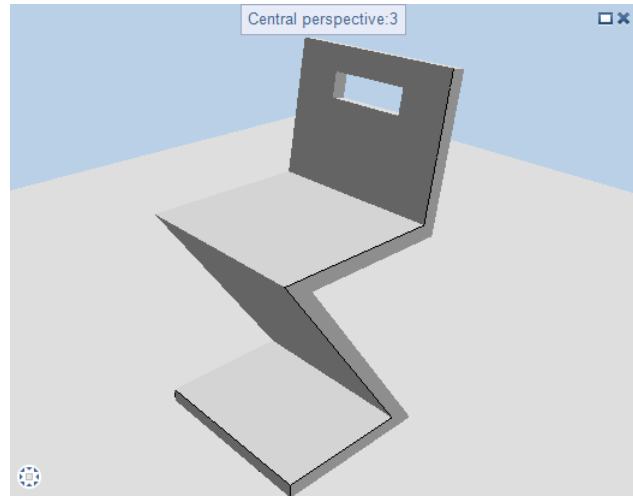
Excursus: design check, color, and texture

To check the design

- 1 Select the F4 key.

A viewport of the **Animation** view type opens, showing the chair.

 **Navigation Mode** is selected.



See also: You can find more information about using the mouse in animation in the Allplan Help. See "Sphere mode, camera mode".

- 2 Select and hold one of the mouse buttons, then drag; You move around the virtual model in **sphere mode**, which is the default setting.
Select the Ctrl key at the same time to switch to **camera mode**.
- 3 Experiment with the navigation modes; start trying things out on your own.

- 4 The chair's color is boring? What about red?

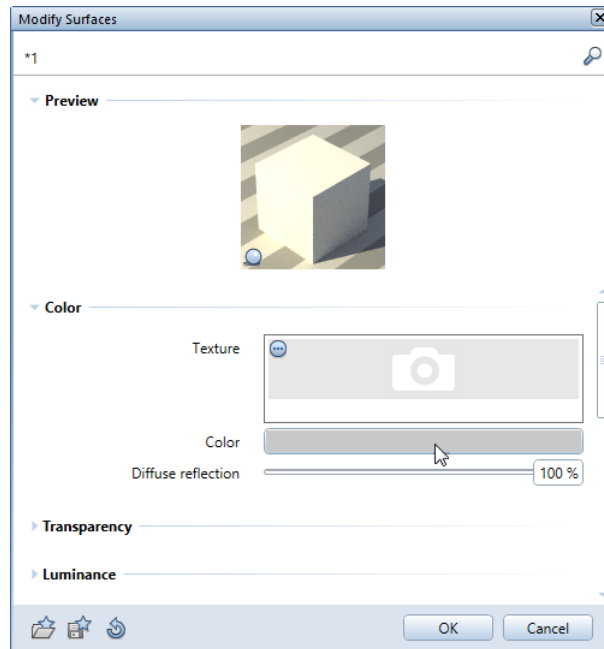
 **Navigation Mode** is still selected in the animation viewport. Right-click the chair; the shortcut menu opens.

Click  **Set Surface**.

The **Modify Surfaces** palette opens.

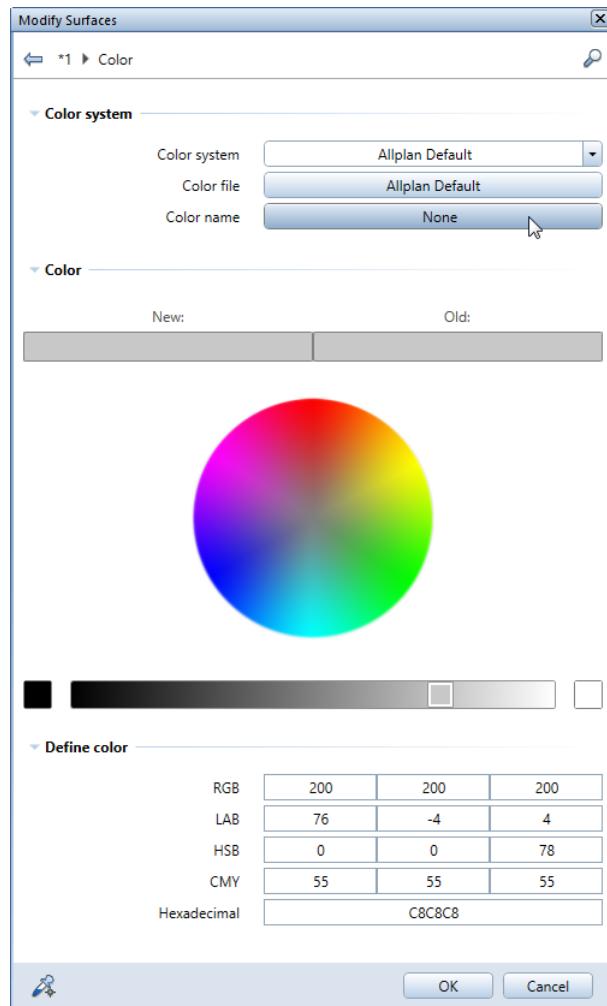
You can check and change the surface properties of the chair.

- 5 Click the button in the **Color** area.

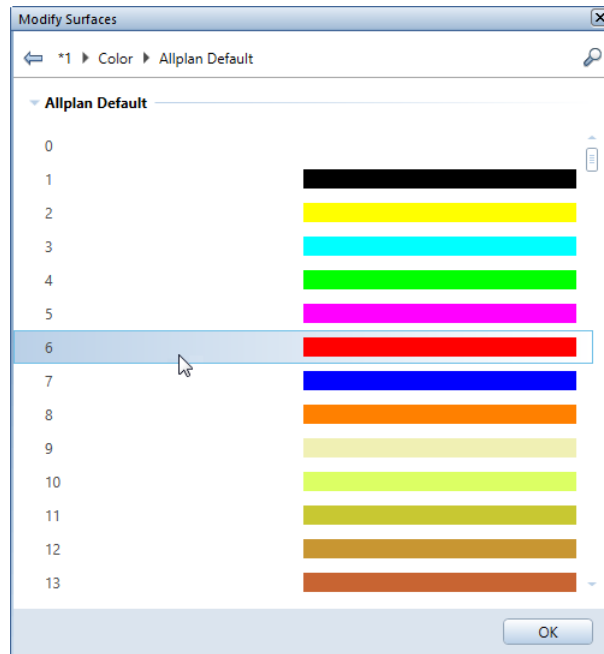


The palette shows the current color of the chair.

- 6 Go to the **Color system** area and click the button next to **Color name**.

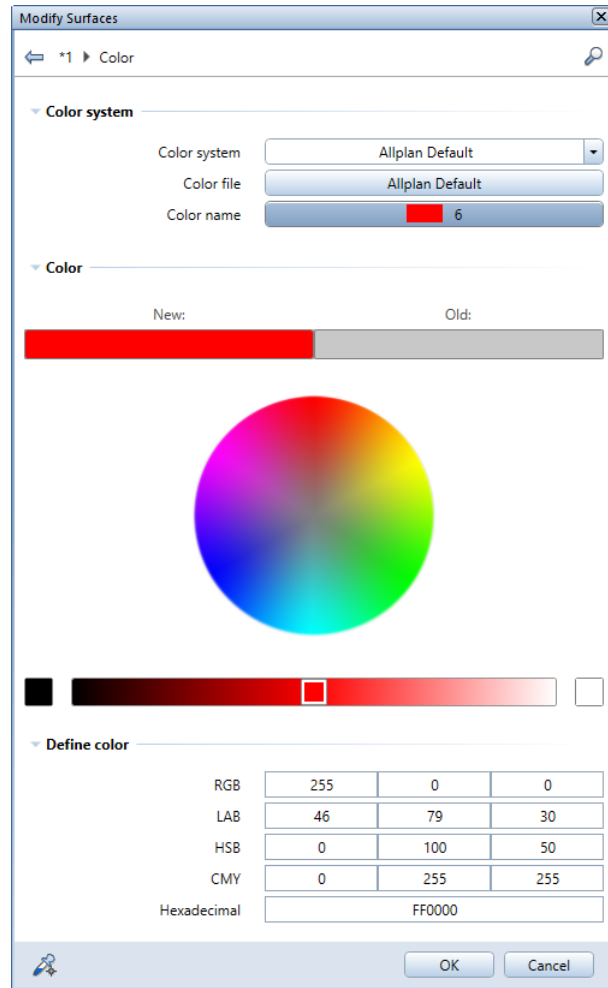


7 Select a color by clicking it.



The palette shows the old color and the new color.

Tip: You can also select a color in the color circle, use the slider, or enter values in the **Define color** area.

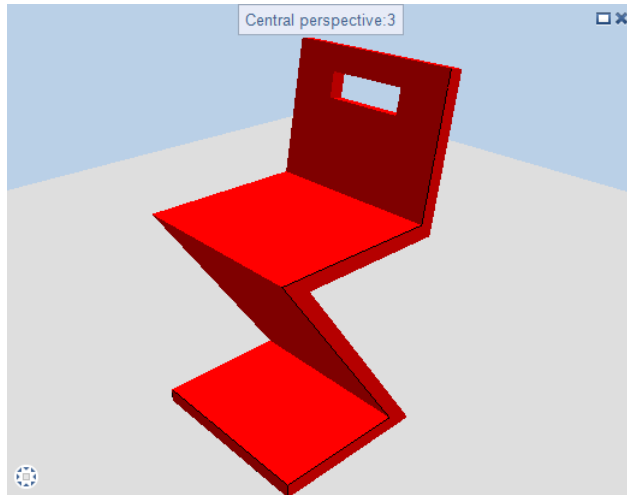


8 Click **OK** to confirm the palette.

You can now customize the surface by defining more properties.

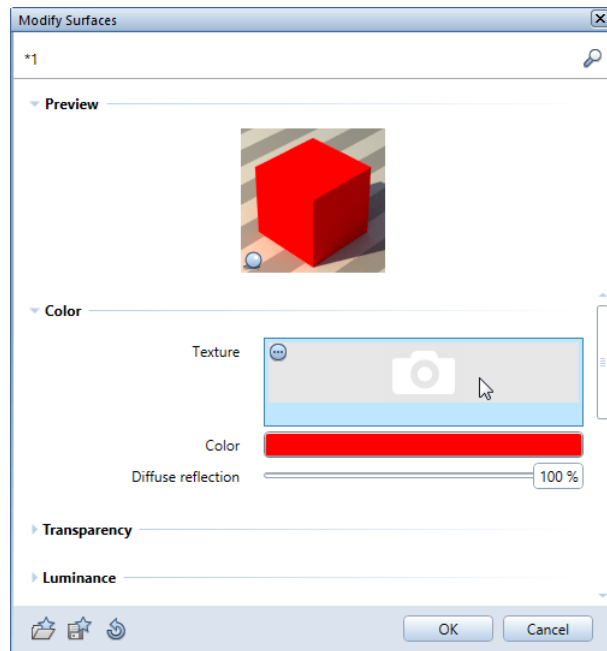
9 Finally, click **OK** to confirm the palette.

The chair appears in the selected color.



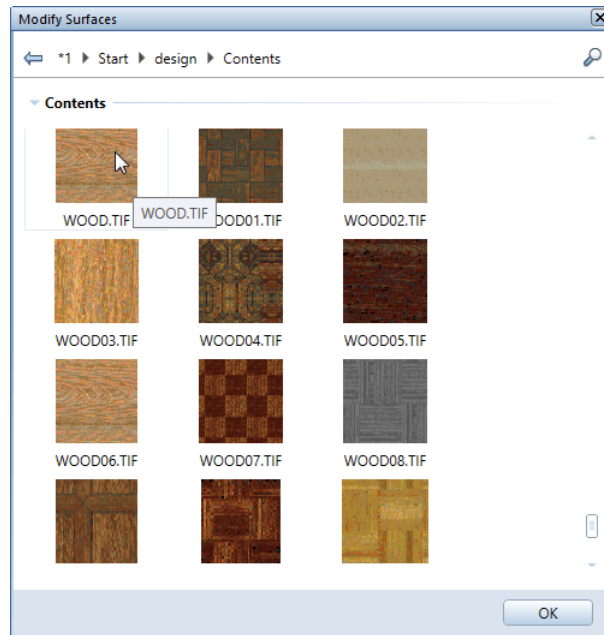
- 10 Or what about the wood grain? Would you like to see it?
Right-click the chair in the animation viewport and select  **Set Surface** on the shortcut menu.

In the **Modify Surfaces** palette, go to the **Color** area and click the button next to **Texture**.



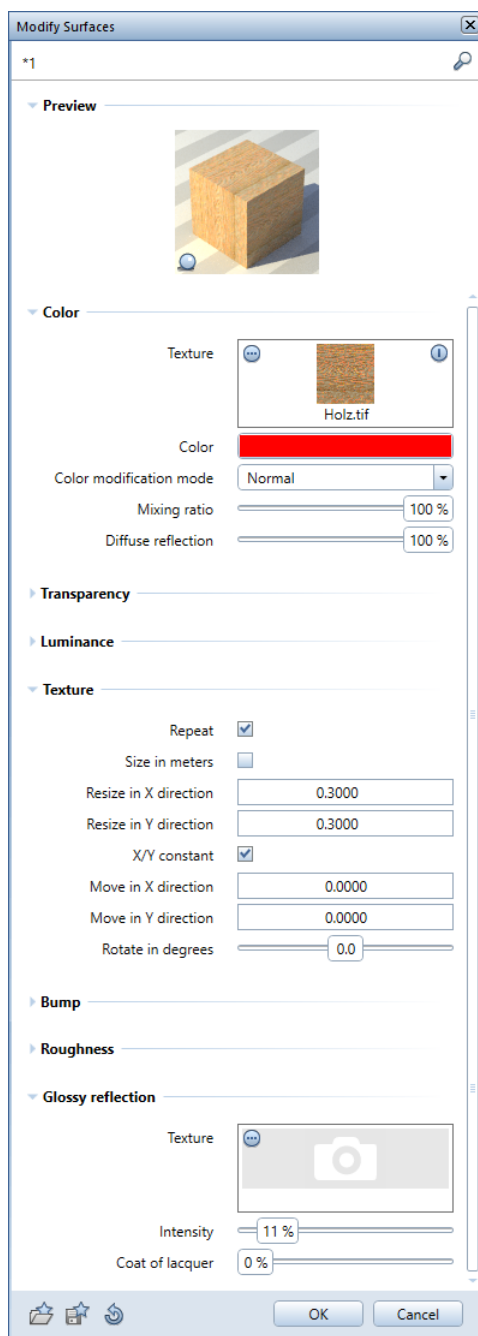
- 11 Open the contents of the design folder in the office standard.
To do this, click the **Office** folder and then the **Contents** folder.

- 12 Select a texture and click **OK** to confirm.

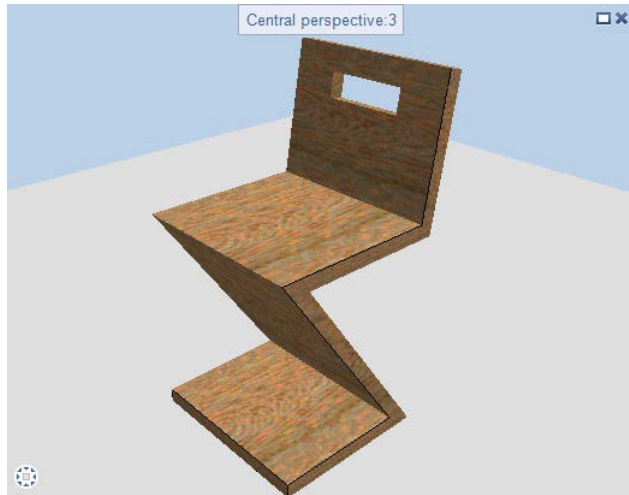


- 13 Continue to modify the surface. You can use the settings that are shown in the illustration. When you have finished, click **OK** to confirm.

Tip: You can find the  **Save as a favorite** tool at the bottom of the **Modify surfaces** palette. By means of this tool, you can save the surface under a new name (for example, `wood_chair.surf`).



The result might look like this:



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